

MTH1002: MATHEMATICAL METHODS (PART 1)

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1 An Introduction to Conic Sections

Families of Numbers

We will deal with various families of numbers in mathematics.

Natural Numbers

Numbers $1, 2, 3, \dots$. We write this *set* as $\mathbb{N}$.

Integers

The set of natural numbers and negative numbers and zero, $0, -1, -2, \dots$ have the set of integers $\mathbb{Z}$.

Rational Numbers

If we consider all quantities of the form $\frac{a}{b}$ where a and b are integers. This set is written $\mathbb{Q}$.

Real Numbers

Some numbers cannot be written as a/b , for example $\sqrt{2}$, and $\pi = 3.141592653\dots$. These numbers, as well as those that can be written as a/b are the real numbers. The set of real numbers is denoted as $\mathbb{R}$.

These $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R})$ are examples of *sets* or collections of objects called ‘elements’. You will be looking into Set Theory in more detail in MTH1001 Mathematical Structures.

We say $A \subseteq B$ if every $x \in A$ is also in B . For example, $A \subseteq \mathbb{R}$ means that A consists only of real numbers. So now we have the following

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R}$$

We can view the set $\mathbb{R}$ as a ‘number line’ and look now at some important examples of subsets of $\mathbb{R}$; so-called *intervals*.

1.1 Prerequisites - Open and Closed Intervals

Open interval (a, b) is the set of numbers

$$(a, b) := \{x : a < x < b\}.$$

Closed interval

$$[a, b] := \{x : a \leq x \leq b\}.$$

Half-open/half-closed interval

$$(a, b] := \{x : a < x \leq b\}.$$

Interval with infinite end points

$$[a, \infty) := \{x : a \leq x\}, \quad (-\infty, a) := \{x : x < a\}.$$

For example, we can write

$$\{x : x^2 > 4\} = (-\infty, -2) \cup (2, \infty).$$

Remember, ∞ is NOT A NUMBER, it's a concept. So whenever ∞ is part of an interval, it is surrounded by an open interval.

*

*'Eternity is pretty boring really, especially towards the end' – Woody Allen

Absolute Value

If $x \in \mathbb{R}$ we define

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

Note that $|x| \geq 0$ for all x .

The *distance* between two real numbers a and b is the quantity

$$|a - b|.$$

Points in a Plane

Note that we can represent points in the plane as pairs $(x, y) \in \mathbb{R}^2$ and use set notation similar. For instance

$$\{(x, y) \in \mathbb{R}^2 : f(x, y) = 0\}$$

means the set of points that satisfy $f(x, y) = 0$. We represent (x, y) with respect to a horizontal x -axis and a vertical y -axis, meeting at an origin $(0, 0) = O$. We say (x, y) is an ordered pair because (x, y) is different from (y, x) - compare with sets where $\{1, 2\}$ is the same as $\{2, 1\}$.

The point with coordinates (a, b) is a units perpendicular from the y -axis and b -units perpendicular from the x -axis (resp. left/right, above/below as appropriate).

There may be many ways of describing the same set in $\mathbb{R}^2$;
for instance

$$\{(x, y) : x - y = 0\}$$

is the same as

$$\{(x, y) : y = x\}, \quad \{(x, x) : x \in \mathbb{R}\}.$$

1.2 Conic Sections

All linear equations in x and y correspond to lines; but what about general quadratics? In other words, what is the geometry of the set of (x, y) such that

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0.$$

for some fixed numbers A, B, C, D, E, F ? There are a number of possibilities, and all of these correspond to examples of “conic sections”.

The circle

We show that a circle has this form: a circle of radius r and center (a, b) is the set of (x, y) such that

$$(x - a)^2 + (y - b)^2 = r^2 \quad (1.1)$$

or equivalently

$$x^2 + y^2 - 2ax - 2by + (a^2 + b^2 - r^2) = 0.$$

Note that we can write a parametric equation for the circle as

$$(x, y) = (a + r \cos \theta, b + r \sin \theta).$$

For example, the set of points (x, y) which are a fixed distance 4 from the point $(2, 3)$ satisfy $(x - 2)^2 + (y - 3)^2 = 4^2$. Thus the equation of a circle of radius 4, center $(2, 3)$ has equation:

$$x^2 + y^2 - 4x - 6y - 3 = 0.$$

In the special case that a circle has centre at the origin, then it can be written $x^2 + y^2 = r^2$.

Example 1.1 Find the equation of the circle with radius 2 centred at $(2, 1)$.

Solution: Using equation 1.1, with $a = 2$, $b = 1$, $r = 2$ we get

$$(x - 2)^2 + (y - 1)^2 = 4.$$

Example 1.2 Find the equation of the circle with centre $(1, -2)$ passing through $(4, 2)$.

Solution: First we determine its radius r which is the distance between $(1, -2)$ and $(4, 2)$. Using Pythagoras' Theorem we get

$$r^2 = (1 - 4)^2 + (-2 - 2)^2 = 9 + 16 = 25$$

which gives $r = 5$. Now we use 1.1 with $a = 1$, $b = 2$, $r = 5$ and get

$$(x - 1)^2 + (y + 2)^2 = 25.$$

In general, the equation of a conic section is given by

$$r = \frac{l}{1 + e \cos \theta}$$

where l is a positive constant, and $e \geq 0$. The constant e is known as the *eccentricity*. When $0 \leq e < 1$ we have an *ellipse*, when $e = 1$ we have a *parabola*, and when $e > 1$ we have a *hyperbola*. Let's look at these *types of conics* in more detail.

The Ellipse

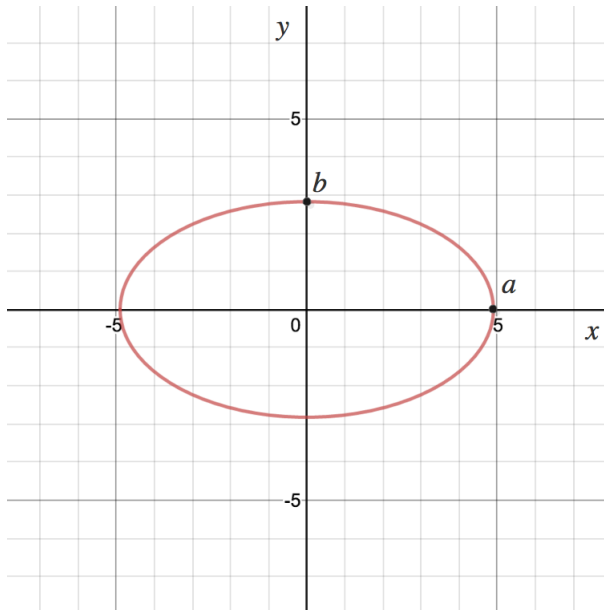
An *ellipse* is given by

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

or any translation or rotation thereof.

When $a = b$, the ellipse reduces to a circle. We can write a parametric equation for the standard ellipse as

$$(x, y) = (a \cos \theta, b \sin \theta).$$

Figure 1.1: *Ellipse*

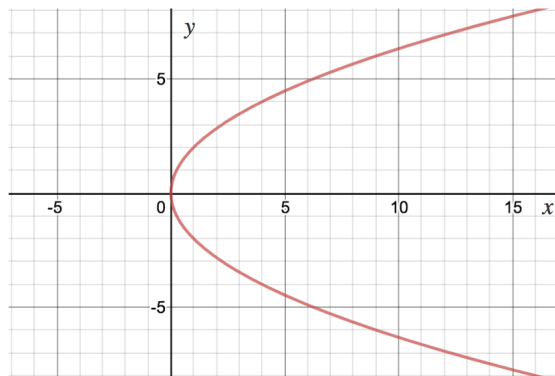
The Parabola

Similarly, a standard *parabola* satisfies

$$y^2 = 4ax$$

with parametric equation

$$(x, y) = (at^2, 2at).$$

Figure 1.2: *Parabola*

The Hyperbola

A standard *hyperbola* satisfies

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Note that we can write a parametric equation for the standard hyperbola as

$$(x, y) = (\pm a \cosh \theta, b \sinh \theta).$$

This hyperbola has asymptotes $y = \pm \frac{b}{a}x$.

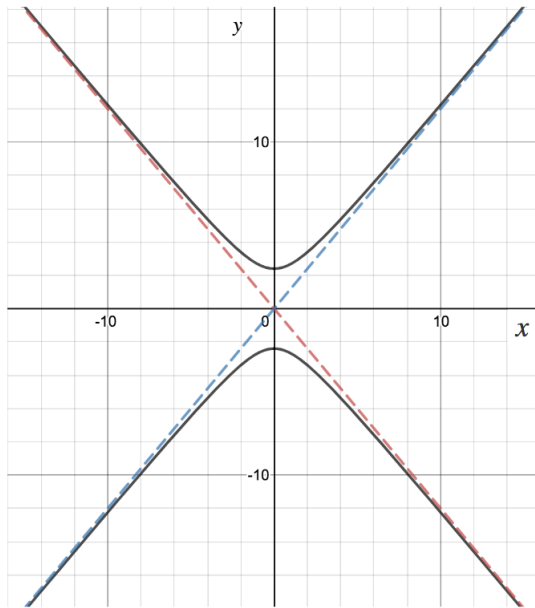


Figure 1.3: *Hyperbola*

It is possible to show that any quadratic curve on the plane can be transformed into just one of the above by a translation and a rotation. In fact for

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

it is possible to show that this curve is

- an ellipse if $B^2 - 4AC < 0$
- a parabola if $B^2 - 4AC = 0$
- a hyperbola if $B^2 - 4AC > 0$

The ellipse is a circle if $B = 0$ and $A = C$.

There are other methods of characterising conic sections using the directrix/focus/eccentricity classification: namely, given any line D (the directrix), point F (the focus) and number $e > 0$ (the eccentricity) the set of points $P : (x, y)$ such that

$$|PF|^2 = e^2|PN|^2.$$

where the point N is the closest point to P on D .

1.3 Intersections and Tangents

Intersection of a line with a circle

Given the equations of a line and a circle we can find the points (if any) by solving simultaneously the equation of a circle and the equation of a straight line. We do this by eliminating one of the variables and thus obtain an equation involving only just one variable.

Example 1.3 Where does the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ meet the line $y = x + 2$?

Solution: We get $x^2 + (x + 2)^2 - 4x - 6(x + 2) - 3 = 0$ and this reduces to the quadratic $2x^2 - 6x - 11 = 0$. The roots are $x = 1/2(3 \pm \sqrt{31})$. We can then get corresponding values for y : $y = 1/2(7 \pm \sqrt{31})$.

In general, a line will cross a circle at two points, tangent to the circle, or not meeting the circle at all.

How can we distinguish between these cases? Substituting in the equation for the line, the intersections (x, y) must satisfy $Ax^2 + Bx + C = 0$ for some A, B, C . This equation has either two roots, one root, or no roots depending on the sign of the discriminant $D = B^2 - 4AC$ from the quadratic formula. If $D > 0$ then there are two roots, $D = 0$ one root, and no roots if $D < 0$.

For the case of tangent intersection, the radius of the circle is the distance of the circle's center to the line and the radial line is perpendicular to the tangent.

More general tangencies

The method above can be generalised to more general tangencies of curves and lines by finding an intersection that corresponds to a multiple root of the equation for x (or y). However more general methods that use calculus are typically more adaptable—using differentiation to obtain the slope of the curve at the tangency.

Example 1.4 Find the equation of the tangent to the circle $x^2 + y^2 = 2$ through the point $P : (-1, 1)$ on the circle.

Solution: OP has gradient -1 . The tangent is perpendicular to this and therefore has gradient 1 . Hence the equation of the line is $y = x + 2$.

Formula for tangent to circle through a point on the circle

Given a circle with center $C : (a, b)$ and a point $P : (x_0, y_0)$ on it, then one can find the equation of the tangent to the circle through P . The gradient of CP is $(y_0 - b)/(x_0 - a)$, thus the equation of the tangent (after some algebra) is given by:

$$y = -\frac{(x_0 - a)}{(y_0 - b)}x + y_0 + \frac{(x_0 - a)}{(y_0 - b)}x_0.$$

For example, consider $P : (-1, 2)$ on the circle of radius 5 , center $(3, -1)$. We obtain $y = \frac{4}{3}x + \frac{10}{3}$.

1.4 Polar Coordinates

We can represent curves via functions, equations or parametrically. Another way is through polar coordinates. We simply replace Cartesian coordinates x and y with polar coordinates r and θ , where r is the distance to the origin and θ is the angle measured anti clockwise from the positive x -axis.

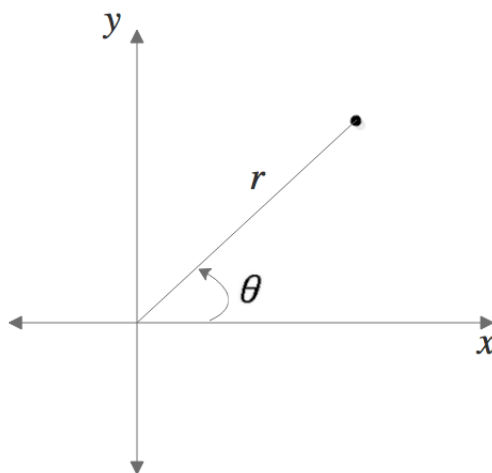


Figure 1.4: Point on coordinate system showing r and θ

Note that replacing θ by $\theta + 2\pi$ has not effect on the position, and hence coordinates, of the point.

The relationship between the Cartesian and polar coordinates is given by

$$x = r \cos \theta$$

$$y = r \sin \theta.$$

Now, using this relationship, we can represent curves using equations involving r and θ instead of x and y .

For example, consider $r = C$, where C is a constant such that $C > 0$. This represents a circle since substituting $r = C$ into $x = r \cos \theta$ and $y = r \sin \theta$ gives $x = C \cos \theta$ and $y = C \sin \theta$. On squaring both x and y and adding the results yields $x^2 + y^2 = C^2 \cos^2 \theta + C^2 \sin^2 \theta = C^2$. This is the Cartesian form for an equation of a circle centred at the origin with radius C . If $C = 1$, we have a *unit circle*.

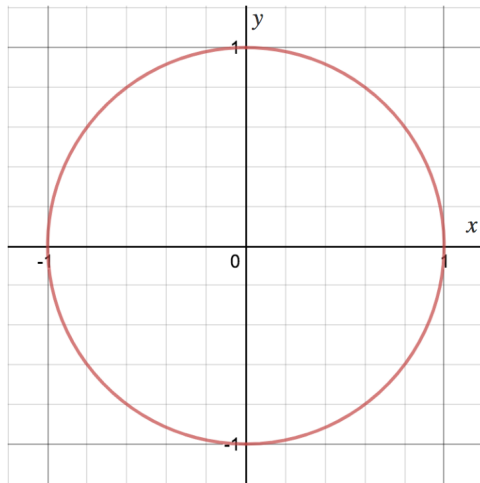


Figure 1.5: A unit circle

Notice that the constant coordinate curves for the Cartesian system are $x = \text{constant}$ which gives straight lines parallel to the y -axis, and $y = \text{constant}$ which gives straight lines parallel to the x -axis. Whereas, the constant coordinate curves for the polar coordinates are $r = \text{constant}$ which gives circles centred at the origin, and $\theta = \text{constant}$ which gives rays emanating from the origin. In both cases, the two families of constant coordinate curves intersect at right angles. Such coordinate systems are called *orthogonal*. Thus, the Cartesian and polar coordinate systems are both orthogonal. *

*Have you noticed that we are capitalising the C in Cartesian but not the p in polar? This is because the word Cartesian bears the Latin form of the name Rene Descartes (Cartesius), who was a famous French philosopher and mathematician. Among others, one of his most renowned legacies was the development of using algebra to describe geometry.

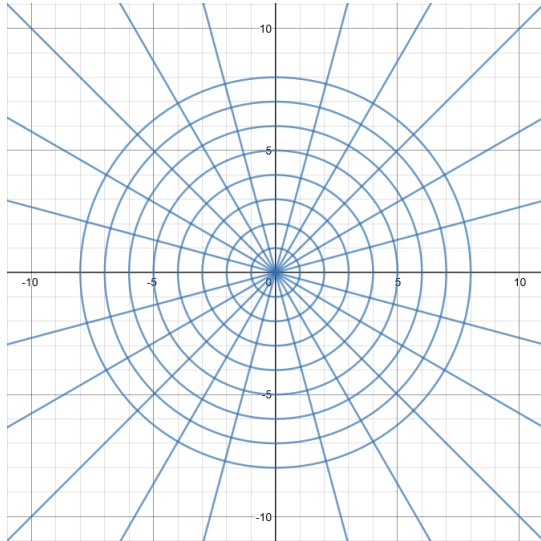


Figure 1.6: A polar coordinate system

Plotting Curves in Polar Coordinates

The following steps will help with plotting curves which are given in polar coordinates.

- (i) Set up a table of r values at a selection of θ values in the given range.*
- (ii) Mark in the polar points. Remember that r is distance from the origin and the polar angle θ increases anticlockwise.
- (iii) If for a given value of θ , say $\theta = \theta^*$, r has a negative value, say $r = -r^*$, then reverse the direction along the ray $\theta = \theta^*$ and plot the point on the ray $\theta = \theta^* + \pi$ instead. Thus, $(-r^*, \theta^*)$ is plotted as $(r^*, \theta^* + \pi)$. The following image may help.

*You should be using Radians as most topics you will come across which use angles will also be using Radians.

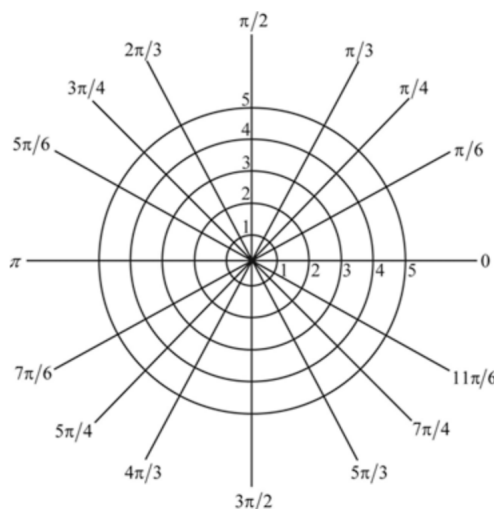


Figure 1.7: Angles in a polar coordinate system

Example 1.5 (a) Sketch the curve given by the equation $r = 2 \sin \theta$.

(b) Prove that this is a circle.

Solution:

(a) We will first fill out a table with the θ values and the corresponding r values.

θ	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
r	0	1	1.41	1.73	2	1.73	1.41	1	0

θ	$\frac{7\pi}{6}$	$\frac{5\pi}{4}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{7\pi}{4}$	$\frac{11\pi}{6}$	2π
r	-1	-1.41	-1.73	-2	-1.73	-1.41	-1	0

Note that the negative values of r are plotted in the reverse direction along the appropriate ray. In this case, these points coincide with points obtained for positive values of r . For example, when $\theta = \frac{7\pi}{6}$, $r = -1$ giving the same point as at $\theta = \frac{\pi}{6}$, $r = 1$.

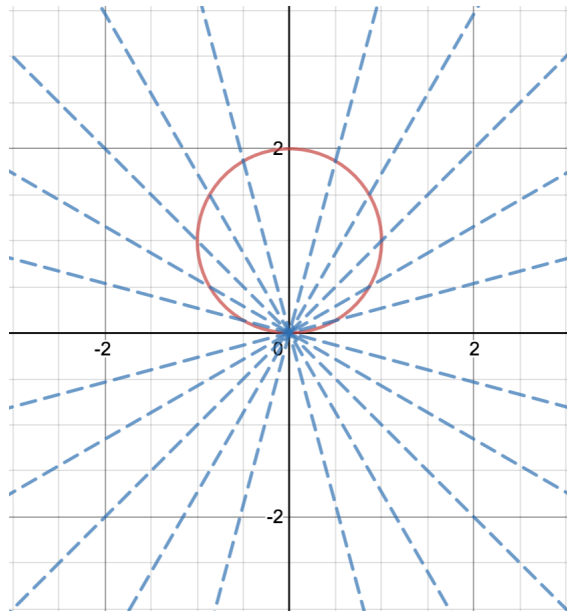


Figure 1.8: $r = 2 \sin \theta$

(b) To prove that this is a circle we re-write the equation of the curve in Cartesian form. If (x, y) are the Cartesian co-

ordinates of a point on the curve, then since $r = 2 \sin \theta$ we have

$$x = r \cos \theta = (2 \sin \theta) \cos \theta = \sin 2\theta$$

and

$$y = r \sin \theta = (2 \sin \theta) \sin \theta = 1 - \cos 2\theta.$$

Now, squaring both sides of these two equations and adding them yields

$$x^2 + (y - 1)^2 = \sin^2 2\theta + (-\cos 2\theta)^2 = 1.$$

This is the equation of a circle with radius 1 and centre $(0, 1)$.

Given a curve in Cartesian form, how do we write it in polar form? Also, given a curve in polar form, how do we write it in Cartesian form? To answer such questions, we need to use the formulas $x = r \cos \theta$, $y = r \sin \theta$ and $x^2 + y^2 = r^2$.

Example 1.6 Write the following polar curves in Cartesian form.

- (i) $r = 2$. This is a circle of radius 2, so we have $x^2 + y^2 = 4$.
- (ii) $r \cos \theta = 1$. This is simply $x = 1$ since $x = r \cos \theta$.
- (iii) $r = 3 \sin \theta$. For this, we substitute $\frac{y}{r}$ for $\sin \theta$, this yields $r^2 = 3y$. Now, we use $r^2 = x^2 + y^2$, and we have $x^2 + y^2 = 3y$. However, we can go further and rewrite this so that it *clearly* looks like a conic section. By completing the square, we can write this as

$$x^2 + \left(y - \frac{3}{2}\right)^2 = \frac{9}{4}.$$

Exercise 1.7 Write the following polar curve in Carte-

sian form

$$r = \frac{1}{1 + 2 \sin \theta}.$$

Example 1.8 Write the following Cartesian curves in polar form. For this question, we simply replace x with $r \cos \theta$ and y with $r \sin \theta$, make r the subject and using $r^2 = x^2 + y^2$ where necessary.

(i) $y = 5$. We have $r = \frac{5}{\sin \theta}$.

(ii) $x^2 + y^2 = 25$. We have $r^2 = 25$, so $r = 5$.

(iii) $x^2 = 2y$. We have $r^2 \cos^2 \theta = 2r \sin \theta$, after simplifying and rearranging, we have

$$r = 2 \tan \theta \sec \theta.$$

Exercise 1.9 Write the following Cartesian curve in polar form

$$y = 2x - 1.$$

2 Functions

The modern idea of a function dates back to the 18th century and the work of Euler*.

2.1 Definition of a Function

Suppose we have two sets D and C . A rule that assigns to each element of one D a unique element in C is called a function from D to C .

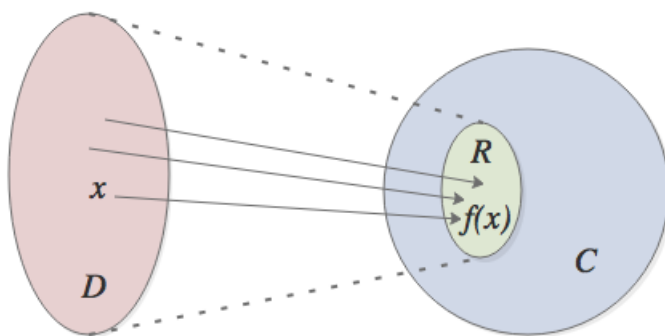
The set D is called the *domain* of the function and C is called the *codomain*. If $x \in D$ and $y \in R$ are related by the function f , we write

$$y = f(x)$$

and we say that x is the *independent variable* and y is the *dependent variable*; namely its value depends on x .

If $R = f(D)$ (i.e. we exclude all points in codomain that are not images) we say R is the *range* or image of f .

We can in fact “shrink-wrap” the codomain of the function to consider a subset that is the range. This is a subtle but important point: given a function and a domain, there are many choices for the codomain but only one for the range.



*Leonard Euler (1707-1783) was a Swiss mathematician who spent most of his career between St-Petersburg and Berlin. He was one of the towering lights of 18th century mathematics. He has been the most prolific mathematician of his exceptional calibre and was nearly blind for many years still producing papers on all areas of mathematics. Euler claimed that he made some of his greatest mathematical discoveries while holding a baby in his arms with other children playing round his feet (he had 13 of them, although only 5 did not die in infancy).

Figure 2.1: Set D is the domain, and set C is the codomain. We write $f : D \rightarrow C$ and say “ f maps from D to C ”. We can write $D(f)$ for the domain of f and $C(f)$ for the codomain of f .

Remark 2.1

The codomain and range are both on the output side, but are subtly different. The codomain is the set of values that could possibly come out and is actually part of

the definition of the function. However, the range is the set of values that actually do come out.

For example, let $f(x) = 2x$ with a domain and codomain of integers. The range (actual output values) would be just the even integers.

Therefore, we see that the range is a subset of the codomain. For a real function defined by a formula, we say the *maximal domain* is the largest subset of $\mathbb{R}$ for which the formula can be evaluated to give a finite result. Unless explicitly mentioned, we will always consider such a function to be defined on its maximal domain.

The maximal domain of a real function $f(x)$ can be found by excluding all values that lead to a root of a negative number, or that lead to dividing anything by zero. Note however that this depends on exactly how the formula is written, for example x^2/x cannot be evaluated at $x = 0$ while cancelling out the x can be evaluated at $x = 0$.

For example, the circle $x^2 + y^2 = 1$ is not a function because for each value of $x \in (-1, 1)$ there are two values of y : $y = \pm\sqrt{1 - x^2}$ but a function must be single-valued.

For the remainder of the module we will be dealing primarily with functions between subsets of $\mathbb{R}$ that are intervals.

Remark 2.2

Functions do not always have nice formulae; for example we could define a function $f(x)$ to be the value of the 100th place in the decimal expansion of x . However, here we will mostly be dealing with functions defined by algebraic formulae.

For example, consider

$$y = f(x) = x^2$$

$D(f)$ is the set of real numbers, $(-\infty, \infty)$ and $R(f)$ is the set of

all non-negative numbers (all squares are positive), i.e. $[0, \infty)$.

Example 2.3 Find the maximum set of real numbers for which the following rule creates a function $g(x)$ where

$$y = g(x) = \sqrt{x-2} + 3.$$

Solution: We can only compute this (as a real number) for $x \geq 2$. Hence $D(g) = [2, \infty)$, $R(g) = [3, \infty)$.

2.2 Properties of Functions

Odd and Even Functions

Definition 2.4 Suppose that f is defined on some subset of $\mathbb{R}$ which is symmetric about 0, i.e. such that $x \in D(f)$ implies that $-x \in D(f)$. If

$$f(x) = -f(-x)$$

for all $x \in D(f)$, we say that f is an **odd function**. Similarly, if

$$f(x) = f(-x)$$

we say it is an **even function**.

Graphically, an odd function has a graph that is symmetric under rotation by π about the origin. An even function is symmetric under reflection in the y -axis. *

Most functions are neither even nor odd, but **any** function $f(x)$ can be decomposed into a sum of an even part $E(f)(x) = (f(x) + f(-x))/2$ and an odd part $O(f)(x) = (f(x) - f(-x))/2$.

To prove that a function f is *even*, we must find an expression for $f(-x)$ and show that it is equal to $f(x)$. Similarly, to prove that a function f is *odd*, we must find an expression for $f(-x)$ and show that it is equal to $-f(x)$.

For example, the function

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$$

*A function being even or odd has nothing to do with being divisible by two.

is even because

$$f(-x) = (-x)^2 = x^2 = f(x).$$

While the function

$$g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = x^3$$

is odd because

$$g(-x) = (-x)^3 = -x^3 = -g(x).$$

To prove that a function is *not even* it is sufficient to find one number a with $f(-a) \neq f(a)$. If there is one value a where $f(-a) \neq f(a)$, then it cannot be true that for every value x , $f(-x) = f(x)$. Similarly, to prove that a function is *not odd* it is sufficient to find one number a with $f(-a) \neq -f(a)$. If there is one value a where $f(-a) \neq -f(a)$ then it cannot be true that for every value x , $f(-x) = -f(x)$.

For example, the function

$$h : \mathbb{R} \rightarrow \mathbb{R}, h(x) = x^2 + x$$

is neither even nor odd because $h(1) = 2$ and $h(-1) = 0$ so $h(1) \neq h(-1)$ and $h(1) \neq -h(-1)$.

Example 2.5 State whether the following are odd, even or neither. In the third case, find the odd and even parts of the function.

Solution:

- (a) $f(x) = \sin(x)$. This is the rule of the odd part of the function.
- (b) $g(x) = x^2 + 2x^4$. This is the rule of the even part of the function.
- (c) $h(x) = (\sin x + x^2)e^{-x^2}$. Here, $O(h)(x) = \sin x e^{-x^2}$ and $E(h)(x) = x^2 e^{-x^2}$.

Periodic functions

A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is *periodic* if there is a $T > 0$ such that

$$f(x + T) = f(x)$$

for all x , where T is the period of $f(x)$ if it is the smallest such T .

Example 2.6 Find the period of $f(x) = \sin(x\sqrt{2})$.

Solution:

$$\sin x\sqrt{2} = \sin(x\sqrt{2} + 2\pi) = \sin((x + \pi\sqrt{2})\sqrt{2})$$

holds for all x . Hence $f(x) = f(x + \pi\sqrt{2})$ and so the period is $\pi\sqrt{2}$.

Note however that, for example, $g(x) = \sin x + \sin(\sqrt{2}x)$ and $f(x) = \sin x^2$ are not periodic.

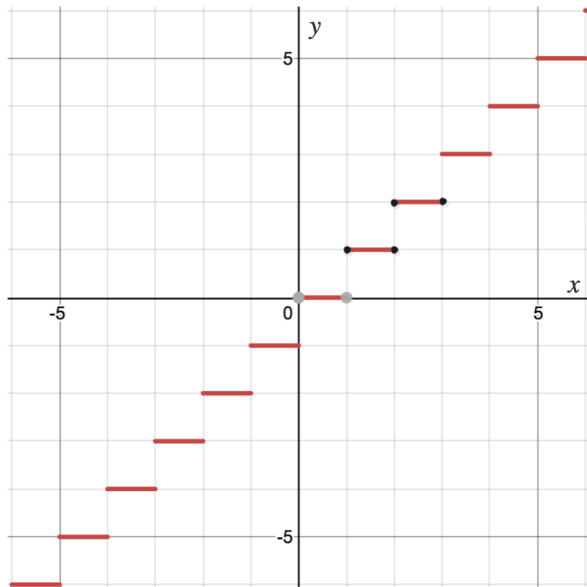
Monotonic functions

Definition 2.7 We say that $f : D \rightarrow \mathbb{R}$ where D is a subset of the real numbers, is **monotonic increasing** if $f(x) \leq f(y)$ for all $x < y$ in the domain of f .

Similarly, it is **monotonic decreasing** if $f(x) \geq f(y)$ for all $x < y$.

For example e^x , x , $\ln(x)$ are monotonic increasing, while e^{-x} , $-x^3$ are rules of functions that are monotonic decreasing.

A function is *strictly monotonic increasing* if $f(x) < f(y)$ for all $x < y$, and *strictly monotonic decreasing* if $f(x) > f(y)$ for all $x < y$. For example $f(x) = \lfloor x \rfloor$, the largest integer less than or equal to x (also known as the *floor function*) is monotonic increasing but not strictly monotonic increasing.

Figure 2.2: $f(x) = \lfloor x \rfloor$

Example 2.8 Which of the following are (a) even (b) odd (c) periodic (d) monotonic.

- (a) $f(x) = x + 7$. Solution: $f(x)$ is monotonic.
 (b) $g(x) = x^3 - 4x$. Solution: $g(x)$ is odd.
 (c) $h(x) = \cos(x^2)$. Solution: $h(x)$ is even.
 (d) $k(x) = (\cos x)^2$. Solution: $k(x)$ is odd and periodic with period π .

2.3 The Absolute Value Function

There are many ways to define functions. We can define perfectly valid functions using different formulae in different regions of the domain. For example,

$$|x| = \begin{cases} x & \text{for } x \geq 0 \\ -x & \text{for } x < 0 \end{cases}$$

Or

$$f(x) = \begin{cases} -x & \text{for } x < 0 \\ x^2 & \text{for } 0 \leq x < 1 \\ 0 & \text{for } 1 \leq x \end{cases}$$

Note that such functions may or may not have ‘jumps’.

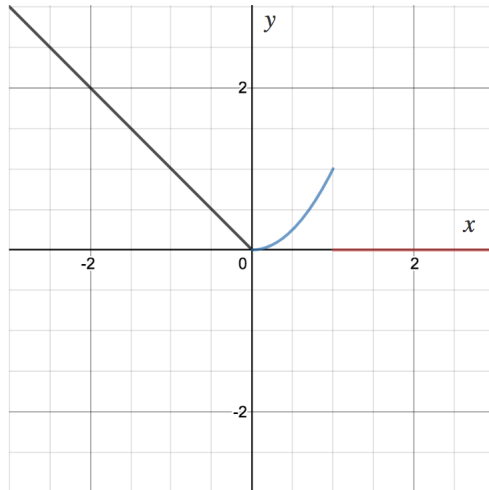


Figure 2.3: Graph of the function $f(x)$

Piecewise definitions often appear when we are dealing with functions involving absolute values. Remember that

$$|f(x)| = \begin{cases} f(x) & \text{if } f(x) \geq 0 \\ -f(x) & \text{if } f(x) < 0 \end{cases}$$

The graph of $|f(x)|$ is obtained by taking the graph of $f(x)$ and reflecting the parts where $f(x)$ is negative in the x -axis.

Example 2.9 Rewrite $f(x) = |5 - 2x|$ in piecewise form.

Solution: Applying the definition of $|f(x)|$ gives

$$f(x) = \begin{cases} 5 - 2x & \text{if } 5 - 2x \geq 0, \\ 2x - 5 & \text{if } 5 - 2x < 0 \end{cases} = \begin{cases} 5 - 2x & \text{if } x \leq 5/2 \\ 2x - 5 & \text{if } x > 5/2 \end{cases}$$

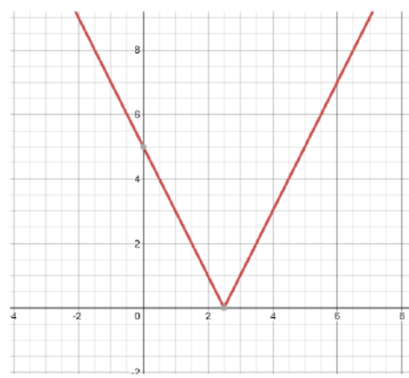
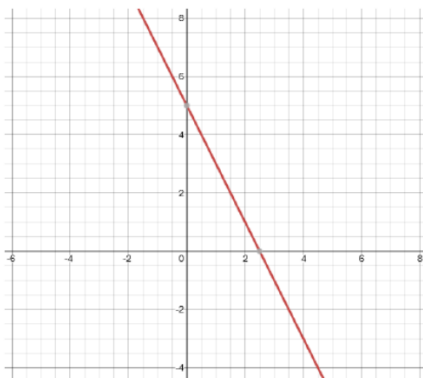


Figure 2.4: Graph of $5 - 2x$ (left) and $|5 - 2x|$ (right)

2.4 Combining Functions

Combining functions by adding, subtracting, multiplying or dividing them allows us to build up a vast array of functions from the basic ones.

Definition 2.10 Let f and g be two functions with a common domain D and codomain $\mathbb{R}$. Then

- (i) $f + g$ is the function with domain $D(f) \cap D(g)$, codomain $\mathbb{R}$ and

$$(f + g)(x) = f(x) + g(x)$$

- (ii) $f - g$ is the function with domain $D(f) \cap D(g)$, codomain $\mathbb{R}$ and

$$(f - g)(x) = f(x) - g(x)$$

- (iii) fg is the function with domain $D(f) \cap D(g)$, codomain $\mathbb{R}$ and

$$(fg)(x) = f(x)g(x)$$

- (iv) f/g is the function with domain $(D(f) \cap D(g)) \setminus \{x \in D(g) : g(x) = 0\}$, codomain $\mathbb{R}$ and

$$(f/g)(x) = \frac{f(x)}{g(x)}$$

What we have done is to define new functions with names $f + g$, fg , f/g and f/g . Their domains and codomains are defined precisely. Their rules should be what you would expect. Notice the use of brackets when we apply $f + g$ to x . We write $(f + g)(x)$ because $f + g(x)$ would mean add the number f to what we get when we apply g to x .

Example 2.11 Find the domain, codomain and rule of $f + g$ and f/g when

$$f : [-4, 9] \rightarrow \mathbb{R}, f(x) = x + 1$$

and

$$g : [2, \infty) \rightarrow \mathbb{R}, g(x) = \sqrt{x - 2}.$$

Solution: The rule for $f + g$ is defined to be

$$(f + g)(x) = f(x) + g(x) = x + 1 + \sqrt{x - 2}.$$

The domain of $f + g$ is defined to be

$$D(f) \cap D(g) = [-4, 9] \cap [2, \infty) = [2, 9].$$

The codomain of $f + g$ is $\mathbb{R}$. The rule for f/g is defined to be

$$(f/g)(x) = \frac{f(x)}{g(x)} = \frac{x + 1}{\sqrt{x - 2}}.$$

The domain of f/g is defined to be

$$(D(f) \cap D(g)) \setminus \{x \in D(g) : g(x) = 0\}.$$

In other words, it is the same as $D(f + g)$ except that we must remove the values where $g(x) = 0$. Now $g(x) = 0$ implies $\sqrt{x - 2} = 0$ which occurs when $x = 2$. So $D(f/g) = [2, 9] \setminus \{2\} = (2, 9]$. The codomain of f/g is $\mathbb{R}$. *

*In set theory, the backslash is used to exclude values. So, for example $A \setminus \{2, 3\}$ means all the elements in set A except 2 and 3.

Example 2.12 Find the domain, codomain and rule of $f + g$ and f/g when

$$f : \mathbb{R} \setminus \{3\} \rightarrow \mathbb{R}, f(x) = \frac{1}{x - 3}$$

and

$$g : (1, \infty) \setminus \{3\} \rightarrow \mathbb{R}, g(x) = \frac{1}{\sqrt{x - 1}(x - 3)}.$$

Solution: The codomain is $\mathbb{R}$. The domain of $f + g$ is

$$(\mathbb{R} \setminus \{3\}) \cap ((1, \infty) \setminus \{3\}) = (1, \infty) \setminus \{3\}.$$

The domain of f/g is the same as the domain of $f + g$ because $g(x) \neq 0$ for all $x \in D(g)$. We have

$$(f + g)(x) = \frac{1}{x - 3} + \frac{1}{\sqrt{x - 1}(x - 3)} \quad \text{and} \quad (f/g)(x) = \sqrt{x - 1}.$$

Example 2.13 Find the domain, codomain and rule of $f + g$, fg and f/g when

$$f : [0, \infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x}$$

and

$$g : [0, \infty) \rightarrow \mathbb{R}, g(x) = \sqrt{x}.$$

Solution: The codomain is $\mathbb{R}$. The domain of $f + g$ and fg is $[0, \infty)$. We have $g(x) = 0$ if and only if $x = 0$, so the domain of f/g is $(0, \infty)$. We have

$$(f + g)(x) = 2\sqrt{x}, \quad (fg)(x) = x \quad \text{and} \quad (f/g)(x) = 1.$$

Example 2.14 Find the domain, codomain and rule of $f + g$, fg and f/g when

$$f : \mathbb{R} \setminus \{-1, 1\} \rightarrow \mathbb{R}, f(x) = \frac{1}{x^2 - 1}$$

and

$$g : [2, \infty) \rightarrow \mathbb{R}, g(x) = \sqrt{x - 2}.$$

Solution: The codomain is $\mathbb{R}$. The domain of $f + g$ and fg is $[2, \infty)$. We have $g(x) = 0$ if and only if $x = 2$, so the domain of f/g is $(2, \infty)$. We have

$$(f + g)(x) = \frac{1}{x^2 - 1} + \sqrt{x - 2},$$

$$(fg)(x) = \frac{\sqrt{x - 2}}{x^2 - 1},$$

$$(f/g)(x) = \frac{1}{\sqrt{x - 2}(x^2 - 1)}.$$

Composition of Functions

We saw that we could *combine* functions by adding, subtracting, multiplying or dividing them. Now, we will study combining functions in another way, by *composing* them. If we have two functions f and g then we can define a new function by first applying g to the input x and then applying f to the result $g(x)$.

We write

$$(f \circ g)(x) = f(g(x)).$$

The domain of $f \circ g$ is the set of $x \in D(g)$ such that $g(x) \in D(f)$.

Finding $f(g(x))$ is not the same as $g(f(x))$. Generally they are completely different functions. Changing the order in which we do things usually makes a difference to the final result.*

*For example, you could revise then do the exam, or you can do the exam then revise.

Remark 2.15

Note that $(f \circ f)(x)$ is sometimes written $f^2(x)$. You are advised to avoid this notation as it can be confused with $(f(x))^2$.

For example, if $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = x + 1$, then to find $f(g(3))$, we calculate $g(3) = 4$, so that $f(g(3)) = f(4) = 16$.

Example 2.16 Let $f(x) = \sqrt{x}$ and $g(x) = \sin x$. Compute the domain of the following.

Solution:

(a) $f \circ g$

Note that $(f \circ g)(x) = \sqrt{\sin x}$ is only defined for values of x such that $\sin x \geq 0$. Hence the domain is

$$\dots[-2\pi, -\pi] \cup [0, \pi] \cup [2\pi, 3\pi]..$$

(b) $g \circ f$

Here, $(g \circ f)(x) = \sin(\sqrt{x})$ is defined for all $x \geq 0$. Hence the domain is $[0, \infty)$.

(c) $f \circ f$

Here, $(f \circ f)(x) = x^{1/4}$ is defined for all $x \geq 0$, hence the domain is $[0, \infty)$.

(d) $g \circ g$

Here, $(g \circ g)(x) = \sin(\sin(x))$ is defined for all x , so the domain is $\mathbb{R}$.

A useful technique is to be able to spot that a function is a composition and be able to split it up into its constituent parts.

Example 2.17 For each of the functions f , express f as a composition. In other words, find g and h such that $f = g \circ h$.

(a) $f(x) = \sqrt{x+2}$

(b) $f(x) = |x^2 - 2x - 8|$

(c) $f(x) = \frac{3}{5 + \cos x}$

Solution: Possible solutions are

(a) Let $h(x) = x + 2$ and $g(x) = \sqrt{x}$. Then $f = g \circ h$.

(b) Let $h(x) = x^2 - 2x - 8$ and $g(x) = |x|$. Then $f = g \circ h$.

(c) Let $h(x) = \cos x$ and $g(x) = \frac{3}{5+x}$. Then $f = g \circ h$.

2.5 Hyperbolic Functions

We define the *hyperbolic sine* and *cosine* by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}.$$

Note that $\sinh$ and $\cosh$ are respectively the odd and even parts of e^x .

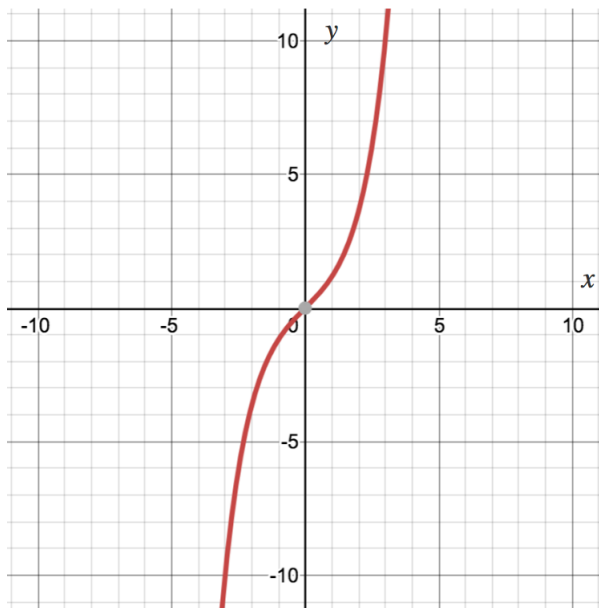


Figure 2.5: Graph of $\sinh(x)$ (an odd function)

The reason for the name is that $x = \cosh t$, $y = \sinh t$

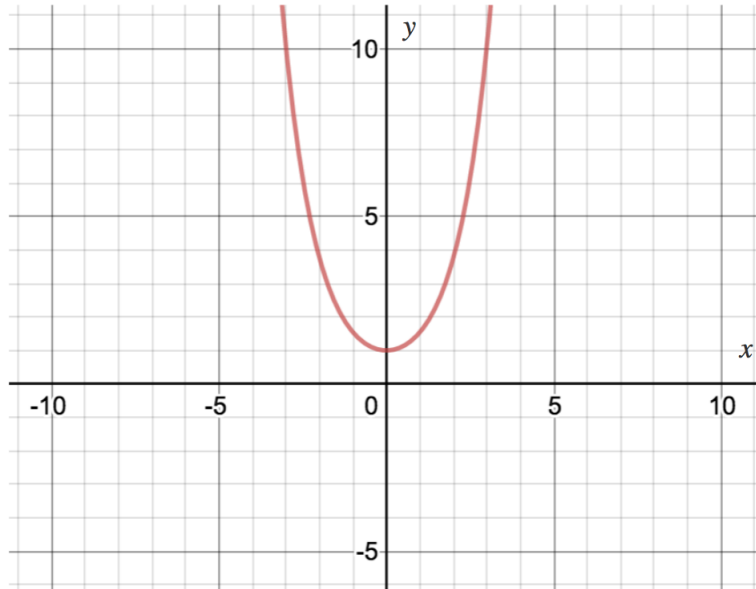


Figure 2.6: Graph of $\cosh(x)$ (an even function)

draws out a standard hyperbola $x^2 - y^2 = 1$, just as $(x, y) = (\cos t, \sin t)$ is a unit circle.

We define

$$\tanh x = \frac{\sinh x}{\cosh x}, \quad \coth x = \frac{\cosh x}{\sinh x}, \quad \operatorname{sech} x = \frac{1}{\cosh x}, \quad \operatorname{csch} x = \frac{1}{\sinh x}.$$

Exercise 2.18 Sketch the graphs of $\tanh x$, $\coth x$, $\operatorname{csch} x$ and $\operatorname{sech} x$.

2.6 Inverse Functions

Definition 2.19 Given $f : D \rightarrow C$, we say an *inverse* of f is a function $g : C \rightarrow D$ such that

$$(g \circ f)(x) = x$$

for all $x \in D$ and

$$(f \circ g)(y) = y$$

for all $y \in C$. We write $g = f^{-1}$.

Such a function will exist as long as f is one-to-one (also called *injective*), this means the graph only intersects the horizontal line at most once in the domain. This is the ‘Horizontal

Line test'. We will write this result formally as follows.

Theorem 2.20 The function f has an inverse if and only if f is one-to-one and onto.

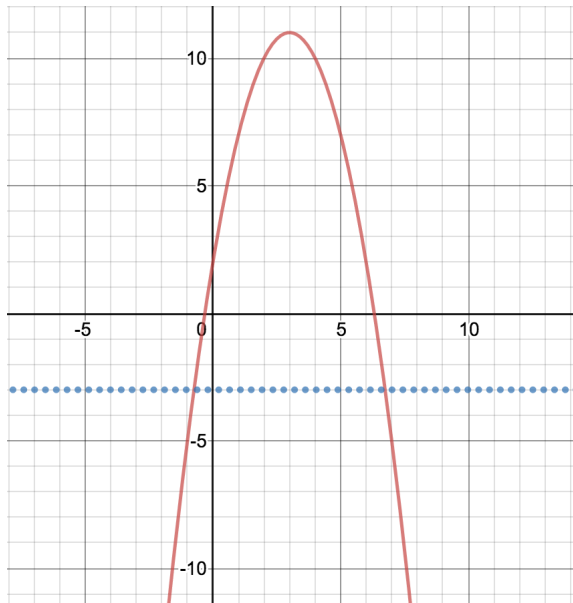


Figure 2.7: This function is not one-to-one since there is a horizontal line (in fact infinitely many in this example) that intersects the curve more than once.

Proposition 2.21 Suppose f has an inverse f^{-1} . Then

- (i) the domain of f^{-1} equals the range of f ;
- (ii) the range of f^{-1} equals the domain of f .

Hint for proof: The proposition states that two *sets are equal*, so we use the usual method of proving that the left-hand side is a subset of the right-hand side and then proving that the right-hand side is a subset of the left-hand side. By doing so, we have proved that the sets are equal.

Method of Finding the Inverse

To obtain f^{-1} , we solve $y = f(x)$ for x in terms of y then interchange x and y to get $y = f^{-1}(x)$.

Remark 2.22

Remember that $f^{-1}(x)$ is used as a notation for the inverse of the function $f(x)$. This is not to be confused with $(f(x))^{-1} = \frac{1}{f(x)}$

Example 2.23 Find the inverse of $f : [1, \infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x-1}$.

Solution: Let $y = \sqrt{x-1}$. Then $x-1 = y^2$, so that $x = 1 + y^2$. Hence $f^{-1}(y) = 1 + y^2$ and so $f^{-1}(x) = 1 + x^2$.

Now, we have to be careful while thinking about the domain of f^{-1} since its rule is defined for all x . However, the domain is not $\mathbb{R}$. To find the domain, we must use the preceding proposition which says that the domain of f^{-1} equals the range of f which is $[0, \infty)$.

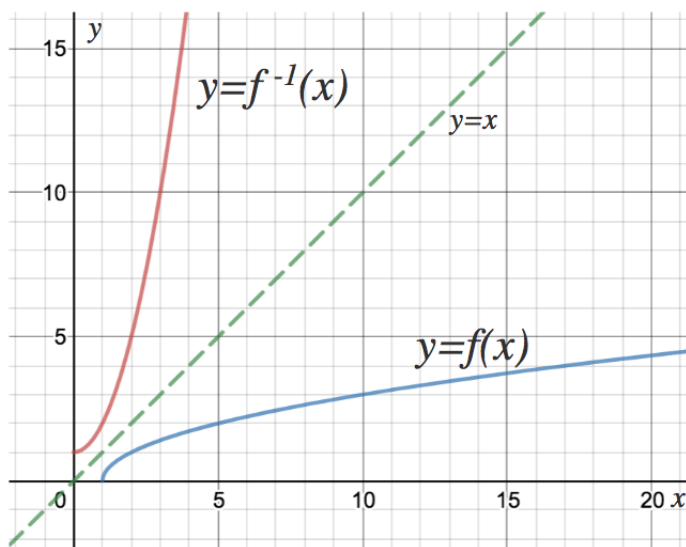


Figure 2.8: Graph of the function $f(x) = \sqrt{x-1}$ for $x \geq 0$, and its inverse.

Remark 2.24

In the case that the function is not one-to-one on the domain, we need to restrict to a domain in which it is one-to-one. Unfortunately there is not a unique way to determine such a domain, so we have to impose conventions. For example $\sqrt{x}$ is the inverse of $y = x^2$ on the domain $x \geq 0$.

Inverse Trigonometric Functions

We can find an inverse to $\cos x$ if we restrict to $x \in [0, \pi]$, and an inverse to $\tan x$ if we take $x \in [-\pi/2, \pi/2]$. We compute

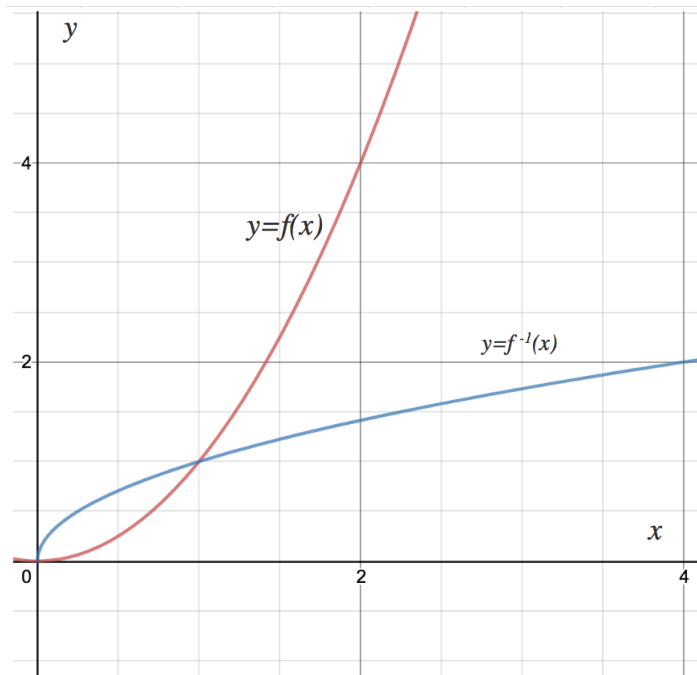


Figure 2.9: Graph of the function $f(x) = x^2$ for $x \geq 0$, and its inverse.

inverse trig functions on the following standard domains:

	Domain	Range
cos	$\mathbb{R}$	$[-1, 1]$
arccos	$[-1, 1]$	$[0, \pi]$
sin	$\mathbb{R}$	$[-1, 1]$
arcsin	$[-1, 1]$	$[-\pi/2, \pi/2]$
tan	$\mathbb{R} \setminus \{k\pi/2, k \in \mathbb{Z}\}$	$\mathbb{R}$
arctan	$\mathbb{R}$	$(-\pi/2, \pi/2)$
cot	$\mathbb{R}$	$\mathbb{R} \setminus \{k\pi, k \in \mathbb{Z}\}$
arccot	$\mathbb{R}$	$(0, \pi)$

We can use these standard domains to find inverses for other domains; for example $f(x) = \cos x$ is a function from $[\pi, 2\pi]$ to $[-1, 1]$ with inverse $f^{-1}(x) = 2\pi - \arccos(x)$.

The Exponential Function and its Inverse

Definition 2.25 An *exponential function* is one of the form $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = a^x$ where a is a strictly positive constant.

For example, when $a = 2$, we have $f(x) = 2^x$ and the graph

is given in Fig. 2.10. If $a = \frac{1}{2}$, we have $f(x) = \left(\frac{1}{2}\right)^x$. The graph is formed by reflecting the graph of $f(x) = 2^x$ in the y -axis. This is expected since if $f(x) = 2^x$, then

$$f(-x) = 2^{-x} = \left(2^{-1}\right)^x = \left(\frac{1}{2}\right)^x.$$

A special case of the exponential function is when $a = e \approx 2.71828$. The function is then $f(x) = e^x$ and this is known as *the* exponential function. It's one of the most important functions that arises in many topics in mathematics. It is sometimes denoted as $\exp(x)$.

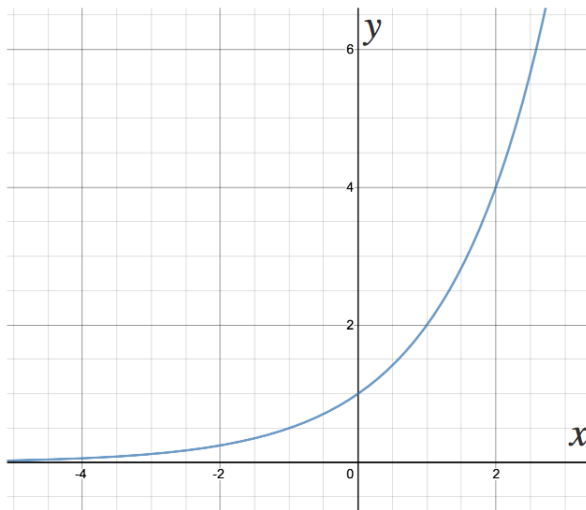


Figure 2.10: Graph of the function $f(x) = 2^x$.

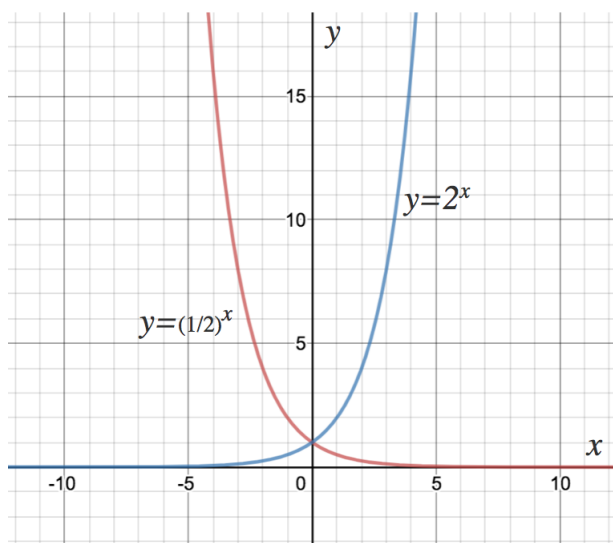


Figure 2.11: Graph of the functions $f(x) = 2^x$ and $f(x) = \left(\frac{1}{2}\right)^x$.

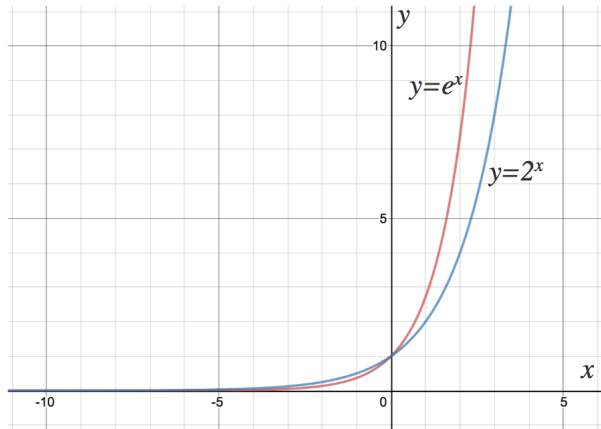


Figure 2.12: Graph of the function $f(x) = e^x$.

Definition 2.26 If $y = a^x$, for $a > 0$ and $a \neq 1$, we say that $x = \log_a y$ is the **logarithm of y** to the base a . Moreover, the function $\log_a$ is the inverse of the function $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = a^x$.

If $y = e^x$ then we say $x = \ln y = \log y = \log_e y$ is the **natural logarithm** of y .

Remark 2.27

Can we say that the exponential function has an inverse? Yes since if $a > 1$ the function is increasing, and if $a < 1$ the function is decreasing. We know that a function has an inverse if and only if it is one-to-one, and a function which is monotonic is one-to-one and therefore has an inverse.

Proposition 2.28 For any $a > 0$ such that $a \neq 1$, the domain of $\log_a$ is $(0, \infty)$ and the range of $\log_a$ is $\mathbb{R}$.

This proposition follows from the properties of an inverse function which we saw earlier. We know that if f and g are inverses of each other, then the domain of f is the range of g and the range of f is the domain of g .

Example 2.29 Find the inverse of $f(x) = 1 + 3 \ln(x)$, $x > 0$.

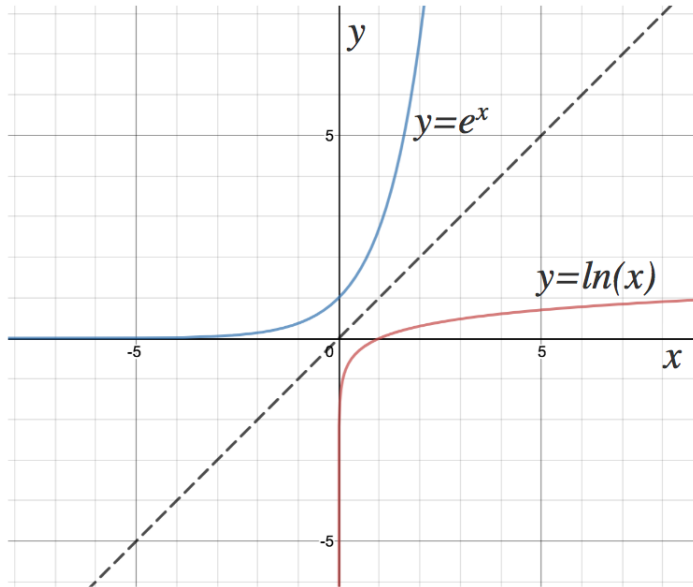


Figure 2.13: Graph of the function $f(x) = e^x$ and its inverse.

Solution: We have

$$y = 1 + 3 \ln x.$$

Rearranging gives

$$\frac{y-1}{3} = \ln x,$$

and so

$$x = e^{(y-1)/3}.$$

Hence

$$y = f^{-1}(x) = e^{(x-1)/3}$$

We can check if this is correct by finding $f \circ f^{-1}(x)$ and showing that it gives x . We have

$$\begin{aligned} f \circ f^{-1}(x) &= 1 + 3 \ln(e^{(x-1)/3}) \\ &= 1 + 3(x-1)/3 \\ &= 1 + x - 1 \\ &= x. \end{aligned}$$

Example 2.30 If $y = \sinh(x)$, find an expression for x in terms of y .

Solution: We write

$$y = \frac{e^x - e^{-x}}{2}$$

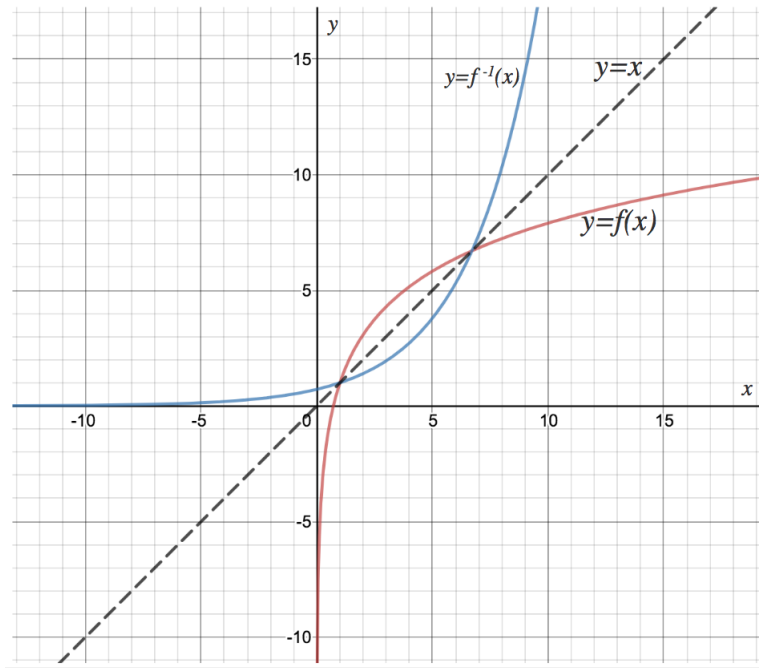


Figure 2.14: Graph of the function $f(x) = 1 + 3\ln(x)$ and its inverse.

and so

$$e^{2x} - 2ye^x - 1 = 0$$

implying that

$$e^x = \frac{2y \pm \sqrt{4y^2 + 4}}{2} = y + \sqrt{y^2 + 1}$$

as negative root gives $e^x < 0$; impossible. Hence

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1}).$$

Example 2.31 (a) Show that $\cos \arcsin x = \sqrt{1 - x^2}$, for $x \in [-1, 1]$.

(b) Show that $\arcsin x = \arctan \frac{x}{\sqrt{1 - x^2}}$, for $x \in [-1, 1]$.

(c) Show that $\arccos x + \arcsin x = \pi/2$.

Solution:

(a) Let $x = \sin y$; then

$$\cos \arcsin x = \cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2}.$$

(b) Let $x = \sin y$. Then

$$\tan y = \frac{\sin y}{\cos y} = \frac{x}{\sqrt{1 - x^2}}.$$

Hence

$$\arcsin x = y = \arctan \frac{x}{\sqrt{1-x^2}}.$$

(c) Let $\alpha = \arcsin x$, $\beta = \arccos x$ and look at

$$\begin{aligned}\sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ &= x \cdot x + (\sqrt{1-x^2})^2 = 1.\end{aligned}$$

Hence

$$\arcsin x + \arccos x = \alpha + \beta = \arcsin 1 = \pi/2.$$

Example 2.32 Find all solutions of $\cos y = \frac{\sqrt{3}}{2}$ such that $2\pi < y < 5\pi$.

Solution: We note that $\arccos(\sqrt{3}/2) = \frac{\pi}{6}$. By periodicity, other solutions are $\pi/6 + 2\pi k$ and by evenness and periodicity $-\pi/6 + 2\pi k$ for any $k \in \mathbb{Z}$. We therefore see that the solutions in the required range are precisely

$$\frac{13}{6}\pi, \frac{23}{6}\pi, \frac{25}{6}\pi.$$

2.7 Limits of Functions

Almost all of calculus is based on the theory of limits so this section is incredibly important. Limits of functions are used to describe how a function behaves as the variable gets close to a particular value. Consider

$$f(x) = \frac{x^2 - 1}{x - 1}.$$

How does this behave near $x = 1$? Note that the domain of f is $x \neq 1$. However, by simplifying, we have

$$\frac{(x-1)(x+1)}{x-1} = x+1$$

which can be evaluated for all x . * Hence, even though $f(1)$ is not defined, it is clear that we can make f arbitrarily close to 2 if x is sufficiently close to 1. We write

$$\lim_{x \rightarrow 1} f(x) = \frac{x^2 - 1}{x - 1} = 2.$$

*In such a case we say that f has a *removable singularity* at $x = 1$.

Definition 2.33 Suppose that $f(x)$ is a real function defined for x arbitrarily close to some a . We say $f(x) \rightarrow L$, or the **limit of $f(x)$** is L , as $x \rightarrow a$ if for ANY infinite sequence $\{x_n\}$ then $x_n \rightarrow a$ implies that $f(x_n) \rightarrow L$. We also write

$$\lim_{x \rightarrow a} f(x) = L.$$

Equivalently, we say: Suppose that for any $\epsilon > 0$ there exists a $\delta > 0$ such that for any x with $0 < |x - a| < \delta$ we have $|f(x) - L| < \epsilon$. Then $\lim_{x \rightarrow a} f(x) = L$. *

*In MTH1001, you will be using this precise definition of a limit to prove limits by obtaining a δ .

Taking a limit is like trying to make a prediction of the value of $f(a)$ based on the values of $f(x)$ for x close to but not equal to a . In the previous example, we tried to predict the value of $f(1)$ for a function f equal to $\frac{x^2 - 1}{x - 1}$ whenever $x \neq 1$.

How do we make such predictions?

- By substituting values of x close to a into the rule of f . We will see that this is often a bad method.
- By looking at the graph of f . This is an effective way to get an idea of what is happening, but does not give a rigorous answer.
- By applying theorem and propositions that we will look at in the remaining of this section.

Remark 2.34

It is not necessary for the function to be defined at a to have a limit there. Sometimes the limit will equal $f(a)$, and sometimes it will not. However, the value of $f(a)$ never comes into the calculations when determining the limit.

Example 2.35 Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2 - 2x + 4$. What is $\lim_{x \rightarrow 1} f(x)$?

Solution: By making a table of values for x and the corre-

sponding values of $f(x)$, you will notice that as x approaches $x = 1$, the limit is 3.

x	0	0.5	0.9	0.95	0.99	0.999
$f(x)$	4	3.25	3.01	3.0025	3.0001	3.000001

x	1.001	1.01	1.05	1.1	1.5	2
$f(x)$	3.000001	3.0001	3.0025	3.01	3.25	4

Consequently, $\lim_{x \rightarrow 1} f(x) = 3$. This is an example where the limit is equal to $f(a)$.

Let's change this function slightly.

Example 2.36 Consider the function

$$g : \mathbb{R} \rightarrow \mathbb{R}, g(x) = \begin{cases} 100 & \text{if } x = 1 \\ x^2 - 2x + 4 & \text{if } x \neq 1 \end{cases}$$

What is $\lim_{x \rightarrow 1} g(x)$?

Solution: The function g is equal to the function f in the previous example except when $x = 1$, i.e. $f(x) = g(x)$ providing $x \neq 1$. So the graphs of f and g are identical except at one point (draw the graphs as an exercise). Therefore, our previous answer of 3 holds because we never used $f(1)$ in the previous example. Our predictions for $f(1)$ and $g(1)$ were based on the same information. Consequently, $\lim_{x \rightarrow 1} g(x) = 3$.

Let's change the function again.

Example 2.37 Consider the function

$$h : \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}, h(x) = x^2 - 2x + 4.$$

What is $\lim_{x \rightarrow 1} h(x)$?

Solution: The function h is again equal to the function f in the earlier example except when $x = 1$. So the graphs of f and h are identical except at one point. Consequently, $\lim_{x \rightarrow 1} h(x) = 3$.

In what follows, we will start to develop a toolbox of techniques which we can use to evaluate limits. Why do we need a

toolbox of techniques? In most cases, the value of the variable is one where the function behaves strangely. In these cases, simply substituting the value a into $f(x)$ will not help us compute the limit.

Once we have our toolbox of techniques, it will not be sufficient or correct to evaluate limits by substituting values or drawing graphs. We can still use graphs to make conjectures or guesses about limits, but we must back them up by using theorems to establish them properly.

Laws of Limits

Let c be a constant and let $\lim_{x \rightarrow a} f(x)$, $\lim_{x \rightarrow a} g(x)$ exist. Then the following properties hold

1. $\lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$
2. $\lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$
3. $\lim_{x \rightarrow a} cf(x) = c \lim_{x \rightarrow a} f(x)$
4. $\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \lim_{x \rightarrow a} g(x)$
5. $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$ if $\lim_{x \rightarrow a} g(x) \neq 0$
6. $\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n$. This also applies when n is not an integer.

Some Useful Basic Limits

Let k be a constant, and a a real number. Then

$$\begin{aligned} \lim_{x \rightarrow a} k &= k, & \lim_{x \rightarrow -\infty} k &= k, & \lim_{x \rightarrow \infty} k &= k, \\ \lim_{x \rightarrow a} x &= a, & \lim_{x \rightarrow -\infty} x &= -\infty, & \lim_{x \rightarrow \infty} x &= \infty, \\ \lim_{x \rightarrow a} x^n &= (a)^n \\ \lim_{x \rightarrow 0^+} \frac{1}{x} &= \infty, & \lim_{x \rightarrow 0^-} \frac{1}{x} &= -\infty & \text{we say the limits do not exist} \\ \lim_{x \rightarrow -\infty} \frac{1}{x} &= 0, & \lim_{x \rightarrow \infty} \frac{1}{x} &= 0, \\ \lim_{x \rightarrow \pm\infty} \frac{1}{x^n} &= \left(\lim_{x \rightarrow \pm\infty} \frac{1}{x} \right)^n = 0^n = 0, & n &\in \mathbb{N}, \end{aligned}$$

$$\lim_{x \rightarrow \infty} x^n = \infty, \quad n \in \mathbb{N},$$

$$\lim_{x \rightarrow -\infty} x^n = \begin{cases} \infty & n = 2, 4, 6, \dots \\ -\infty & n = 1, 3, 5, \dots \leq x \end{cases}$$

Techniques to evaluate limits

In many cases, we will come across limits of the type “ $\frac{0}{0}$ ”, “ $\frac{\infty}{\infty}$ ”, “ $\infty - \infty$ ”, or other cases where we can not conclude what the limit is. In such cases, it is often very useful to write the function rule in a different way.

If we are computing the limit of $\frac{p(x)}{q(x)}$ as $x \rightarrow a$ where $p(x)$ and $q(x)$ are polynomials, and a is finite, and this is a limit of the type “ $\frac{0}{0}$ ”, $(x - a)$ must be a factor of the numerator and denominator. Hence, we proceed by factoring out the numerator and denominator and see if the expression simplifies.

Example 2.38 Let $f : \mathbb{R} \setminus \{-3\} \rightarrow \mathbb{R}, f(x) = \frac{x^2 + x - 6}{x + 3}$. Find $\lim_{x \rightarrow -3} f(x)$.

Solution: Substituting -3 into the $f(x)$ would lead to a “ $\frac{0}{0}$ ” problem. So, we simplify the expression and then substitute -3 . This yields

$$\lim_{x \rightarrow -3} \frac{x^2 + x - 6}{x + 3} = \lim_{x \rightarrow -3} \frac{(x - 2)(x + 3)}{x + 3} = \lim_{x \rightarrow -3} x - 2 = -5$$

Example 2.39 Find $\lim_{x \rightarrow -1} \frac{x^2 + x}{x^2 + 3x + 2}$

Solution: we see that denominator is zero when $x = -1$,

$$\lim_{x \rightarrow -1} \frac{x^2 + x}{x^2 + 3x + 2} = \lim_{x \rightarrow -1} \frac{x(x + 1)}{(x + 1)(x + 2)} = \lim_{x \rightarrow -1} \frac{x}{x + 2} = -1.$$

Example 2.40 Compute the limit $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$.

Solution: This is also a “ $\frac{0}{0}$ ” problem. Using the *difference of two squares*, $a^2 - b^2 = (a - b)(a + b)$, we have

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1} = \lim_{x \rightarrow 1} \frac{(\sqrt{x} - 1)(\sqrt{x} + 1)}{\sqrt{x} - 1} = \lim_{x \rightarrow 1} \sqrt{x} + 1 = 2.$$

Now, we will look at some examples of limits of $\frac{p(x)}{q(x)}$ as $x \rightarrow \pm\infty$. The technique here is to try dividing each term in the numerator and denominator by the highest power of x that occurs in the denominator. Then we let $x \rightarrow \pm\infty$ and all negative powers of x will vanish.

Example 2.41 Find $\lim_{x \rightarrow \infty} \frac{5-x}{x^2+1}$.

Solution: We divide each term in the numerator and denominator by x^2 . This yields

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{5-x}{x^2+1} &= \lim_{x \rightarrow \infty} \frac{\frac{5}{x^2} - \frac{1}{x}}{1 + \frac{1}{x^2}} \\ &= \frac{\lim_{x \rightarrow \infty} \left(\frac{5}{x^2} - \frac{1}{x} \right)}{\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x^2} \right)} \\ &= \frac{0}{1} = 0.\end{aligned}$$

Example 2.42 Find $\lim_{x \rightarrow \infty} \frac{x^3+5x^2}{x^2+2}$.

Solution: Again, the highest power in the denominator is 2 so we divide each term in the numerator and denominator by x^2 . We have

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{x^3+5x^2}{x^2+2} &= \lim_{x \rightarrow \infty} \frac{x+5}{1+\frac{2}{x^2}} \\ &= \frac{\lim_{x \rightarrow \infty} (x+5)}{\lim_{x \rightarrow \infty} \left(1 + \frac{2}{x^2} \right)} \\ &= \frac{\infty}{1} = \infty.\end{aligned}$$

the limit does not exist.

Example 2.43 Find

$$\lim_{x \rightarrow \infty} \frac{3x^2+8x-3}{2x^2+2}.$$

Solution:

$$\lim_{x \rightarrow \infty} \frac{3x^2+8x-3}{2x^2+2} = \lim_{x \rightarrow \infty} \frac{3+8/x-3/x^2}{2+2/x^2} = \frac{3}{2}.$$

Now, we will look at an example where “substituting ∞ ” leads to an “ $\infty - \infty$ ” problem. In such a case, it is often useful to turn it into a “ $\frac{0}{0}$ ” or “ $\frac{\infty}{\infty}$ ” and deal with it then.

Example 2.44 Find $\lim_{x \rightarrow \infty} \sqrt{x} (\sqrt{x+1} - \sqrt{x})$.

We can see that “substituting ∞ ” into x leads to “ $\infty - \infty$ ” and we cannot make any conclusions about this directly. We turn this into a fraction by multiplying by 1 in the form of a fraction. We have

$$\begin{aligned} \lim_{x \rightarrow \infty} \sqrt{x} (\sqrt{x+1} - \sqrt{x}) &= \lim_{x \rightarrow \infty} \sqrt{x} (\sqrt{x+1} - \sqrt{x}) \frac{\sqrt{x+1} + \sqrt{x}}{\sqrt{x+1} + \sqrt{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{x} ((x+1) - x)}{\sqrt{x+1} + \sqrt{x}} \\ &= \lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\sqrt{x+1} + \sqrt{x}} = \frac{1}{2}. \end{aligned}$$

2.8 Left-Hand and Right-Hand Limits

In the previous examples, we found that $f(x)$ has the same limit independent of which side we approach from. However this is not necessarily always the case. For example

$$g(x) = \frac{x}{|x|} = \text{sgn}(x)$$

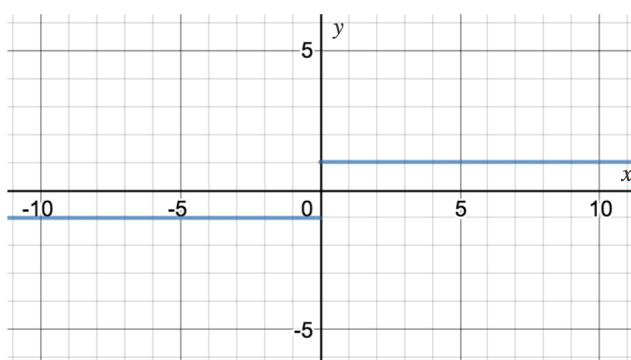


Figure 2.15: Graph of $g(x) = \frac{x}{|x|}$

Approaching $x = 0$ from the left or right will give different limits. More precisely, in this case we say *the* limit doesn't exist but the left-hand and right-hand limits do. We can use a similar approach to calculate infinite limits.

We write the *left hand limit* and *right hand limit* respectively

as

$$\lim_{x \rightarrow 0^-} g(x) = -1$$

and

$$\lim_{x \rightarrow 0^+} g(x) = 1$$

and we say that

$$\lim_{x \rightarrow 0} g(x)$$

does not exist, or fails to converge.

Theorem 2.45 If $\lim_{x \rightarrow a} g(x) = L$ converges, then it follows that

$$\lim_{x \rightarrow a^-} g(x) = \lim_{x \rightarrow a^+} g(x) = L.$$

Conversely, if $\lim_{x \rightarrow a^-} g(x) \neq \lim_{x \rightarrow a^+} g(x)$ then $\lim_{x \rightarrow a} g(x)$ does not exist.

Remark 2.46

Precise definition: we say $\lim_{x \rightarrow a} f(x) = L$ if for any $\{x_n\}$ sequence with $x_n < a$ and $x_n \rightarrow a$ we have $f(x_n) \rightarrow L$.

Example 2.47 Find $\lim_{x \rightarrow 1^-} \frac{1}{1-x}$ and $\lim_{x \rightarrow 1^+} \frac{1}{1-x}$.

Solution: Note that

$$\lim_{x \rightarrow 1^+} \frac{1}{1-x} = \lim_{e \rightarrow 0^+} \frac{1}{-e} = -\infty.$$

whereas

$$\lim_{x \rightarrow 1^-} \frac{1}{1-x} = \lim_{e \rightarrow 0^-} \frac{1}{-e} = \infty.$$

Example 2.48 Let

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right) & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases} \quad (2.1)$$

Find the set of values of a such that $\lim_{x \rightarrow a} f(x)$ exists.

Solution:

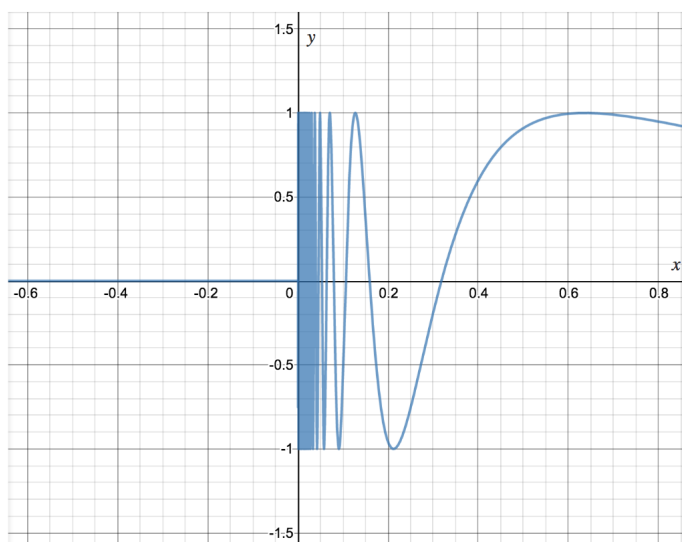


Figure 2.16: Graph of $f(x)$ given by 2.1

The limit exists everywhere except at $a = 0$. At this point, the limit from the left is 0 and the limit from the right does not exist. The graph of $f(x) = \sin\left(\frac{1}{x}\right)$ is given in Figure 2.17.

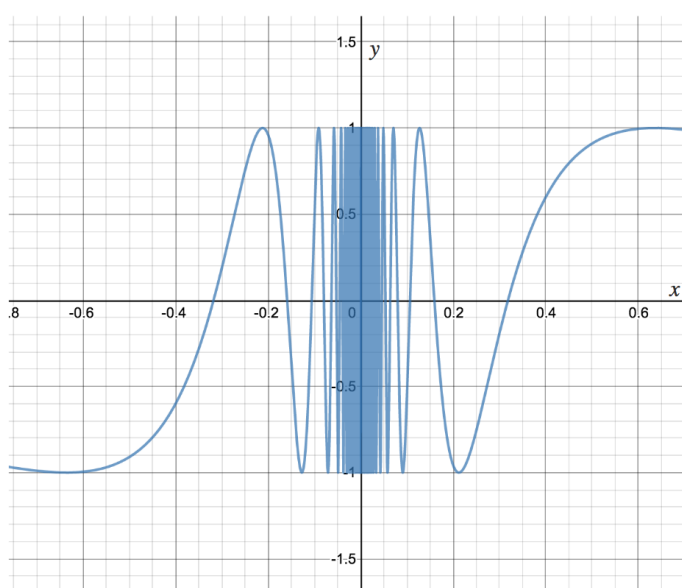


Figure 2.17: Graph of $f(x) = \sin\left(\frac{1}{x}\right)$. Approaching $x = 0$, the graph oscillates infinitely between 1 and -1

We often need to find the limit of a piecewise function. If the point we are approaching is not the "breakpoint" of the piecewise function, then we evaluate the limit of the function where that point lies, and ignore all other parts. For instance consider the following example.

Example 2.49 Find $\lim_{x \rightarrow 5} f(x)$ where

$$f(x) = \begin{cases} x^3 + 5x & x < 8 \\ \frac{100 - x}{x^2 + 7} & 8 \leq x \leq 10 \\ 0 & x > 10 \end{cases}$$

Solution: we are approaching $x = 5$, since this point lies strictly in the first part, $x^3 + 5x$, we ignore all other parts and simply evaluate $\lim_{x \rightarrow 5} x^3 + 5x$. This gives 150.

However, it may be that the point we are approaching, $x \rightarrow a$, is the breakpoint of the piecewise function. In this case, we need to find the right-hand limit, and the left-hand limit and see if they are equal. Remember, we are not interested at all what the value of the function is at the point $x = a$. Consider the following example.

Example 2.50 Find $\lim_{x \rightarrow 2} f(x)$ where

$$f(x) = \begin{cases} \frac{x^2 - x + 1}{x^2 + 2} & x < 2 \\ 9 & x = 2 \\ \frac{x^2 + 2x - 8}{x^2 - 2x} & x > 2 \end{cases}$$

Solution: We need to find the right-hand limit and left-hand limit. Respectively, we have

$$\begin{aligned} \lim_{x \rightarrow 2^+} f(x) &= \lim_{x \rightarrow 2^+} \frac{x^2 + 2x - 8}{x^2 - 2x} \\ &= \lim_{x \rightarrow 2^+} \frac{(x - 2)(x + 4)}{x(x - 2)} \\ &= \lim_{x \rightarrow 2^+} \frac{x + 4}{x} = 3 \end{aligned}$$

and

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} \frac{x^2 - x + 1}{x^2 + 2} = \frac{1}{2}$$

Since the right-hand and left-hand limits are not equal, $\lim_{x \rightarrow 2} f(x)$ does not exist. Notice that here, we did not use the value of the function at $x = 2$.

The graph of the function in the previous example is given in Fig. 2.18. We can see that at $x = 2$, the function “jumps” to 9 then jumps back down again. This is an example of a *discontinuity* where the function is not *continuous*. We will now dedicate the rest of this chapter to the fundamental topic of *continuity*.

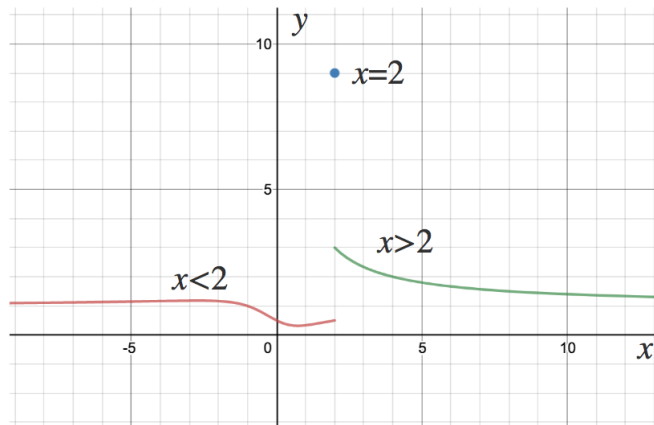


Figure 2.18: Graph of $f(x)$.

2.9 Continuity

We now relate the intuitive idea of a continuous function being one that can be drawn ‘without taking the pencil from the paper’ with a precise definition of continuity.

Definition 2.51 A function $f(x)$ is *continuous* at a if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

If $\lim_{x \rightarrow (a)^+} f(x) = f(a)$ then we say it is *continuous from the right*, (respectively left if $x \rightarrow a^-$), otherwise we say it is *discontinuous*.

We say a function is *continuous* on an interval if it is continuous at all points in this interval.

Any point a where a function is not continuous is called a *point of discontinuity* of the function.

This gives a rigorous definition of the familiar idea that a continuous function is one whose graph can be drawn without taking the pencil off the page; discontinuities are precisely points where

the pencil must be removed to continue drawing.

Types of Discontinuities

Discontinuities can come in different forms. We will look at these through examples.

Example 2.52 Let $f(x) = \frac{x^2 - 1}{x + 1}$. Here, $f(-1)$ is not defined. A function cannot be continuous at a point where it is not defined. At this point, $x = -1$, there is a break in the graph of $f(x)$. Thus we say f is discontinuous at $x = -1$. The graph is given below.

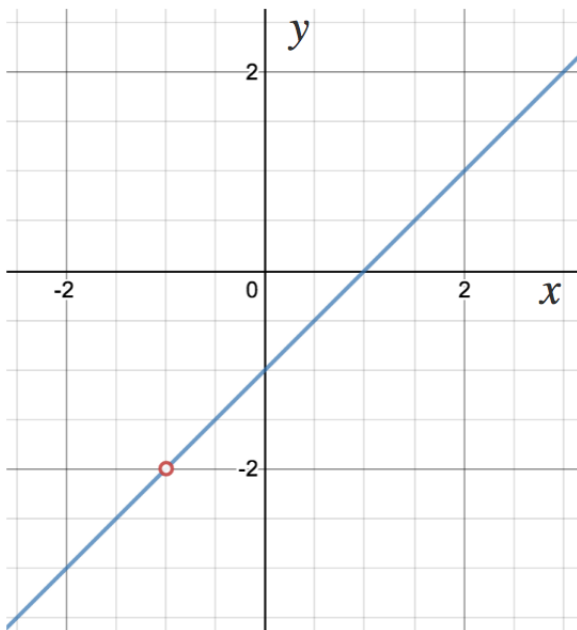


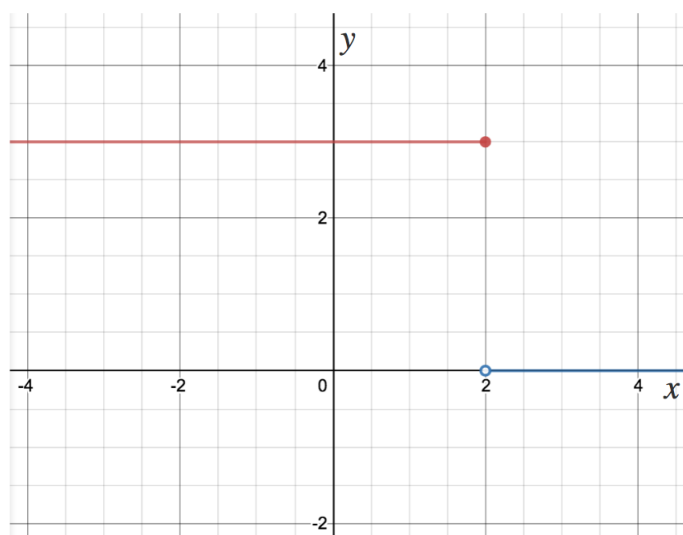
Figure 2.19: Graph of $f(x) = \frac{x^2 - 1}{x + 1}$. Usually, we use an open circle to draw points of discontinuities

Example 2.53 Let $f(x) = \begin{cases} 3 & x \leq 2 \\ 0 & x > 2 \end{cases}$

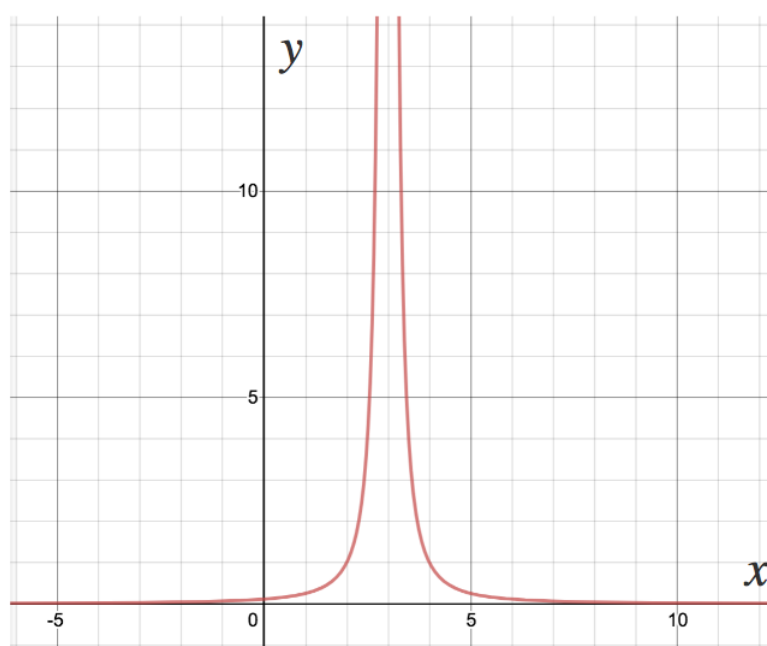
Here, f has a *jump* discontinuity at $x = 2$. The right-hand limit as $x \rightarrow 2$ is 0 and the left-hand limit is 3, so the limit at $x = 2$ does not exist. Therefore, f is not continuous.

Example 2.54 Let $f(x) = \frac{1}{(x - 3)^2}$.

Here, $f(3)$ is not defined, so the function cannot be contin-

Figure 2.20: Graph of $f(x)$

uous at $x = 3$. We call this kind of discontinuity an “infinite” discontinuity at $x = 3$.

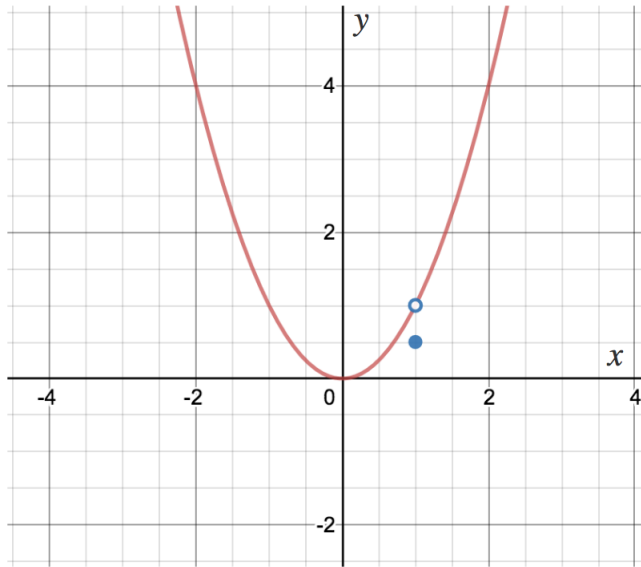
Figure 2.21: Graph of $f(x) = \frac{1}{(x-3)^2}$

Example 2.55 Let's revisit the function $f(x) = \sin\left(\frac{1}{x}\right)$, see Fig. 2.17. The function is not defined at $x = 0$, so has a discontinuity at $x = 0$.

Example 2.56 Let

$$f(x) = \begin{cases} x^2 & x \neq 1 \\ \frac{1}{2} & x = 1. \end{cases}$$

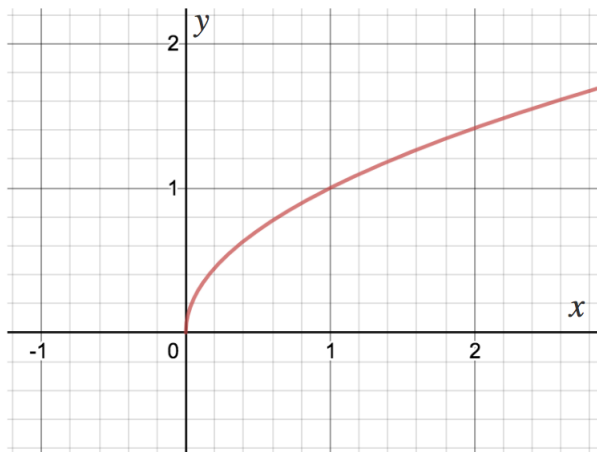
The function is defined at $x = 1$, $f(1) = \frac{1}{2}$, and the limit of f as $x \rightarrow 1$ exists and is equal to 1 (since the right-hand and left-hand limits both exist and are equal to 1), however, this limit does not equal to the function value at that point. So, we have a jump discontinuity at $x = 1$.

Figure 2.22: Graph of $f(x)$

Remark 2.57

For a function to be called continuous, it must be continuous at all points in its domain.

For example, $f : [0, \infty) \rightarrow \mathbb{R}, f(x) = \sqrt{x}$ is continuous on $(0, \infty)$ but since the left-hand limit at $x = 0$ is undefined, by definition, the function is not continuous there.

Figure 2.23: Graph of $f(x) = \sqrt{x}$

Any function x^k is continuous for $k = 0, 1, 2, \dots$. Moreover, $\sin$, $\cos$, $\exp$, $\ln$ are all continuous on their domains of definition. Hence any polynomial $a_n x^n + \dots + a_0$ is continuous at all x , as is any ratio of functions except at zeros of the denominator. For example $\tan(x) = \sin(x)/\cos(x)$ is discontinuous when $\cos x = 0$.

Also, one can prove that composing, adding, multiplying and dividing continuous functions will give new continuous functions *except when dividing by 0*. Let's write this "properly".

Theorem 2.58 Let f and g be continuous functions at $x = a$. Then

- (i) $f + g$ is continuous at $x = a$
- (ii) $f - g$ is continuous at $x = a$
- (iii) fg is continuous at $x = a$
- (iv) $\frac{f}{g}$ is continuous at $x = a$ provided $g(a) \neq 0$.

Example 2.59 For which values of x are the following continuous?

- (a) $f(x) = \sec^2(x)$ This function is continuous at all x except $x = \frac{\pi}{2} + k\pi$ with $k \in \mathbb{Z}$.

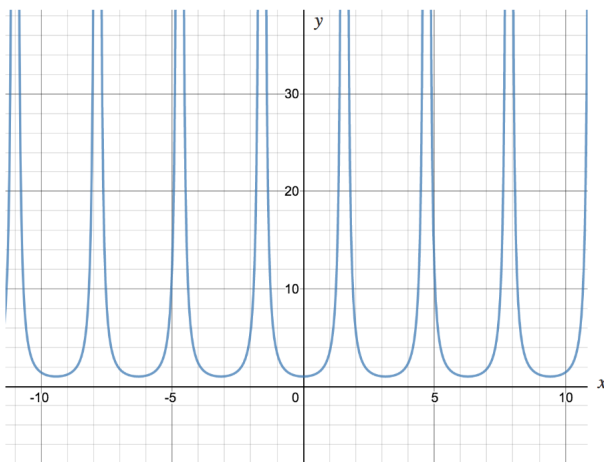


Figure 2.24: Graph of $f(x) = \sec^2(x)$.

(b)

$$f(x) = \begin{cases} 1 & x \leq 0 \\ \cos(x) & 0 \leq x \leq 2\pi \\ \sin(x) & x > 2\pi \end{cases}$$

As we saw, for piecewise defined functions, we need to check continuity at places where the pieces join. To that end, we need to see that the limits from the left and right are defined and are equal. So, the function in this example is continuous everywhere except possibly at $x = 0, 2\pi$. At 0 we have

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 1 = 1,$$

and

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \cos x = 1.$$

and both these are equal to $f(0) = 1$. Hence it is continuous at $x = 0$. At 2π we have

$$\lim_{x \rightarrow 2\pi^-} f(x) = \lim_{x \rightarrow 2\pi^-} \cos x = 1,$$

and

$$\lim_{x \rightarrow 2\pi^+} f(x) = \lim_{x \rightarrow 2\pi^+} \sin x = 0.$$

which are not equal. Hence it is not continuous at $x = 2\pi$.

By definition of continuity, if a function is continuous at a point where we want to find the limit then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

So, knowing that a function is continuous at a point, we can easily evaluate the limit there by knowing the function value at that point. For example,

$$\lim_{x \rightarrow 1} \frac{e^{x^2}}{2 - x^2} = \frac{e}{1} = e.$$

Example 2.60 Calculate the limits where possible. If not possible, see if the left hand and right hand limits exist.

(a) $\lim_{x \rightarrow 0} \frac{x^2 + x}{x^2 + 3x + 2}$

Here the function is continuous. We have

$$\lim_{x \rightarrow 0} \frac{x^2 + x}{x^2 + 3x + 2} = \frac{0^2 + 0}{0^2 + 3 \cdot 0 + 2} = 0.$$

(b) $\lim_{x \rightarrow 0} \frac{e^x - 1}{|x|}$

This limit does not exist since the right hand limit does not equal the left hand limit. We have

$$\lim_{x \rightarrow 0^-} \frac{e^x - 1}{|x|} = \lim_{x \rightarrow 0^-} \frac{e^x - 1}{-x} = -1$$

and

$$\lim_{x \rightarrow 0^+} \frac{e^x - 1}{|x|} = \lim_{x \rightarrow 0^+} \frac{e^x - 1}{x} = 1.$$

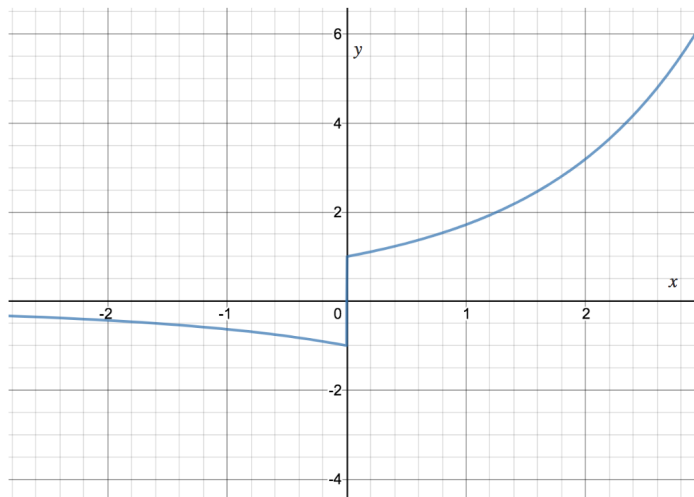


Figure 2.25: Graph of $f(x) = \frac{e^x - 1}{|x|}$.

An important consequence of continuity is the following.

The Intermediate Value Theorem

Theorem 2.61 If f is continuous on the closed interval $[a, b]$ and k is any number strictly between $f(a)$ and $f(b)$ then there exists $c \in (a, b)$ such that $f(c) = k$.

This theorem tells us that there is a number $c \in (a, b)$ such that $f(c) = k$. It doesn't tell us any more about c , nor does it prevent there being two or more such values.

Example 2.62 Show that the equation $x - \sin x = 1$ has at least one root in the interval $(0, 3)$.

Solution: Let $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x - \sin(x)$. Then f is continuous on $[0, 3]$. Also, $f(0) = 0$ and $f(3) = 3 - \sin 3 \geq 2$ so by the Intermediate Value Theorem, there exists $c \in (0, 3)$ such that $f(c) = 1$.

Example 2.63 Let

$$f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \begin{cases} 2 & \text{if } x < -1 \\ x^3 + x - 1 & \text{if } -1 \leq x \leq 1 \\ -2 & \text{if } x > 1 \end{cases}.$$

Prove that $f(x) = 0$ has a root in the interval $[-3, 3]$.

Solution: f is not continuous on $[-3, 3]$ so we cannot immediately apply the Intermediate Value Theorem. However, it is continuous on $[-1, 1]$ since in that range it is equal to a polynomial. We have $f(1) = 1$ and $f(-1) = -3$ and so the Intermediate Value Theorem shows that $f(x) = 0$ has a root in $[-1, 1]$ and hence it has one in $[-3, 3]$.

Remark 2.64

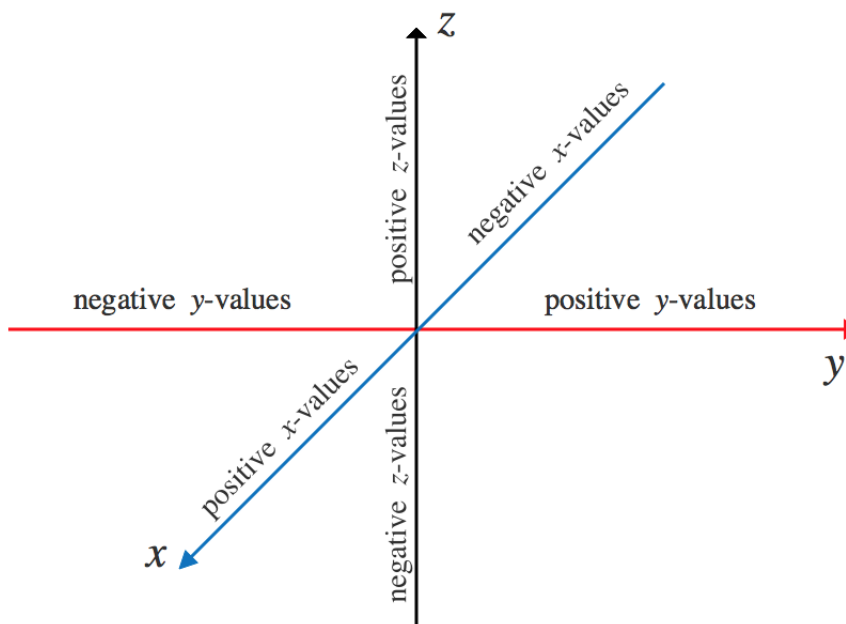
Quite often we will not be able to prove directly that an equation has a root in a given interval. We may be able to find a smaller interval on which the equation has a root and deduce that it has a root in the larger interval.

3 Vectors

You will have looked at vectors, functions and graphs in 2-dimensional space. In 2-dimensional space, $\mathbb{R}^2$, your coordinate system has two axes, usually an x -axis and a y -axis. All points and vectors in $\mathbb{R}^2$ will have two components, for example an x -component and a y -component. We will now extend this idea to 3-dimensional space, $\mathbb{R}^3$. Firstly, (x, y) -axes only represent $\mathbb{R}^2$, so we are going to need another axis, the z -axis. *

*You will usually come across 3-dimensional axes where the z -axis represents the vertical axis. However, you may wish to swap the axes over as long as you use the axes correctly.

Figure 3.1: Axes in 3-dimensional space



Points in $\mathbb{R}^3$ have three coordinates, for example $P = (2, 3, 4)$, $Q = (1, 2, -1)$. Vectors in $\mathbb{R}^3$ have three coordinates, for example $\vec{PQ} = (q_1 - p_1, q_2 - p_2, q_3 - p_3) = (1 - 2, 2 - 3, -1 - 4) = (-1, -1, -5)$.

Remark 3.1

Vectors in $\mathbb{R}^2$, written as an 'ordered pair' uses the same notation as an open interval; (a, b) where a, b are real numbers. Which is meant should be obvious from the context.

Definition 3.2 The length of a vector $\vec{v}$ is often called the *norm, modulus, length, or magnitude* of $\vec{v}$. It is denoted $|\vec{v}|$. A vector of norm 1 is called a unit vector.

If $\vec{v} = (v_1, v_2, v_3)$ then

$$|\vec{v}| = \sqrt{v_1^2 + v_2^2 + v_3^2}.$$

This definition can also be generalised to vectors in any dimension.

Example 3.3 Let $\vec{v} = (3, -2, 4)$, then

$$|\vec{v}| = \sqrt{3^2 + (-2)^2 + 4^2} = \sqrt{29}.$$

Properties of the Norm

1. $|\vec{v}| \geq 0$ for all $\vec{v} \in \mathbb{R}^3$.
2. $|k\vec{v}| = |k||\vec{v}|$ for all $\vec{v} \in \mathbb{R}^3$ and all $k \in \mathbb{R}$.
3. If $\vec{v} \neq \vec{0}$, then the unit vector that points in the same direction as $\vec{v}$ is $\frac{\vec{v}}{|\vec{v}|}$.
4. For all $\vec{u}, \vec{v} \in \mathbb{R}^3$,

$$|\vec{u} + \vec{v}| \leq |\vec{u}| + |\vec{v}|.$$

This is known as the *triangle inequality*.

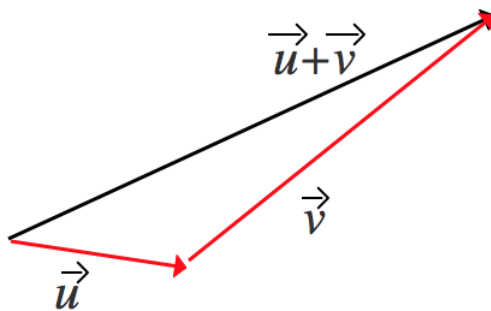


Figure 3.2: Sum of two vectors

5. For all $\vec{u}, \vec{v} \in \mathbb{R}^3$,

$$|\vec{u} - \vec{v}| \geq ||\vec{u}| - |\vec{v}||.$$

This property can be proved using the triangle inequality.

3.1 Scalar Product of Vectors

Let $\vec{u}$ and $\vec{v}$ be two vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$. To understand what is meant geometrically by “the angle between $\vec{u}$ and $\vec{v}$ ”, imagine positioning them so that their initial points coincide.

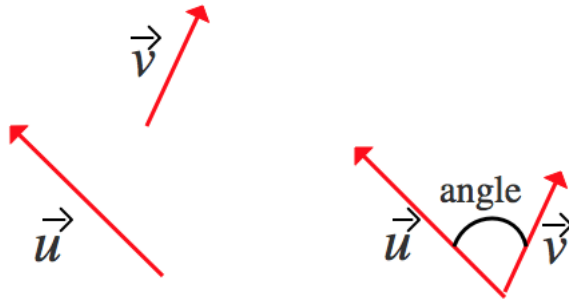


Figure 3.3: Angle between two vectors

Definition 3.4 Let $\vec{u}$ and $\vec{v}$ be two vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ with angle θ between them. The **scalar product** (also known as the *dot product*) of $\vec{u}$ and $\vec{v}$ is

$$\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$$

If $\vec{u}$ or $\vec{v}$ is the zero vector, then the angle between them is undefined. In this case, we define $\vec{u} \cdot \vec{v}$ to be 0.

Example 3.5 Let $\vec{u} = \left(\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right)$ and $\vec{v} = \left(\frac{1}{\sqrt{2}}, 1, \frac{1}{\sqrt{2}} \right)$. the angle between $\vec{u}$ and $\vec{v}$ is $\frac{\pi}{4}$. Compute the scalar product.

Solution: We have

$$\begin{aligned} \vec{u} \cdot \vec{v} &= |\vec{u}| |\vec{v}| \cos \frac{\pi}{4} \\ &= \sqrt{\frac{1}{2} + \frac{1}{2}} \sqrt{\frac{1}{2} + 1 + \frac{1}{2}} \frac{1}{\sqrt{2}} \\ &= 1 \end{aligned}$$

Remark 3.6

The scalar product of two vectors is a *scalar*. It is not a vector. It does not have different components like a vector.

There is an easy way to compute the scalar product of two

vectors.

Theorem 3.7 If $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$, then

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + u_3v_3.$$

(A similar result applies for vectors in $\mathbb{R}^2$ or in a higher dimension).

Exercise 3.8 Prove this theorem.

Example 3.9 Let $\vec{u} = (1, 4, -2)$ and $\vec{v} = (-3, 1, 0)$. Find $\vec{u} \cdot \vec{v}$.

Solution: We have

$$\begin{aligned}\vec{u} \cdot \vec{v} &= u_1v_1 + u_2v_2 + u_3v_3 \\ &= -3 + 4 + 0 = 1\end{aligned}$$

Example 3.10 Let $\vec{u} = (1, 2, 2)$ and $\vec{v} = (3, 4, 0)$. Find the angle θ between $\vec{u}$ and $\vec{v}$.

Solution: We know that

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + u_3v_3 = |u||v| \cos \theta.$$

So

$$\vec{u} \cdot \vec{v} = (1, 2, 2) \cdot (3, 0, 4) = 3 + 0 + 8 = 11.$$

Also, $|\vec{u}| = \sqrt{1^2 + 2^2 + 2^2} = 3$ and $|\vec{v}| = \sqrt{3^2 + 0^2 + 4^2} = 5$. Substituting these values into our two expressions for $\vec{u} \cdot \vec{v}$, we have

$$|\vec{u}||\vec{v}| \cos \theta = u_1v_1 + u_2v_2 + u_3v_3 + 3$$

which implies

$$15 \cos \theta = 11.$$

Solving for the angle gives

$$\theta \approx 0.748 \text{ rad.}$$

Properties of the Scalar Product

Let $\vec{u}$, $\vec{v}$, $\vec{w}$ be vectors in $\mathbb{R}^n$ and let k be a scalar. Then

1. $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$ (commutative law).
2. $\vec{u} \cdot (\vec{v} + \vec{w}) = \vec{u} \cdot \vec{v} + \vec{u} \cdot \vec{w}$ (distributive law).
3. $k(\vec{u} \cdot \vec{v}) = (k\vec{u}) \cdot \vec{v} = \vec{u} \cdot (k\vec{v})$.
4. If $\vec{v} \neq \vec{0}$ then $\vec{v} \cdot \vec{v} > 0$ (and if $\vec{v} = \vec{0}$ then $\vec{v} \cdot \vec{v} = 0$.)

Exercise 3.11 Prove these properties.

Definition 3.12 If two non-zero vectors are at right angles to each other then they are said to be *orthogonal* or perpendicular. This is denoted as

$$\vec{u} \perp \vec{v}.$$

The following theorem shows the relationship between the scalar product of vectors and whether or not they are orthogonal.

Theorem 3.13 Let $\vec{u}$ and $\vec{v}$ be non-zero vectors. Then $\vec{u} \perp \vec{v}$ if and only if $\vec{u} \cdot \vec{v} = 0$.

Proof. Since $\vec{u}$ and $\vec{v}$ are non-zero vectors, $|\vec{u}| > 0$ and $|\vec{v}| > 0$. Now, $\vec{u} \cdot \vec{v} = 0$ if and only if $|\vec{u}||\vec{v}|\cos\theta = 0$, and this occurs if and only if $\cos\theta = 0$. Now, $\cos\theta = 0$ if and only if $\theta = \frac{\pi}{2}$ or $\frac{3\pi}{2}$, which occurs if and only if $\vec{u} \perp \vec{v}$. ■

Example 3.14 Determine which of the following pairs of vectors are orthogonal.

1. $\vec{u} = (3, 2, 5)$ and $\vec{v} = (4, 1, -3)$.

We simply calculate the scalar product. Here, $\vec{u} \cdot \vec{v} = -1$ so $\vec{u}$ and $\vec{v}$ are not orthogonal.

2. $\vec{u} = (1, 3, -7)$ and $\vec{v} = (8, 2, 2)$.

The scalar product is $\vec{u} \cdot \vec{v} = 0$ so $\vec{u}$ and $\vec{v}$ are orthogonal.

3. $\vec{u} = (2, 4, 5)$ and $\vec{v} = (-7, 3, 2)$.

The scalar product is $\vec{u} \cdot \vec{v} = 8$ so $\vec{u}$ and $\vec{v}$ are not orthogonal.

4. $\vec{u} = (-7, 3, 2)$ and $\vec{v} = (-1, -1, -2)$.

The scalar product is $\vec{u} \cdot \vec{v} = 0$ so $\vec{u}$ and $\vec{v}$ are orthogonal.

Example 3.15 In $\mathbb{R}^2$, consider the line given by the equation $2x - 4y + 6 = 0$ and the vector $\vec{n} = (2, -4)$. Show that $\vec{n}$ is orthogonal to the line.

Solution: In order to show that $\vec{n}$ is orthogonal to the line, we need to show that $\vec{n}$ is orthogonal to a vector that points in the same direction as the line.

Let's choose a vector that points in the same direction: we do this by choosing two different points on the line, P and Q , then $\vec{PQ}$ points in the same direction as the line.

Let $P = (-3, 0)$ and $Q = \left(0, \frac{3}{2}\right)$, then $\vec{PQ} = \left(3, \frac{3}{2}\right)$.

Now we check that $\vec{PQ}$ is orthogonal to $\vec{n}$. We have $\vec{PQ} \cdot \vec{n} = \left(3, \frac{3}{2}\right) \cdot (2, -4) = 0$. Therefore, by definition, $\vec{PQ} \perp \vec{n}$ which means $\vec{n}$ is orthogonal to the line.

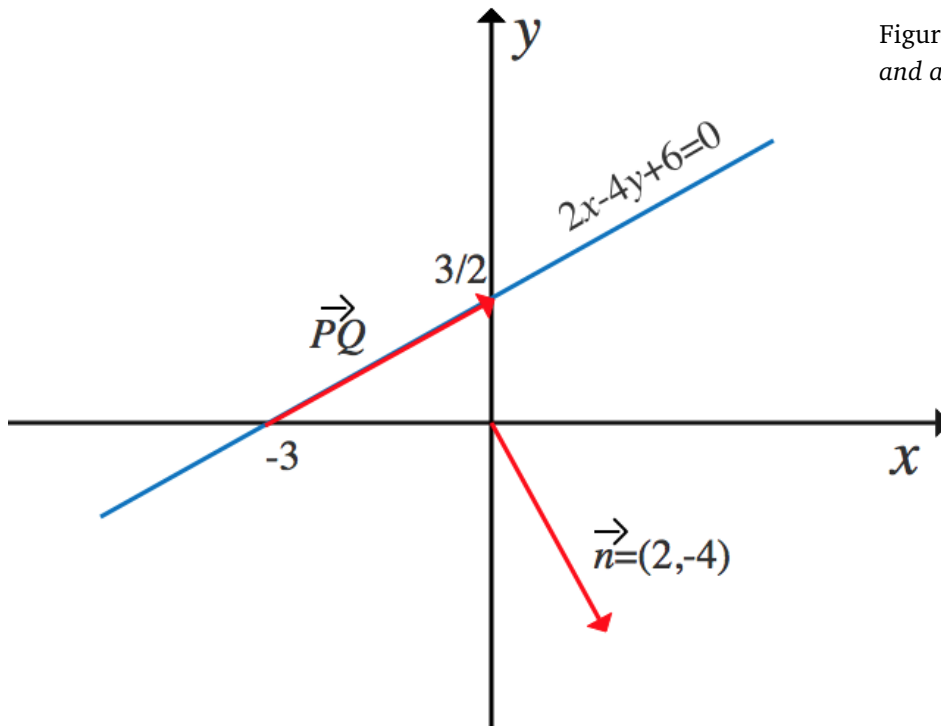
Example 3.16 Extend the result of the previous example to the general case. That is, show that the line $ax + by + c = 0$ is orthogonal to the vector $\vec{n} = (a, b)$.

Solution: Let $\vec{P} = (p_1, p_2)$ and $\vec{Q} = (q_1, q_2)$ be two points on the line. Then, $ap_1 + bp_2 + c = 0$ and $aq_1 + bq_2 + c = 0$. Subtracting the second equation from the first yields

$$a(q_1 - p_1) + b(q_2 - p_2) = 0. \quad (3.1)$$

The vector $\vec{PQ}$ points in the direction of the line, so we need to check if it is orthogonal to $\vec{n} = (a, b)$. To that end,

$$\begin{aligned} \vec{PQ} \cdot \vec{n} &= (q_1 - p_1, q_2 - p_2) \cdot (a, b) \\ &= a(q_1 - p_1) + b(q_2 - p_2) = 0 \quad \text{from (3.1)} \end{aligned}$$

Figure 3.4: Graph showing line and $\vec{n}$

Therefore, $\vec{n}$ is orthogonal to the line $ax + by + c = 0$.

Definition 3.17 The standard unit vectors, also known as the *canonical unit vectors*, are

$$\hat{i} = (1, 0, 0), \quad \hat{j} = (0, 1, 0), \quad \hat{k} = (0, 0, 1).$$

They each have norm 1 and point in the directions of the x , y , and z -axes respectively.

We can decompose any vector into its i , j , k components. For example, $(3, 4, -6) = 3\hat{i} + 4\hat{j} - 6\hat{k}$, and $-2\hat{i} + 7\hat{j} - 3\hat{k} = (-2, 7, -3)$. Furthermore, $\hat{i}$, $\hat{j}$, and $\hat{k}$ are orthogonal to each other. Let's check this using the scalar product. We have

$$\hat{i} \cdot \hat{j} = (1, 0, 0) \cdot (0, 1, 0) = 0 \quad \text{so} \quad \hat{i} \perp \hat{j},$$

$$\hat{i} \cdot \hat{k} = (1, 0, 0) \cdot (0, 0, 1) = 0 \quad \text{so} \quad \hat{i} \perp \hat{k}$$

and

$$\hat{j} \cdot \hat{k} = (0, 1, 0) \cdot (0, 0, 1) = 0 \quad \text{so} \quad \hat{j} \perp \hat{k}.$$

3.2 The Vector Cross Product

Definition 3.18 Let $\vec{u}, \vec{w} \in \mathbb{R}^3$. The **vector cross product** of $\vec{u}$ with $\vec{w}$ is given by

$$\begin{aligned}\vec{u} \times \vec{w} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ u_1 & u_2 & u_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \\ &= (u_2w_3 - u_3w_2, -u_1w_3 + u_3w_1, u_1w_2 - u_2w_1)\end{aligned}$$

Example 3.19 Let $\vec{u} = (1, 1, -1)$ and $\vec{w} = (-2, 3, 2)$. Find $\vec{u} \times \vec{w}$ and $\vec{w} \times \vec{u}$. Solution:

$$\begin{aligned}\vec{u} \times \vec{w} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & -1 \\ -2 & 3 & 2 \end{vmatrix} \\ &= \hat{i}(2 + 3) - \hat{j}(2 - 2) + \hat{k}(3 + 2) = (5, 0, 5)\end{aligned}$$

$$\begin{aligned}\vec{w} \times \vec{u} &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 3 & 2 \\ 1 & 1 & -1 \end{vmatrix} \\ &= \hat{i}(-3 - 2) - \hat{j}(2 - 2) + \hat{k}(-2 - 3) = (-5, 0, -5)\end{aligned}$$

From this example, we can see that the relationship between $\vec{u} \times \vec{w}$ and $\vec{w} \times \vec{u}$ is $\vec{u} \times \vec{w} = -(\vec{w} \times \vec{u})$. This relationship is true for all possible pairs of vectors. In other words, if $\vec{u} = (u_1, u_2, u_3)$ and $\vec{w} = (w_1, w_2, w_3)$ then $\vec{u} \times \vec{w} = -(\vec{w} \times \vec{u})$.

Theorem 3.20 Let $\vec{u}$ and $\vec{w}$ be vectors in $\mathbb{R}^3$. Then $\vec{u} \times \vec{w}$ is orthogonal to both $\vec{u}$ and $\vec{w}$.

Proof. To prove this theorem, we will recall that two vectors $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$.

Now, to show that $\vec{u}$ is orthogonal to $\vec{u} \times \vec{w}$, we will show

that $\vec{u} \cdot (\vec{u} \times \vec{w})$ is 0. We have

$$\begin{aligned}
 \vec{u} \cdot (\vec{u} \times \vec{w}) &= (u_1, u_2, u_3) \cdot (u_2w_3 - u_3w_2, -u_1w_3 + u_3w_1, u_1w_2 - u_2w_1) \\
 &= u_1(u_2w_3 - u_3w_2) + u_2(-u_1w_3 + u_3w_1) + u_3(u_1w_2 - u_2w_1) \\
 &= u_1u_2w_3 - u_1u_3w_2 - u_2u_1w_3 + u_2u_3w_1 + u_3u_1w_2 - u_3u_2w_1 \\
 &= 0.
 \end{aligned}$$

Similarly, we can show that $\vec{w} \cdot (\vec{u} \times \vec{w}) = 0$. Therefore, $\vec{u}$ and $\vec{w}$ are both orthogonal to $\vec{u} \times \vec{w}$ as required. ■

Theorem 3.21 Let $\vec{u}$ and $\vec{w}$ be vectors in $\mathbb{R}^3$, and let θ be the angle between $\vec{u}$ and $\vec{w}$. Then

$$\begin{aligned}
 |\vec{u} \times \vec{w}|^2 &= |\vec{u}|^2 |\vec{w}|^2 - (\vec{u} \cdot \vec{w})^2 \\
 &= |\vec{u}|^2 |\vec{w}|^2 \sin^2 \theta.
 \end{aligned}$$

Proof. We will prove this theorem by showing that the expansion of $|\vec{u} \times \vec{w}|^2$ is equal to the expansion of $|\vec{u}|^2 |\vec{w}|^2 - (\vec{u} \cdot \vec{w})^2$. Then we will show that $|\vec{u}|^2 |\vec{w}|^2 - (\vec{u} \cdot \vec{w})^2$ is equal to $|\vec{u}|^2 |\vec{w}|^2 \sin^2 \theta$. We have

$$\begin{aligned}
 |\vec{u} \times \vec{w}|^2 &= (u_2w_3 - u_3w_2)^2 + (u_3w_1 - u_1w_3)^2 + (u_1w_2 - u_2w_1)^2 \\
 &= u_2^2w_3^2 - 2u_2u_3w_2w_3 + u_3^2w_2^2 + u_3^2w_1^2 - 2u_1u_3w_1w_3 + u_1^2w_3^2 \\
 &\quad + u_1^2w_2^2 - 2u_1u_2w_1w_2 + u_2^2w_1^2 \\
 &= (u_1w_2 - u_2w_1)^2 + (u_1w_3 - u_3w_1)^2 + (u_2w_3 - u_3w_2)^2
 \end{aligned}$$

Now, for the right hand side, we have

$$\begin{aligned}
 |\vec{u}|^2 |\vec{w}|^2 - (\vec{u} \cdot \vec{w})^2 &= (u_1^2 + u_2^2 + u_3^2)(w_1^2 + w_2^2 + w_3^2) - (u_1w_1 + u_2w_2 + u_3w_3)^2 \\
 &= u_1^2w_1^2 + u_1^2w_2^2 + u_1^2w_3^2 + u_2^2w_1^2 + u_2^2w_2^2 + u_2^2w_3^2 \\
 &\quad + u_3^2w_1^2 + u_3^2w_2^2 + u_3^2w_3^2 \\
 &\quad - u_1^2w_1^2 - 2u_1w_1u_2w_2 - u_2^2w_2^2 - 2u_1w_1 - 2u_2w_2u_3w_3 - u_3^2w_3^2 \\
 &= u_2^2w_3^2 - 2u_2u_3w_2w_3 + u_3^2w_2^2 + u_3^2w_1^2 - 2u_1u_3w_1w_3 + u_1^2w_3^2 \\
 &\quad + u_1^2w_2^2 - 2u_1u_2w_1w_2 + u_2^2w_1^2 \\
 &= (u_1w_2 - u_2w_1)^2 + (u_1w_3 - u_3w_1)^2 + (u_2w_3 - u_3w_2)^2
 \end{aligned}$$

Now, using the relationship between the scalar product and the angle between two vectors, we have

$$\begin{aligned} |\vec{u}|^2 |\vec{w}|^2 - (\vec{u} \cdot \vec{w})^2 &= |\vec{u}|^2 |\vec{w}|^2 - |\vec{u}|^2 |\vec{w}|^2 \cos^2 \theta \\ &= |\vec{u}|^2 |\vec{w}|^2 (1 - \cos^2 \theta) \\ &= |\vec{u}|^2 |\vec{w}|^2 \sin^2 \theta \end{aligned}$$

as required. ■

The vector $\vec{u} \times \vec{w}$ is orthogonal to both $\vec{u}$ and $\vec{w}$. From Theorem 3.21, it follows that

$$|\vec{u} \times \vec{w}| = |\vec{u}| |\vec{w}| \sin \theta.$$

Geometrically, this quantity represents the area of a parallelogram formed by vectors $\vec{u}$ and $\vec{w}$. Specifically,

$$\begin{aligned} \text{Area of parallelogram} &= (\text{length of base side}) \cdot (\text{height}) \\ &= |\vec{u}| |\vec{w}| \sin \theta \\ &= |\vec{u} \times \vec{w}|. \end{aligned}$$

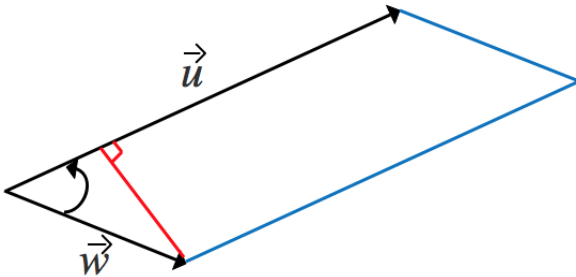


Figure 3.5: A parallelogram. The height is equal to $|\vec{w}| \sin \theta$.

Right-Handed and Left-Handed Systems

The (x, y, z) — axes form a right-handed system. In a right-handed coordinate system, the (x, y, z) — axes are oriented so that if the forefinger of the right hand points in the direction of positive y , and the second finger points in the direction of positive z , then the thumb points towards positive x , as shown in Figure 3.6 (b). With the vector cross product, $\vec{u}$, $\vec{w}$, $\vec{u} \times \vec{w}$ is a right-handed system (in that order). This means if the thumb points towards $\vec{u}$, the second finger towards $\vec{w}$, then the forefinger points towards $\vec{u} \times \vec{w}$.

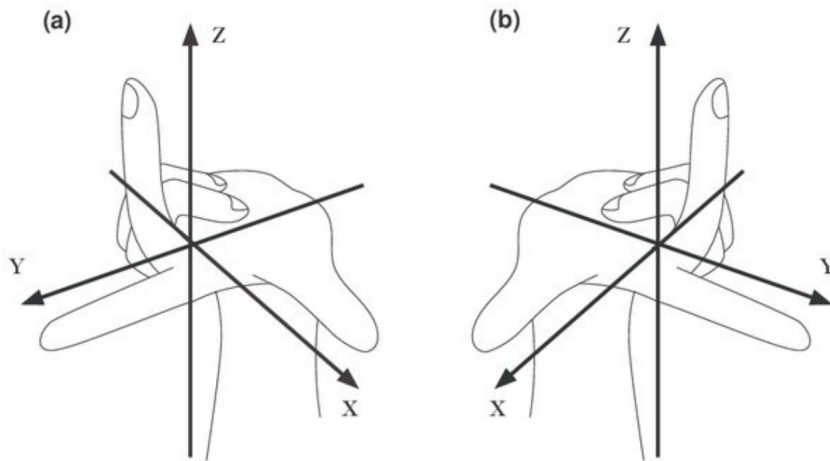


Figure 3.6: A left-handed system (a), and a right-handed system (b)

The image in Figure 3.6 was taken from this paper^{*}. However, the left-handed system (Figure 3.6 (a)) is genuinely different to a right-handed system. You cannot rotate a right-handed system to turn it into a left-handed system. If we have vectors $\vec{u}$ and $\vec{w}$, then geometrically we can build the vector $\vec{u} \times \vec{w}$ because we know the length and direction of $\vec{u} \times \vec{w}$. This means that $\vec{u} \times \vec{w}$ does not depend on the coordinate system used.

^{*}Tsuneo I., Confusion in electromagnetism and implications of CPT symmetry – system of units associated with symmetry, *Information and Communications Research Unit*, 2010

Properties of the Vector Cross Product

For all vectors $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ and scalar $c \in \mathbb{R}$,

- (i) $\vec{u} \times \vec{w} = -(\vec{w} \times \vec{u})$ (Not commutative)
- (ii) $\vec{u} \times (\vec{v} + \vec{w}) = \vec{u} \times \vec{v} + \vec{u} \times \vec{w}$ (distributive)
- (iii) $(\vec{u} + \vec{v}) \times \vec{w} = \vec{u} \times \vec{w} + \vec{v} \times \vec{w}$ (distributive)
- (iv) $c(\vec{u} \times \vec{w}) = (c\vec{u}) \times \vec{w} = \vec{u} \times (c\vec{w})$
- (v) $\vec{u} \times \vec{0} = \vec{0} \times \vec{u} = \vec{0}$
- (vi) $\vec{u} \times \vec{u} = \vec{0}$

Exercise 3.22 Prove the above properties.

3.3 The scalar triple product and vector triple product

Definition 3.23 Let $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$. Then the *scalar triple product* of these three vectors (in the given or-

der) is given by

$$\vec{u} \cdot (\vec{v} \times \vec{w}).$$

Exercise 3.24 a Use the definitions of the scalar product and the cross product to write the scalar triple product in terms of the components of the vectors.

b What is the relationship between $\vec{u} \cdot (\vec{v} \times \vec{w})$ and $\vec{v} \cdot (\vec{w} \times \vec{u})$ and $\vec{w} \cdot (\vec{u} \times \vec{v})$?

c What is the relationship between $\vec{u} \cdot (\vec{v} \times \vec{w})$ and $\vec{u} \cdot (\vec{w} \times \vec{v})$?

Consider the 3-D solid parallelepiped (a 3-dimensional parallelogram) formed from the vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$. Each face of this solid is a parallelogram formed from two of the vectors.

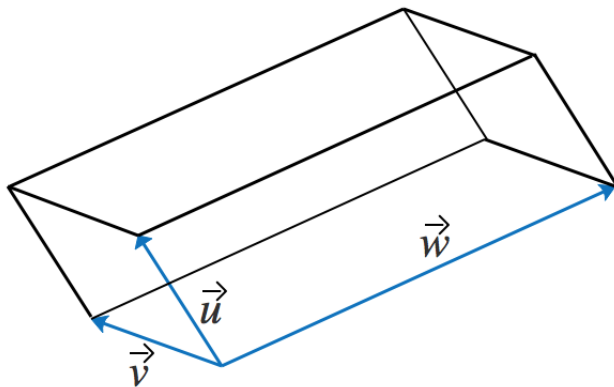


Figure 3.7: A parallelepiped

Proposition 3.25 The volume of the parallelepiped is given by

$$|\vec{u} \cdot (\vec{v} \times \vec{w})|.$$

Proof. The volume of the parallelepiped is equal to the area of the base face times the height. Let the base face be given by $\vec{v}$ and $\vec{w}$. Then the area of the base face is $|\vec{v} \times \vec{w}|$.

Now, the height of the parallelepiped can be determined

by considering a normal vector $\vec{n}$ to the base face.

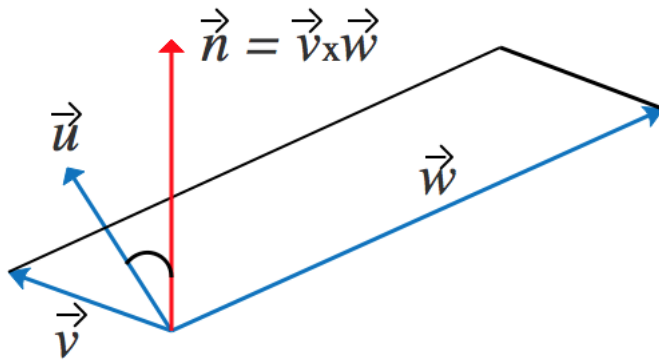


Figure 3.8: Base of a parallelepiped

Let $\vec{n} = \vec{v} \times \vec{w}$ since it is normal to the base face, so it is normal to $\vec{v}$ and $\vec{w}$. Let θ be the angle between $\vec{n}$ and $\vec{u}$. Then the height of the parallelepiped is $|\vec{u}| |\cos \theta|^*$. So the volume of the parallelepiped is

*We took the modulus of $\cos \theta$ here since $\cos \theta$ might be negative.

$$\text{Volume} = (\text{area of base face}) \cdot (\text{height})$$

$$= |\vec{v} \times \vec{w}| |\vec{u}| |\cos \theta|$$

$$= |\vec{u} \cdot (\vec{v} \times \vec{w})|$$

as required. ■

Example 3.26 Calculate the volume of the following parallelepiped.

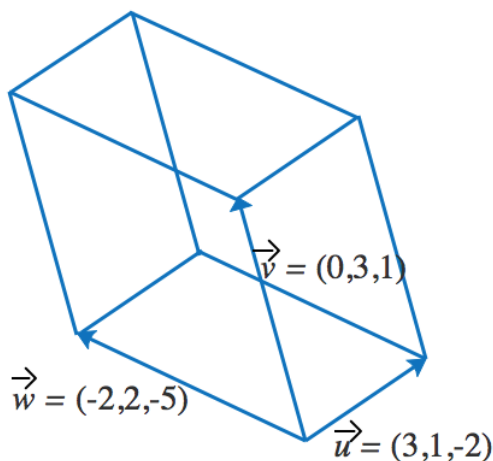


Figure 3.9: Parallelepiped example

Solution: We have

$$\begin{aligned}
 \text{Volume} &= |\vec{u} \cdot (\vec{v} \times \vec{w})| \\
 &= \begin{vmatrix} 3 & 1 & -2 \\ 0 & 3 & 1 \\ -2 & 2 & -5 \end{vmatrix} \\
 &= 3(3)(-5) + (1)(1)(-2) + (-2)(0)(2) - (-2)(3)(-2) - 3(1)(2) - 1(0)(-5) \\
 &= -65.
 \end{aligned}$$

So, the volume of the parallelepiped is 65 units³.

Remark 3.27

In this example, the scalar tripled product $\vec{u} \cdot (\vec{v} \times \vec{w})$ is negative because the vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ (in that order) form a left-handed system. If we had calculated the tripled product of the right-handed system $\vec{v}, \vec{u}, \vec{w}$ (which would be calculating $\vec{v} \cdot \vec{u} \times \vec{w}$) then we would have got 65. Remember: right-handed systems give positive tripled products.

Definition 3.28 Let $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$. Then the **vector triple product** of these three vectors is given by

$$\vec{u} \times (\vec{v} \times \vec{w}).$$

Note: the vector $(\vec{u} \times \vec{v}) \times \vec{w}$ is also a vector triple product (which is not the same as $\vec{u} \times (\vec{v} \times \vec{w})$).

Vectors in a three-dimensional space are *coplanar vectors* if they lie on the same plane. If the scalar triple product of any three vectors is 0, then they are coplanar.

Generally, $\vec{u} \times (\vec{v} \times \vec{w}) \neq (\vec{u} \times \vec{v}) \times \vec{w}$. Notice that $\vec{u} \times (\vec{v} \times \vec{w})$ is a vector coplanar with $\vec{v}$ and $\vec{w}$, whereas $(\vec{u} \times \vec{v}) \times \vec{w}$ is coplanar with $\vec{u}$ and $\vec{v}$.

3.4 Lines and Planes

Lines and Line Segments

Let L be a line in 3-D space passing through a point $P_0 = (x_0, y_0, z_0)$ parallel to a vector $\vec{v}$.

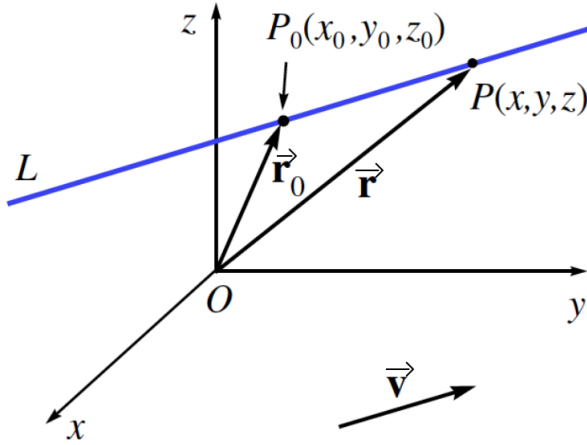


Figure 3.10: line in 3-D space passing through a point $P_0 = (x_0, y_0, z_0)$ parallel to a vector $\vec{v}$

Let $P = (x, y, z)$ be an arbitrary point on L and let $\vec{r}_0$ and $\vec{r}$ be the position vectors of P_0 and P respectively, with respect to the origin, that is

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}, \quad \vec{r}_0 = x_0\hat{i} + y_0\hat{j} + z_0\hat{k}.$$

Then P lies on L if and only if $\overrightarrow{P_0P}$ is a scalar multiple of $\vec{v}$ (since $\overrightarrow{P_0P}$ is parallel to $\vec{v}$). It means that there is a scalar t such that $\vec{r} - \vec{r}_0 = t\vec{v}$. Thus

$$\vec{r} = \vec{r}_0 + t\vec{v}, \quad t \in \mathbb{R}. \quad (3.2)$$

Equation (3.2) represents a *vector equation of the line* L , t is a parameter for the line in the sense that each value of t gives a point on the line, and each point on the line corresponds to a unique value t . We can write the vector $\vec{v}$ that gives the direction of the line L in component form as

$$\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}.$$

Two vectors are equal if and only if corresponding components are equal. Equating the corresponding components of both sides of (3.2) gives three scalar equations involving the parameter t

$$x = x_0 + tv_1 \quad y = y_0 + tv_2, \quad z = z_0 + tv_3, \quad t \in \mathbb{R}.$$

These equations are called parametric equations of the line L through the point $P_0 = (x_0, y_0, z_0)$ parallel to the vector $\vec{v} = v_1\hat{i} + v_2\hat{j} + v_3\hat{k}$.

Now, suppose that $v_1 \neq 0$, $v_2 \neq 0$, and $v_3 \neq 0$, then we have

$$t = \frac{x - x_0}{v_1}, \quad t = \frac{y - y_0}{v_2}, \quad t = \frac{z - z_0}{v_3}.$$

These values are equal to the same value of t . Thus, we have

$$\frac{x - x_0}{v_1} = \frac{y - y_0}{v_2} = \frac{z - z_0}{v_3}.$$

This representation is called the *symmetric representation* of L .

Suppose now that L is a line in 3-D space passing through points P_1 and P_2 . In this case, a point P is on the line if and only if $\overrightarrow{P_1P} = t\overrightarrow{P_1P_2}$ for some real number t . Let $\vec{r}_1$ and $\vec{r}_2$ be the position vectors of P_1 and P_2 . Then,

$$\vec{r} - \vec{r}_1 = t(\vec{r}_2 - \vec{r}_1),$$

that is

$$\vec{r} = \vec{r}_1 + t(\vec{r}_2 - \vec{r}_1), \quad t \in \mathbb{R}$$

or

$$\vec{r} = (1 - t)\vec{r}_1 + t\vec{r}_2.$$

So, to parametrise a line segment joining two points, we first parametrise the line through the points. We then find the t values for the end points and restrict t to lie in the closed interval bounded by the values. The line equations together with this added restriction parametrise the segment.

Example 3.29 Find a parametrisation for the line segment joining the points $A = (1, 2, 3)$ and $B = (3, 2, 1)$.

Solution: A vector passing through both points has coordinates $B - A = (3, 2, 1) - (1, 2, 3) = (2, 0, -2)$. So, the vector equation of the line is

$$\vec{r} = (1, 2, 3) + t(2, 0, -2) \quad t \in [0, 1].$$

Planes

A plane in space is determined by a point $P_0 = (x_0, y_0, z_0)$ in the plane and a vector $\vec{n}$ that is orthogonal to the plane. The orthogonal vector is called a normal vector.

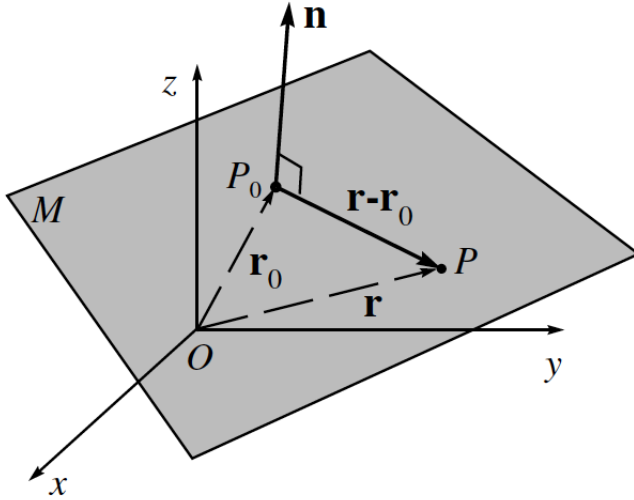


Figure 3.11: Plane M

Let M be a plane. Let $\vec{n} = (a, b, c)$ denote the normal to this plane. Let $P = (x, y, z)$ be an arbitrary point that lies on this plane, and $P_0 = (x_0, y_0, z_0)$ is another point on this plane. Since $\vec{n}$ is orthogonal to $\overrightarrow{P_0P}$, it is also orthogonal to $\vec{r} - \vec{r}_0$. Thus

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0. \quad (3.3)$$

Since

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}, \quad \vec{r}_0 = x_0\hat{i} + y_0\hat{j} + z_0\hat{k}, \quad \vec{n} = a\hat{i} + b\hat{j} + c\hat{k},$$

combining these with the vector equation given by (3.3) gives us the following definition.

Definition 3.30 Let $\vec{n} = (a, b, c)$ be normal to a plane M , and let $P = (x, y, z)$ and $P_0 = (x_0, y_0, z_0)$ be points on this plane. The *scalar equation of the plane* M is given by

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.$$

This equation can be rewritten as $ax + by + cz = ax_0 + by_0 + cz_0$.

Example 3.31 Determine whether the planes $x - 2y + 3z = 1$ and $-2x + 4y - 6z = 5$ are parallel.

Solution: Two planes are parallel if their normal vectors are parallel. So, planes $x - 2y + 3z = 1$ and $-2x + 4y - 6z = 5$ are parallel because their normal vectors are $\vec{n}_1 = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{n}_2 = -2\hat{i} + 4\hat{j} - 6\hat{k}$, and so

$$\vec{n}_2 = -2\vec{n}_1.$$

If two planes are not parallel, then they intersect in a straight line and the angle between the two planes is defined to be the acute angle determined by their normal vectors. Thus, to find the angle between two planes we need to find the (acute) angle between their normal vectors. In order to write an equation of the line of intersection of two planes, we need to know either a vector in the direction of the line and a point on the line, or two distinct points on the line. Let L be the line of intersection of two planes. Since L lies in both planes, its direction must be perpendicular to $\vec{n}_1$ and $\vec{n}_2$, normal vectors of two planes. Therefore, L is parallel to

$$\vec{v} = \vec{n}_1 \times \vec{n}_2.$$

Example 3.32 (a) Find the angle between the planes $x + y + z = 1$ and $x - 2y + 3z = 1$.

(b) Find symmetric equations for the line of intersection L of these two planes.

Solution:

(a) The normal vectors of these planes are $\vec{n}_1 = (1, 1, 1)$ and $\vec{n}_2 = (1, -2, 3)$, and so if θ is the angle between the planes, we have

$$\cos(\theta) = \frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|} = \frac{2}{\sqrt{42}}.$$

So $\theta \approx 1.26$ rads.

- (b) We first find a point on L . For instance, this can be the point where the line intersects the xy -plane by setting $z = 0$ in the equations of both planes. This gives the equations $x + y = 1$ and $x - 2y = 1$, whose solution is $x = 1$, $y = 0$. So the point $(1, 0, 0)$ lies on L .

Now, since L lies in both planes, it is orthogonal to both of the normal vectors. Thus a vector $\vec{v}$ parallel to L is given by the cross product

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = 5\hat{i} - 2\hat{j} - 3\hat{k},$$

so the symmetric equations of L can be written as

$$\frac{x-1}{5} = \frac{y}{-2} = \frac{z}{-3}.$$

Distance from a point to a line

To find the distance from a point S to a line L that passes through a point P_0 parallel to a vector $\vec{v}$, we need to find the length of the component of $\vec{b} = \overrightarrow{P_0S}$ normal to the line.

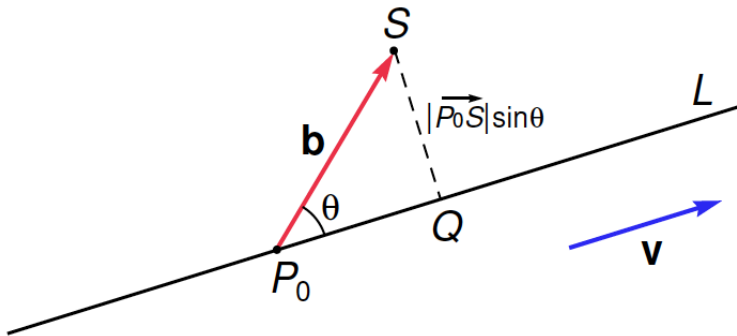


Figure 3.12: Distance from a point to a line

Let SQ be orthogonal to line L , Q be a point on L , and let θ be the angle between the line and $\overrightarrow{P_0S}$. Then d , the distance from S to the line, is $d = |SQ| = |\vec{b}| \sin(\theta)$. Now since $|\vec{b} \times \vec{v}| = |\vec{b}| |\vec{v}| \sin(\theta)$, we get

$$d = \frac{|\vec{b} \times \vec{v}|}{|\vec{v}|} = |\vec{b} \times \hat{v}|,$$

where $\hat{v}$ denotes the unit vector of $\vec{v}$. This is the general formula for the **distance from a point S to a line L** .

3.5 Vector-Valued Functions and Motion in Space

A **position vector** describes the location of a point with respect to a frame of reference. Suppose that a particle P moves in 3-D space such that its position vector, $\vec{r}$, depends only on time t . So with respect to a Cartesian frame of reference

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j} + z(t)\hat{k}, \quad t \geq 0,$$

where at any time t , the point P has Cartesian co-ordinates $(x(t), y(t), z(t))$.

For particles moving in the xy -plane the position vector is

$$\vec{r}(t) = x(t)\hat{i} + y(t)\hat{j}, \quad t \geq 0.$$

Now, suppose that a particle moving along a smooth curve in 3-D space has a position vector $\vec{r}(t)$. At time t , the particle is at point P whilst at time $t + \Delta t$ it moves to point Q .

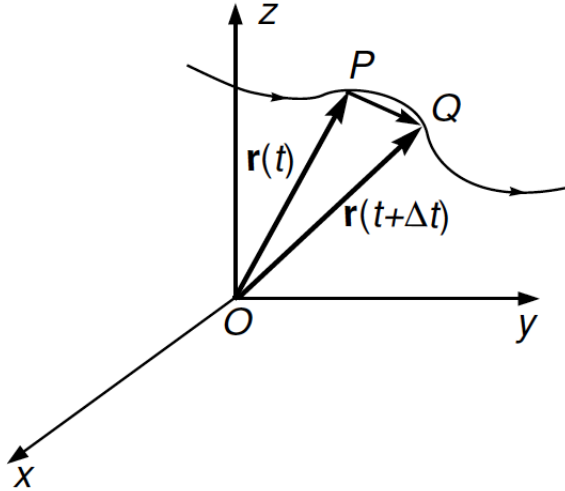


Figure 3.13: Position vector

The average **velocity** of the particle over a time interval of length Δt is

$$\frac{\overrightarrow{PQ}}{\Delta t} = \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}.$$

The instantaneous velocity, v , at time t is the limit of the average velocity as Δt approaches zero, that is

$$\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\vec{r}(t + \Delta t) - \vec{r}(t)}{\Delta t}.$$

On writing the right hand side in component form and using the definition of the derivative, this becomes

$$\vec{v}(t) = \frac{dx(t)}{dt}\hat{i} + \frac{dy(t)}{dt}\hat{j} + \frac{dz(t)}{dt}\hat{k}$$

or simply

$$\vec{v}(t) = \frac{d\vec{r}(t)}{dt} = \dot{\vec{r}}.$$

Note that as Δt approaches zero, the secant line passing through P and Q approaches a limiting position tangent to the curve at P . Hence, the velocity vector, $\vec{v} = \frac{d\vec{r}}{dt}$, when different from $\vec{0}$, is tangential to the particle's path. At any time t , the direction of $\vec{v}$ is the direction of motion. The norm of $\vec{v}$ is the particle's **speed**.

The **acceleration** of a particle is the time derivative of the velocity vector. That is,

$$\vec{a}(t) = \frac{d\vec{v}(t)}{dt} = \frac{d^2\vec{r}(t)}{dt^2} = \frac{d^2x}{dt^2}\hat{i} + \frac{d^2y}{dt^2}\hat{j} + \frac{d^2z}{dt^2}\hat{k}.$$

In general, the acceleration is not tangential to the particle path.

To summarise,

1. Velocity is the derivative of the position: $\vec{v} = \frac{d\vec{r}(t)}{dt}$.
2. Speed is the norm (magnitude) of the velocity.
3. Acceleration is the derivative of the velocity: $\vec{a} = \frac{d\vec{v}(t)}{dt}$.
4. The vector $\frac{\vec{v}}{|\vec{v}|}$ is the direction of motion at time t .
5. The velocity vector $\vec{v}$, is tangential to the particle's path.

Differentiation Rules for Vector Functions

Let $\vec{a}(t)$ and $\vec{b}(t)$ be differentiable vector functions, let $f(t)$ be a differentiable scalar function, let k be a constant, and let $\vec{c}$ be a constant vector. Then

- (a) $\frac{d}{dt}(\vec{c}) = \vec{0}$
- (b) $\frac{d}{dt}(k\vec{a}) = k\frac{d\vec{a}}{dt}$
- (c) $\frac{d}{dt}(\vec{a} + \vec{b}) = \frac{d\vec{a}}{dt} + \frac{d\vec{b}}{dt}$
- (d) $\frac{d}{dt}(f(t)\vec{a}(t)) = \frac{df}{dt}\vec{a} + f\frac{d\vec{a}}{dt}$
- (e) $\frac{d}{dt}(\vec{a}(t) \cdot \vec{b}(t)) = \frac{d\vec{a}}{dt} \cdot \vec{b} + \vec{a} \cdot \frac{d\vec{b}}{dt}$

- (f) $\frac{d}{dt}(\vec{a}(t) \times \vec{b}(t)) = \frac{d\vec{a}}{dt} \times \vec{b} + \vec{a} \times \frac{d\vec{b}}{dt}$
- (g) $\frac{d}{dt}[\vec{a}(f(t))] = \frac{df}{dt} \frac{d\vec{a}}{df}$ (Chain Rule).

Example 3.33 The following equation describes the motion of a particle having the same path, namely the unit circle $x^2 + y^2 = 1$,

$$\vec{r}(t) = \cos(t)\hat{i} + \sin(t)\hat{j}, \quad t \geq 0.$$

- (a) Does the particle have constant speed?
- (b) Is the particle acceleration vector always orthogonal to its velocity vector?

Solution:

- (a) To find the speed, we need to find the velocity vector, then compute its norm. We have

$$\vec{v}(t) = -\sin(t)\hat{i} + \cos(t)\hat{j}, \quad t \geq 0.$$

Thus the speed is $|\vec{v}| = \sqrt{\sin^2(t) + \cos^2(t)} = 1$. Since the speed is independent of t , the particle has constant speed.

- (b) Here, we need to find the scalar product of the velocity vector and acceleration vector, and see whether it is zero. The acceleration is

$$\vec{a}(t) = -\cos(t)\hat{i} - \sin(t)\hat{j}.$$

Thus $\vec{v}(t) \cdot \vec{a}(t) = \sin(t)\cos(t) - \sin(t)\cos(t) = 0$, so the acceleration vector is always orthogonal to its velocity vector.

4 Differentiation

4.1 Differentiation from First Principles

The gradient of a tangent line to the function is very important in many real-life applications as it defines the instantaneous rate of change of the function with respect to one of its variables. However, solving problems graphically or using a set of points is not always easy, imprecise or relies on us being able to draw accurately. Fortunately, we have the following theorem.

Theorem 4.1 The gradient of the tangent line to a function f at a point x is equal to the first derivative $f'(x)$.

This theorem is important as it tells us exactly what a derivative of a function means and so we can derive useful facts about the function itself.

Now, we will use the theory of limits that we have developed in order to establish a rigorous definition of the derivative. This will enable us to derive various features of the derivative which enable us to find key properties of functions and determine some limits which could not be found otherwise.

Suppose $f(x)$ is defined in some interval containing x , then

$$\frac{f(x+h) - f(x)}{h}$$

is the slope of a chord through the points $(x, f(x))$ and $(x+h, f(x+h))$. This is known as *Newton's * difference quotient*.

On taking the limit as $h \rightarrow 0$ we obtain the *derivative* of $f(x)$ with respect to x at $x = a$ (if this limit exists!). We write

$$\frac{df(x)}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Thus we have the following definition.

*Sir Isaac Newton (1642-1727) was a British mathematician, physicist, and philosopher. His inventions in calculus have been widely influential to many notable mathematicians in his era. He is also behind numerous discoveries in physics, such as the theory of gravity and the 3 laws of motion. The ever-growing fields of *Newtonian mechanics* and *Newtonian science* bear his name.

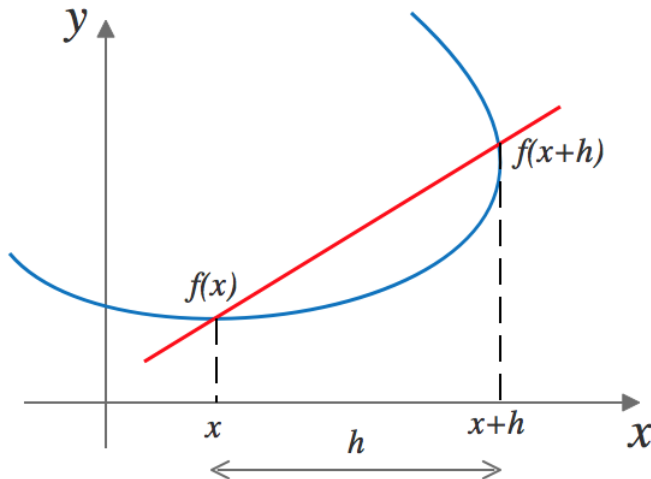


Figure 4.1: The slope of a chord through the points $(x, f(x))$ and $(x+h, f(x+h))$.

Definition 4.2 Given a function f , its *derivative*, denoted f' is a function defined by the rule

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h},$$

providing this limit exists.

This is the standard definition of a derivative, otherwise known as *differentiation from first principles*.

Since in principle this limit can be calculated for any value of x ; it defines a function x . The domain of f' consists of all x in the domain of f for which the limit exists. If the derivative of f' is defined at x , then we say that f is differentiable at x . Notice that $f(x)$ must be defined in order that $f'(x)$ is defined.

Note that both the numerator and denominator vanish in the limit, so we would always need to manipulate the function in order to find the limit.

Remark 4.3

Other notation that you might come across for the derivative are $D_x f(x)$, $f'(x)$, $f_x(x)$.

The Differential Operator

It is often useful to think of differentiation

$$D = \frac{d}{dx}$$

as an *operator*. An operator is simply a function that acts on functions. Given a function $f(x)$ then $\frac{d}{dx}f(x)$ is a new function defined by the derivative.

Example 4.4 Suppose $f(x) = \frac{1}{x}$. Find $f'(x)$ from first principles.

Solution:

$$\frac{f(x+h) - f(x)}{h} = \frac{\frac{1}{x+h} - \frac{1}{x}}{h} = \frac{x - (x+h)}{h(x+h)x} = \frac{-1}{x(x+h)}$$

so if we compute the limit we obtain

$$f'(x) = \lim_{h \rightarrow 0} \frac{-1}{x(x+h)} = -\frac{1}{x^2}.$$

This holds for any $x \neq 0$.

Note that from your previous experience of differentiation, you will know what the answer should be in many cases.

Example 4.5 Find $f'(x)$ and its domain in the following cases.

(a) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x$

(b) $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$

Solution:

(a) In this example, we have a “nice”, well-behaved function.

There are no special places where the definition or behaviour of the function changes, so we should just be able to compute the limit in one go, for every real number.

Formally, for any x ,

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = 1. \end{aligned}$$

Since the limit exists for all x , the domain of f' is $\mathbb{R}$.

- (b) When we deal with absolute values, we should always think about the equivalent piecewise definition. This function has a different definition when $x > 0$ compared with when $x < 0$, so we should expect there to be two separate cases to deal with these values of x . Points on the boundary between two different parts of a piecewise definition always need special care, so we should have a separate case to deal with 0. We will deal with these three cases separately.

If $x > 0$ then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h) - x}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h} = 1. \end{aligned}$$

If $x < 0$ then

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-(x+h) - (-x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h}{h} = -1. \end{aligned}$$

If $x = 0$ then we need to decide if

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$$

exists. To do this we look at the two one-sided limits:

$$\lim_{h \rightarrow 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{h - 0}{h} = 1$$

and

$$\lim_{h \rightarrow 0^-} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{-h - 0}{h} = -1.$$

So $\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h}$ does not exist. Hence $f'(0)$ does not exist and so the domain of f' is $\mathbb{R} \setminus \{0\}$.

Definition 4.6 A function is *differentiable on an open interval* (a, b) if it is differentiable at every point in (a, b) .

Similar definitions exist for a function being differentiable on open intervals of the form (a, ∞) , $(-\infty, a)$ and $(-\infty, \infty)$. If a

function is differentiable on $(-\infty, \infty)$, we say it is differentiable everywhere.

So, from the previous example, we can say that $|x|$ is differentiable on $(-\infty, 0)$ and $(0, \infty)$.

Relating Continuity and Differentiability

Before we present a fundamental theorem which connects continuity and differentiability, we will rewrite the limit in the definition of a derivative slightly differently. If we set $x = a + h$, then $x \rightarrow a$ is equivalent to $h \rightarrow 0$. Consequently,

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Theorem 4.7 If a function f is differentiable at $x = a$ then f is continuous at $x = a$.

Proof. We will prove this by computing $\lim_{x \rightarrow a} (f(x) - f(a))$. We have

$$\begin{aligned} \lim_{x \rightarrow a} (f(x) - f(a)) &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} (x - a) \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \lim_{x \rightarrow a} (x - a) \\ &= f'(a)(a - a) = 0. \end{aligned}$$

Now, if $\lim_{x \rightarrow a} (f(x) - f(a)) = 0$, then $\lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} f(a) = 0$, but $\lim_{x \rightarrow a} f(a) = f(a)$ and so $\lim_{x \rightarrow a} f(x) = f(a)$. Consequently, we have $\lim_{x \rightarrow a} f(x) = f(a)$. Thus by definition of continuity, f is continuous at a . ■

This theorem doesn't say anything about whether a continuous function is differentiable. In other words, the converse is not true, there are functions that are continuous at $x = a$ but not differentiable there. Remember the following picture. An example we have already seen, $|x|$, is the most common, and easily remembered, example of a continuous function that fails to be differentiable.

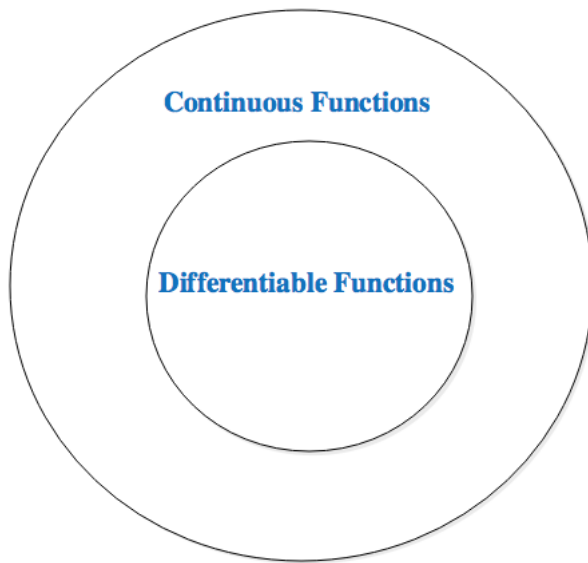


Figure 4.2: *Discontinuous functions are never differentiable but continuous functions may or may not be differentiable.*

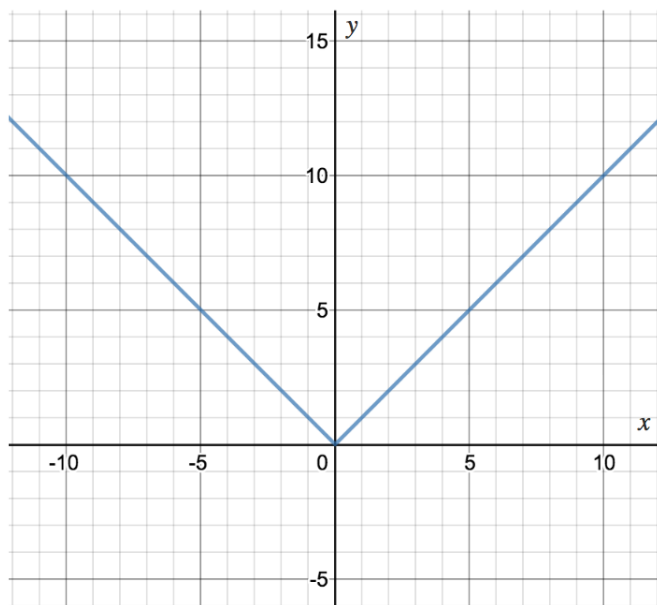


Figure 4.3: *Graph of $f(x) = |x|$.*

Common behaviour that might occur when a continuous function fails to be differentiable include corner points, such as in $|x|$, or points where the tangent becomes vertical such as in $x^{1/3}$. *

*Surprisingly, there is a function that is continuous everywhere but is differentiable nowhere. The graph is awesome! Google-image Weierstrass Function.

Exercise 4.8 Prove, using differentiation from first principles, that $f(x) = x^{1/3}$ is not differentiable at $x = 0$.

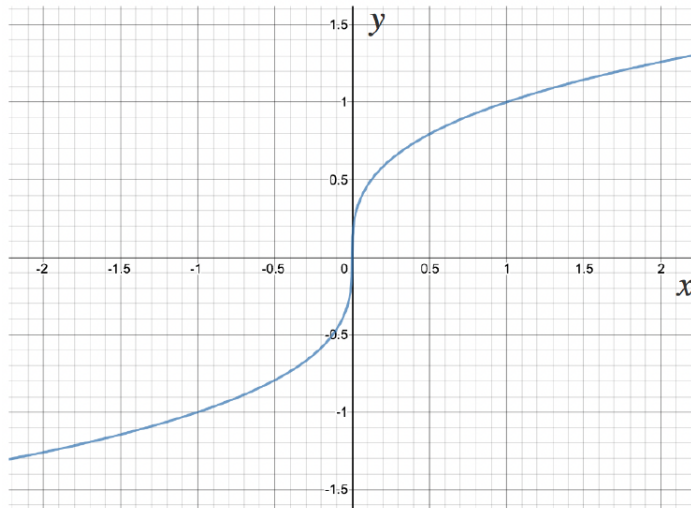


Figure 4.4: Graph of $f(x) = x^{1/3}$.

4.2 Rules for differentiation

Since we have a set of rules that help us find the derivative of a function, we can use those rules, unless specifically stated to differentiate from first principles.

Differentiating Terms of a Polynomial

For any $a \in \mathbb{R}$ we have

$$\frac{d}{dx} x^a = ax^{a-1}.$$

We can prove this result using the binomial expansion. Note that

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{(x+h)^a - x^a}{h} &= \lim_{h \rightarrow 0} \frac{x^a(1 + (h/x))^a - x^a}{h} \\ &= \lim_{h \rightarrow 0} x^a \frac{1 + ah/x + \dots - 1}{h} = ax^{a-1}. \end{aligned}$$

Differentiating the Logarithm and Exponential Functions

Also

$$\frac{d}{dx} \ln x = \frac{1}{x}, \quad \text{and} \quad \frac{d}{dx} e^x = e^x.$$

Differentiating Trigonometric and Hyperbolic Functions

We have

$$\frac{d}{dx} \sin(x) = \cos x, \quad \text{and} \quad \frac{d}{dx} \cos(x) = -\sin x,$$

Given that $\frac{d}{dx} e^x = e^x$ we can show that

$$\frac{d}{dx} \sinh x = \cosh x, \quad \text{and} \quad \frac{d}{dx} \cosh x = \sinh x.$$

Exercise 4.9 Use the standard definition of a derivative to show that $\frac{d}{dx} \sinh x = \cosh x$ and $\frac{d}{dx} \cosh x = \sinh x$.

As a result, we can obtain the derivatives of other functions.

For example

$$\frac{d}{dx} \tanh x = 1 - \tanh^2 x.$$

Given knowledge about these standard derivatives we can use the following to understand more complicated functions. Suppose that $f(x)$ and $g(x)$ are functions. Then:

Linearity

$$(f + g)'(x) = f'(x) + g'(x) = (f' + g')(x).$$

Product Rule

$$(fg)'(x) = f(x)g'(x) + f'(x)g(x) = (fg' + f'g)(x).$$

Quotient rule

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x)^2)} = \frac{(f'g - g'f)(x)}{g^2}.$$

Chain rule (function of a function)

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

We will prove the product rule here, and the proof of linearity, the quotient rule, and the chain rule have been left as an exercise.

Proof of the product rule: We have

$$\begin{aligned} (fg)'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(f(x+h) - f(x))g(x+h) + f(x)(g(x+h) - g(x))}{h} \\ &= g(x)f'(x) + f(x)g'(x) \end{aligned}$$

assuming that all these limits exist.

Exercise 4.10 Prove the property of linearity, the quotient rule, and the chain rule using differentiation from first principles.

Alternative statement of the chain rule: If $y = y(u)$ and $u = u(x)$ then

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}.$$

Example 4.11 Differentiate

$$y = \frac{(x-2)(x^2-3x)}{x^4}.$$

Solution: We could use the quotient rule, but an easier technique is to expand the numerator and then differentiate the simplified result.

$$y = \frac{x^3 - 3x^2 - 2x^2 + 6x}{x^4} = x^{-1} - 5x^{-2} + 6x^{-3}$$

and so

$$\frac{dy}{dx} = -x^{-2} + 10x^{-3} - 18x^{-4}.$$

Example 4.12 Find

(a) $\frac{d}{dx} \sin^4 x$

Using the chain rule, we have

$$\frac{d}{dx} \sin^4 x = 4 \sin^3 x \frac{d}{dx} \cos x = 4 \sin^3 x \cos x.$$

(b) $\frac{d}{dx} (4x^2 + 1)^{-3}$

Similarly, we have

$$\frac{d}{dx} (4x^2 + 1)^{-3} = -3(4x^2 + 1)^{-4} \frac{d}{dx} (4x^2 + 1) = -24x(4x^2 + 1)^{-4}.$$

(c) $\frac{d}{dx} \cosh(x^2 - 3x + 1)$

We have

$$\begin{aligned} \frac{d}{dx} \cosh(x^2 - 3x + 1) &= \sinh(x^2 - 3x + 1) \frac{d}{dx} (x^2 - 3x + 1). \\ &= (2x - 3) \sinh(x^2 - 3x + 1). \end{aligned}$$

(d) $\frac{d}{dx} \tan(5 - \sin^2(3x))$

Using the chain rule repeatedly, we have

$$\begin{aligned} \frac{d}{dx} \tan(5 - \sin^2(3x)) &= \sec^2(5 - \sin^2(3x)) \frac{d}{dx} (5 - \sin^2(3x)) \\ &= \sec^2(5 - \sin^2(3x)) (-2 \sin(3x)) \frac{d}{dx} \sin(3x) \\ &= \sec^2(5 - \sin^2(3x)) (-2 \sin(3x)) 3 \cos(3x) \\ &= -6 \sin(3x) \cos(3x) \sec^2(5 - \sin^2(3x)). \end{aligned}$$

Higher Derivatives

It is often useful to compute derivatives of derivatives repeatedly to give the higher derivatives of a function. In principle there are arbitrarily many derivatives. We write

$$\frac{d}{dx} \frac{dy}{dx} = \frac{d^2y}{dx^2}, \quad \text{and} \quad \frac{d}{dx} \frac{d^2y}{dx^2} = \frac{d^3y}{dx^3},$$

etc. These are computed simply by repeated differentiation. *

*A function where all derivatives can be computed at all points in the domain is called a *smooth function*.

Remark 4.13

The *zeroth derivative* is the function itself,

$$f^{(0)}(x) = f(x).$$

Remark 4.14

Derivatives with respect to t , usually seen in mechanics, are often written with dots as follows:

$$\dot{x} = \frac{dx}{dt}, \quad \ddot{x} = \frac{d^2x}{dt^2}.$$

Example 4.15 Suppose that $y = y(u)$ and $u = u(x)$. Find an expression for

$$\frac{d^2y}{dx^2}.$$

Solution: We note that

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

and so

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} \left(\frac{dy}{du} \frac{du}{dx} \right) \\ &= \frac{d}{dx} \left(\frac{dy}{du} \right) \frac{du}{dx} + \frac{dy}{du} \frac{d}{dx} \frac{du}{dx} \\ &= \frac{d^2y}{du^2} \left(\frac{du}{dx} \right)^2 + \frac{dy}{du} \frac{d^2u}{dx^2}.\end{aligned}$$

One can repeat this to obtain, for $y = y(u)$ and $u = u(x)$:

$$\frac{d^3y}{dx^3} = \frac{d^3y}{du^3} \left(\frac{du}{dx} \right)^3 + 3 \frac{d^2y}{du^2} \frac{du}{dx} \frac{d^2u}{dx^2} + \frac{dy}{du} \frac{d^3u}{dx^3}.$$

4.3 Implicit Differentiation

An equation of the form $y = f(x)$ is said to define y *explicitly* as a function of x since y only appears by itself on one side of the equation. (Of course we should really be writing $y(x)$ instead of y since y is a function of x .) However, we might want to define a function by an equation of the form $f(x, y) = 0$ such as $yx + y + 1 - x = 0$. We say that this equation defines y *implicitly* as a function of x . An equation may not define a unique function because the graph of the equation may fail the vertical line test, in other words, it may not be the graph of a *function*. Sometimes it is not possible to solve the equation we are given for y , such as $x + \cos y = xy$. However, it is not always necessary to obtain an equation for y in terms of x in order to find $y'(x)$. Fortunately, we have something called *Implicit differentiation*.

Implicit differentiation allows one to compute derivatives without finding the explicit inverses. Suppose that

$$f(x, y) = 0$$

for some $f(x, y)$. We consider y as being a function of x , differentiate with respect to x to give an expression involving x , y , $\frac{dy}{dx}$ and solve the result for the derivative. Note that in this case $\frac{dy}{dx}$ is a function on the curve $f(x, y) = 0$ but not just of x or of y .

Example 4.16 Find $\frac{dy}{dx}$ for

$$y^2 = x$$

by implicit differentiation (this equation is equivalent to $y^2 - x = 0$).

Solution: We find

$$\frac{d}{dx}(y^2) = \frac{d}{dx}x \Rightarrow 2y\frac{dy}{dx} = 1$$

and so

$$\frac{dy}{dx} = \frac{1}{2y}$$

as long as $y \neq 0$. Note that this is the same as what we would obtain by differentiating $y = \pm\sqrt{x}$.

Remark 4.17

Implicit differentiation can in fact give many answers all of which are correct. Try and aim for the simplest version. For example, the example above satisfies

$$y^4 = x^2$$

and so

$$4y^3\frac{dy}{dx} = 2x$$

implying that

$$\frac{dy}{dx} = \frac{x}{2y^3}$$

We can substitute $y^2 = x$ to simplify the result and obtain the previous expression $\frac{dy}{dx} = \frac{1}{2y}$.

Example 4.18 Find the slope of the circle

$$x^2 + y^2 = 25$$

at the point $(-3, -4)$. Hence find an equation for its tangent at that point.

Solution: Note that

$$x^2 + y^2 = 25$$

and so

$$2x + 2y \frac{dy}{dx} = 0, \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$$

Hence slope at $(-3, -4)$ is

$$\frac{dy}{dx} = -\frac{-3}{-4} = -\frac{3}{4}.$$

Hence the equation for tangent is

$$(y + 4) = -\frac{3}{4}(x + 3) \Rightarrow y = -\frac{25}{4} - \frac{3}{4}x.$$

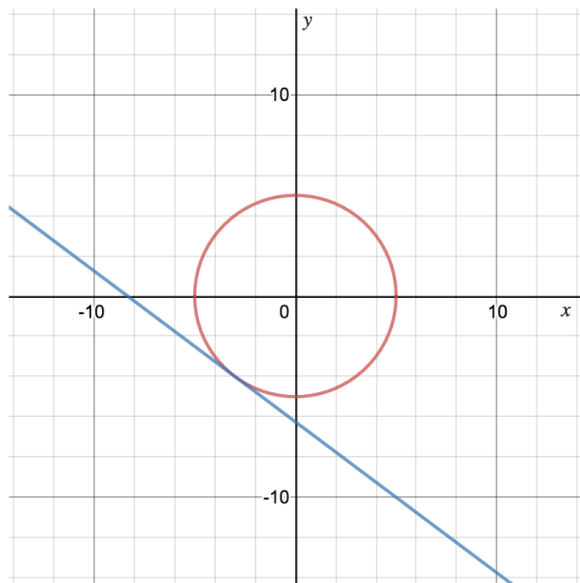


Figure 4.5: Graph of circle $x^2 + y^2 = 25$ and its tangent at $(-3, -4)$.

4.4 Logarithmic Differentiation

If functions of x appear in exponents or factors that are easily spotted, it is sometimes best to take logs and differentiate implicitly; this is called *logarithmic differentiation*.

Example 4.19 Find $\frac{dy}{dx}$ for

(a) $y = a^x$, where a is a constant.

We can write this as

$$\ln y = x \ln a$$

and so

$$\frac{1}{y} \frac{dy}{dx} = \ln a \Rightarrow \frac{dy}{dx} = (\ln a) a^x.$$

(b) $y = x^x$

We can write this as

$$\ln y = x \ln x$$

and so

$$\frac{1}{y} \frac{dy}{dx} = \ln x + \frac{x}{x} = \ln x + 1,$$

implying that

$$\frac{dy}{dx} = y(1 + \ln x) = x^x(1 + \ln x).$$

$$(c) \ y = \frac{(x-4)^7(2x+3)^2}{(4x+7)^3}$$

We have

$$\ln y = 7 \ln(x-4) + 2 \ln(2x+3) - 3 \ln(4x+7)$$

so

$$\frac{1}{y} \frac{dy}{dx} = \frac{7}{x-4} + \frac{4}{2x+3} - \frac{12}{4x+7}.$$

Making dy/dx the subject gives

$$\frac{dy}{dx} = \frac{7(2x+3)^2(x-4)^6}{(4x+7)^3} + \frac{4(x-4)^7(2x+3)}{(4x+7)^3} - \frac{12(x-4)^7(2x+3)^2}{(4x+7)^2}.$$

Differentiation of Inverse Functions

If we can find the inverse explicitly, we can differentiate this.

However, it is often useful to use implicit differentiation to help us find the derivative of inverse functions.

Example 4.20 Find the derivatives of the following with respect to x .

$$(a) \ \sin^{-1} x$$

Note that if $y = \sin^{-1} x$ then $\sin y = x$. Hence differentiating implicitly gives

$$\cos y \frac{dy}{dx} = 1$$

and so

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1 - \sin^2 y}} = \frac{1}{\sqrt{1 - x^2}}.$$

(b) $\cosh^{-1} 3x$ We see $\cosh y = 3x$ and so

$$\frac{dy}{dx} \sinh y = 3 \Rightarrow \frac{dy}{dx} = \frac{3}{\sinh y} = \frac{3}{\sqrt{\cosh^2 y - 1}}$$

and so

$$\frac{dy}{dx} = \frac{3}{\sqrt{9x^2 - 1}}.$$

(c) $\cos^{-1} x$ Putting $y = \cos^{-1} x$ we obtain $x = \cos y$ and so

$$-\sin y \frac{dy}{dx} = 1,$$

and rearranging gives

$$\frac{dy}{dx} = -\frac{1}{\sin y}$$

and so

$$\frac{dy}{dx} = -\frac{1}{\sqrt{1 - x^2}}.$$

Example 4.21 Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ for

$$x^2 + 4xy - y^2 = 8.$$

Solution: Note that

$$2x + 4xy' + 4y - 2yy' = 0$$

so $y' = \frac{x + 2y}{y - 2x}$. Hence

$$\begin{aligned} y'' &= \frac{(y - 2x)(x + 2y)' - (x + 2y)(y - 2x)'}{(y - 2x)^2} \\ &= \frac{(y - 2x)(1 + 2y') - (x + 2y)(y' - 2)}{(y - 2x)^2} \\ &= \frac{5(y - xy')}{(y - 2x)^2} \\ &= \frac{5y}{(y - 2x)^2} - \frac{5x}{(y - 2x)^2} \frac{x + 2y}{y - 2x} \\ &= \frac{5(y^2 - 4xy - x^2)}{(y - 2x)^3}. \end{aligned}$$

Exercise 4.22 Given

$$xy^3 - 3x^2 = xy + 5$$

find $\frac{dy}{dx}$.

Generally, if $y = f(x)$ has inverse $g = f^{-1}$ then

$$g(f(x)) = x$$

and so

$$g'(f(x))f'(x) = 1$$

which implies that

$$g'(y)f'(x) = 1.$$

Therefore, if x and y are related by a formula, then

$$\frac{dx}{dy} = \left(\frac{dy}{dx}\right)^{-1}.$$

4.5 Parametric Differentiation

One can describe quite general curves using a parametrization of the curve; in other words, a pair of functions $(x(t), y(t))$ where t represents a real number that parametrizes the curve.

If

$$(x, y) = (g(t), f(t))$$

then

$$\frac{dy}{dx} = \frac{\frac{df}{dt}}{\frac{dg}{dt}}.$$

To see this, let us compute the derivative, using the standard definition, at some t_0 . We have

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{y(t_0 + h) - y(t_0)}{x(t_0 + h) - x(t_0)} \\ &= \lim_{h \rightarrow 0} \frac{y(t_0 + h) - y(t_0)}{h} \frac{h}{x(t_0 + h) - x(t_0)} \\ &= \frac{\frac{df}{dt}}{\frac{dg}{dt}} \end{aligned}$$

where these are evaluated at t_0 .

We can evaluate higher derivatives of a curve in parametric form. Suppose that $(x, y) = (g(t), f(t))$.

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{d}{dx} \frac{dy}{dx} = \frac{d}{dx} \left(\frac{\frac{df}{dt}}{\frac{dg}{dt}} \right) \\ &= \frac{d}{dt} \left(\frac{\frac{df}{dt}}{\frac{dg}{dt}} \right) \frac{dt}{dx} \\ &= \frac{d}{dt} \left(\frac{\frac{df}{dt}}{\frac{dg}{dt}} \right) \frac{1}{\frac{dx}{dt}} \\ &= \frac{\frac{dg}{dt} \frac{d^2f}{dt^2} - \frac{df}{dt} \frac{d^2g}{dt^2}}{\left(\frac{dg}{dt} \right)^3}.\end{aligned}$$

where we have used the quotient rule to reach the last step.

Example 4.23 Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in parametric form for $x = \cos(t)$, $y = \sin(t)$.

Solution: We see that

$$\frac{dy}{dx} = \frac{\cos(t)}{-\sin(t)} = -\cot(t).$$

For the second derivative, we have

$$\begin{aligned}\frac{d^2y}{dx^2} &= \frac{-\sin(t)(-\sin(t)) - \cos(t)(-\cos(t))}{-\sin^3(t)} \\ &= -\csc^3 t.\end{aligned}$$

4.6 Leibniz' Rule for Repeated Differentiation of Products

Suppose that

$$y(x) = f(x)g(x)$$

Then

$$y' = f'g + g'f$$

and

$$y'' = f''g + f'g' + f'g' + fg'' = f''g + 2f'g' + fg''$$

We can generalise this using the binomial coefficients to show that

$$\frac{d^n y}{dx^n} = \sum_{r=0}^n \binom{n}{r} f^{(r)} g^{(n-r)}.$$

This is known as *Leibniz' rule*, where

$$\binom{n}{r} = \frac{n!}{(n-r)!r!}.$$

Example 4.24 Find

$$\frac{d^4}{dx^4} (x^5 \sin x).$$

Solution: Using Leibniz' rule, we have

$$\begin{aligned} \frac{d^4}{dx^4} (x^5 \sin x) &= \frac{d^4}{dx^4} (x^5) \sin x + 4 \frac{d^3}{dx^3} (x^5) \frac{d}{dx} \sin x + 6 \frac{d^2}{dx^2} (x^5) \frac{d^2}{dx^2} \sin x \\ &\quad + 4 \frac{d}{dx} (x^5) \frac{d^3}{dx^3} \sin x + x^5 \frac{d^4}{dx^4} \sin x \\ &= 120x \sin x + 240x^2 \cos x - 120x^3 \sin x - 20x^4 \cos x + x^5 \sin x. \end{aligned}$$

We can also use induction to compute formulae for higher derivatives.

Example 4.25 Show that

$$\frac{d^n}{dx^n} (e^x \sin x) = 2^{n/2} e^x \sin \left(x + \frac{\pi n}{4} \right).$$

Solution: For $n = 1$ we have

$$\frac{d}{dx} (e^x \sin x) = 2^{1/2} e^x \sin \left(x + \frac{\pi}{4} \right).$$

Assume that the formula holds for $n = k$, so that

$$\frac{d^k}{dx^k} (e^x \sin x) = 2^{k/2} e^x \sin \left(x + \frac{\pi k}{4} \right).$$

Differentiating this and simplifying gives

$$\frac{d^{k+1}}{dx^{k+1}} (e^x \sin x) = 2^{(k+1)/2} e^x \sin \left(x + \frac{\pi(k+1)}{4} \right).$$

Hence, under the assumption for $n = k$, it is true for $n = k + 1$.

Therefore, by induction, it is true for all $n \in \mathbb{N}$.

Differential calculus developed as it has only because it is supremely useful as a source of methods to calculate things people want to know. We look at some methods used for this.

*Gottfried Wilhelm Leibniz (1646 - 1716) was a German mathematician and philosopher who contributed to many different areas of human knowledge. He is famous for his version of calculus and the dispute with Newton. Many of the notation and ideas were made explicit by Leibniz. He used the notation $\int$, gave the product rule for differentiation and the derivative of x^r for integral and fractional r , but he never thought of the derivative as a limit. Other great achievements in mathematics were the development of the binary system of arithmetic, his work on determinants which arose from his developing methods to solve systems of linear equations.

4.7 Linearisation

Differentiation can be used to find the ‘best fit’ straight line to any curve. For example, if

$$y = f(x)$$

then the equation for the tangent line to this curve at any $x = a$ is given by

$$\frac{y - f(a)}{x - a} = f'(a)$$

in other words, the function

$$L(x) = f'(a)(x - a) + f(a)$$

is the *linearisation* of $f(x)$ at $x = a$. If $L(x)$ is a linearisation of $f(x)$ at $x = a$ we can think of

$$L(x) \equiv f(x)$$

for x near a . What does this mean? It means that

$$\lim_{h \rightarrow 0} \frac{f(a+h) - L(a+h)}{h} = 0.$$

Note that the linearisation of any straight line is itself; suppose $f(x) = mx + c$; then for any a , $f'(a) = m$ and so

$$L(x) = m(x - a) + ma + c = mx + c = f(x)$$

This is the only function that has this property.

Example 4.26 Find the linearisation of $f(x) = \cos x$ at $x = \frac{\pi}{2}$.

Solution: We see $f'(x) = -\sin(x)$ so $f'\left(\frac{\pi}{2}\right) = -1$ and $f\left(\frac{\pi}{2}\right) = 0$. Hence

$$L(x) = -\left(x - \frac{\pi}{2}\right) = -x + \frac{\pi}{2}.$$

4.8 Extrema of Functions

Definition 4.27 A function f has a **local maximum** at c if $f(c) \geq f(x)$ for all x in some open interval containing c . Similarly, f has a **local minimum** at c if $f(c) \leq f(x)$ for all x in some open interval containing c .

Example 4.28 Using the graph, extract the local maximum and local minimum of the function

$$f : [-1, 4] \rightarrow \mathbb{R}, f(x) = 3x^4 - 16x^3 + 18x^2.$$

Solution: The points $(0, 0)$ and $(3, -27)$ are local minima.

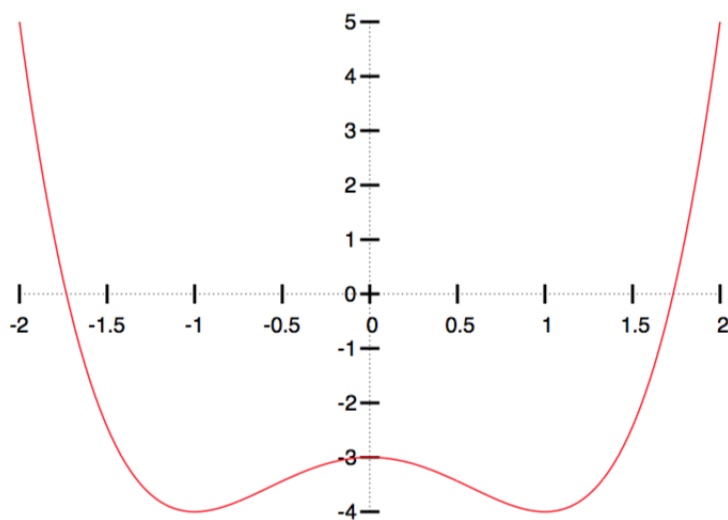


Figure 4.6: $f(x) = 3x^4 - 16x^3 + 18x^2$.

The point $(1, 5)$ is a local maximum. The point $(-1, 37)$ is not a local maximum because it is an endpoint and there is no open interval containing -1 on which f is defined. In other words, f is not defined on both sides of -1 , so there cannot be a local maximum at $(-1, 37)$.

If we were interested in the *absolute minimum* and *absolute maximum*, then these would be $(3, -27)$ and $(-1, 37)$ respectively, as these would be the lowest and highest points the function can take in its range. The idea behind this exercise is to demonstrate the difference between a local maximum and an absolute maximum. For *local*, we are only looking for the turning points, whereas for *absolute*, we want the highest function value possible in the whole range. Similarly for minimum.

Theorem 4.29 — Extreme Value Theorem. If $f(x)$ is continuous at every point on $[a, b]$ then f takes both its maximum and minimum values on this interval. In other words, there are points $x_{\max} \in [a, b]$ and $x_{\min} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for all $x \in [a, b]$.

Definition 4.30 A *stationary point* of a function f is a number $c \in D(f)$ such that $f'(c) = 0$.

Theorem 4.31 If f has a local maximum or local minimum at c , and if $f'(c)$ is defined, then $f'(c) = 0$.

This theorem tell us that if $f'(c)$ is defined at this local minimum/maximum then c is a stationary point.

This means it is possible for a function to have a local minimum/maximum at c when $f'(c)$ is undefined. The most common/easy-to-remember example is $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = |x|$. At $x = 0$, we have a minimum, but $f'(0) \neq 0$ as $f'(0)$ is undefined.

On the other hand, it is possible to have a point c such that $f'(c) = 0$ however c is not a local maximum/minimum, for example the function $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^3$ satisfies $f'(0) = 0$, but it does not have a local maximum or minimum at $x = 0$. As a reminder, these graphs are given below.

What if we are looking for both? Points when $f'(x) = 0$ or when $f'(x)$ is undefined? There's a word for that. We have the following definition.

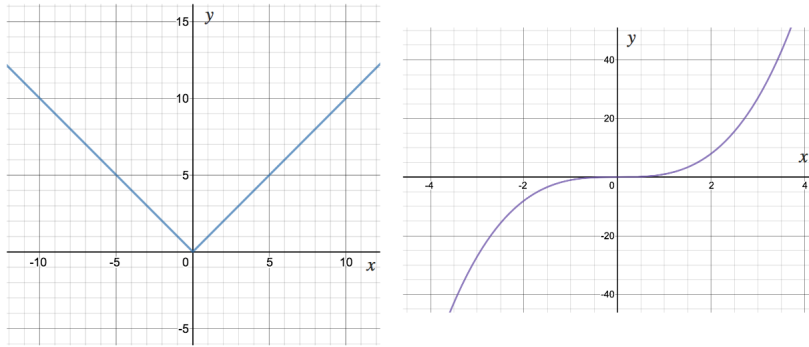


Figure 4.7: $f(x) = |x|$ (left), $f(x) = x^3$ (right).

Definition 4.32 A **critical point** of a function f is a number $c \in D(f)$ such that either $f'(c)$ is undefined or $f'(c) = 0$.

The following proposition follows from the above theorem.

Proposition 4.33 If f has a local maximum or minimum at c , then c is a critical point of f .

This proposition tells how to find a set of points that *includes* all the local minima and maxima. It does not tell us that a particular point really is a maximum or a minimum nor does it tell us how to distinguish a maximum from a minimum.

4.9 Classifying Critical Points

Definition 4.34 We say f is **convex** if any chord joining two points lies above the graph. Similarly it is **concave** if any chord lies below the graph.

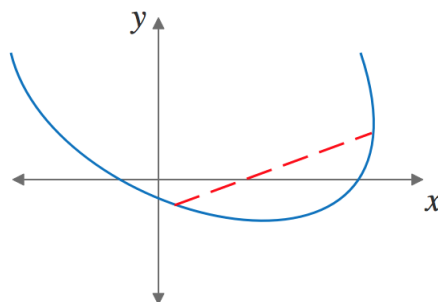


Figure 4.8: An example of a convex function.

One can define f to be concave on an interval if it is concave

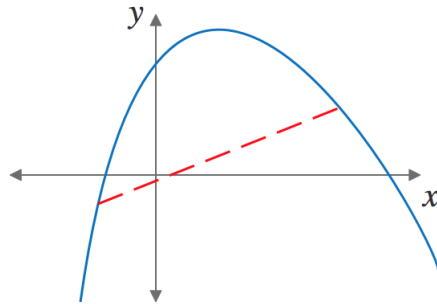


Figure 4.9: An example of a concave function.

when restricted to that interval.

Moreover, we have the following result

- If $f''(x) > 0$ for $x \in I$ (some interval) then f is convex on that interval.
- If $f''(x) < 0$ for $x \in I$ then f is concave on that interval.

This can be used to classify critical points of a curve where $f'(x) = 0$. To that end,

- If $f''(x) > 0$ then x is a *local minimum*.
- If $f''(x) < 0$ then x is a *local maximum*.

If $f'(x) = 0$ and $f''(x) = 0$ then the critical point may be a local minimum, maximum or neither. For example, if $f'''(x) \neq 0$ then x is locally monotonic.

A point where a function changes from convex to concave is called a *point of inflexion*.

If $f''(x) = 0$ and $f'''(x) \neq 0$ then x is a point of inflexion.

Example 4.35 Locate the points of inflexion for $y = x^3 + x$ and $y = \operatorname{sgn}(x) |x|^{1/3}$.

Solution: Both have inflexion points at $x = 0$. The first can be found by taking the derivatives, the second not.

Example 4.36 Find the extrema of $y = x^2 + 3x + 2$ for cases

(a) For all values of x

Note that $y' = 2x + 3$ so there is a critical point at $x = -3/2$. Also, $y'' = 2$ everywhere so the function

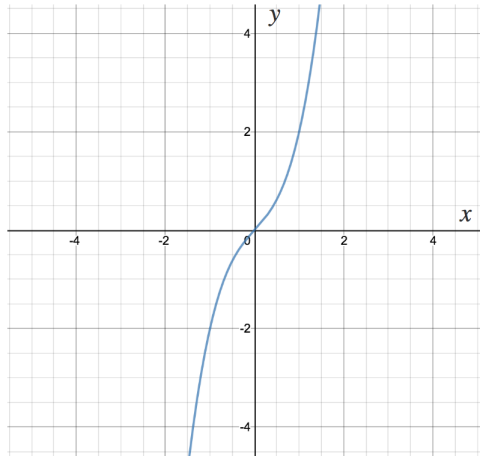


Figure 4.10: Graph of $y = x^3 + x$ showing an inflexion at $x = 0$.

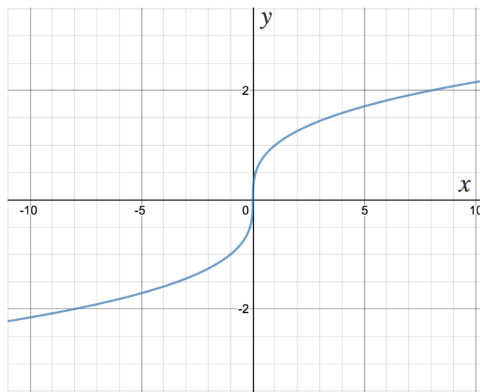


Figure 4.11: Graph of $y = \text{sgn}(x)|x|^{1/3}$ showing an inflexion at $x = 0$.

is convex; hence the critical point at $x = -3/2$ is a minimum. Hence, the critical point is located at $x = -3/2$, $y = -1/4$.

- (b) For $x \geq 1$. Here, we can see that the function is monotonically increasing for $x \geq 1$. So, the minimum lies at $x = 1$, $y = 6$.

Example 4.37 Find the extrema and inflexion points of $y = x^2 e^{-x}$.

Solution: Differentiating gives

$$y' = (2x - x^2)e^{-x}$$

which is 0 at $x = 0$ and $x = 2$. Now, we find the second derivative. We have $y''(x) = (2 - 4x + x^2)e^{-x}$ so that

$$y''(0) = 2, \quad y''(2) = -\frac{2}{e^2}.$$

Hence we have a minimum at $x = 0$, and a maximum at $x = 2$.

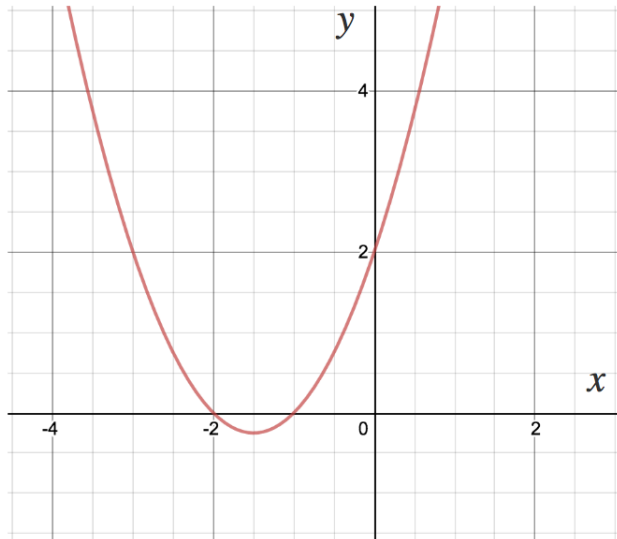


Figure 4.12: Graph of $y = x^2 + 3x + 2$ showing a minimum at $x = -3/2$.

Inflexion points occur at $y''(x) = 0$ so that $2 - 4x + x^2 = 0$ (since e^{-x} is never 0) and solving this gives

$$x = 2 \pm \sqrt{2}.$$

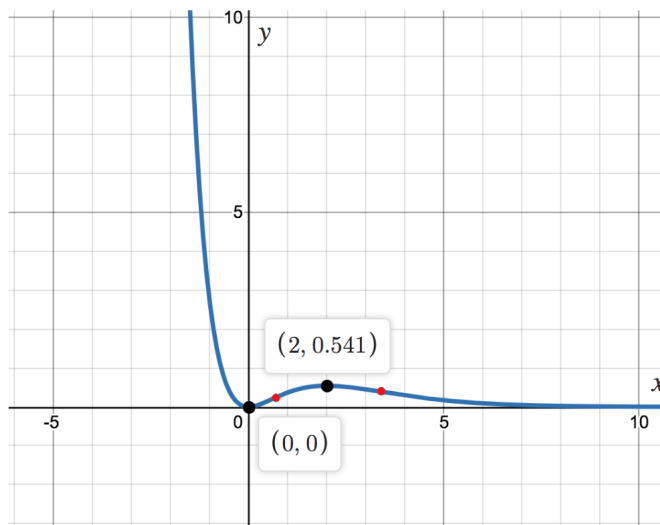
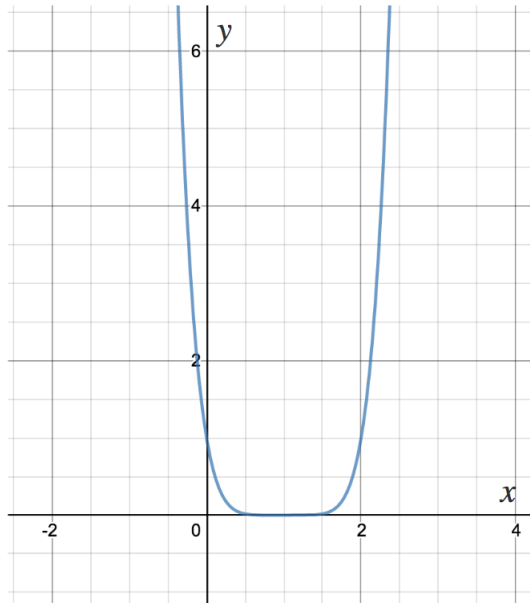


Figure 4.13: Graph of $y = x^2 e^{-x}$ showing the maximum and minimum points in black and the points of inflexion in red.

Example 4.38 Find the nature of the critical point of $y = (x - 1)^6$ at $x = 1$.

Solution: Note that $y' = y'' = y''' = \dots = y^{(6)}(0) = 1$. Hence it is easier to use the substitution $x = 1 + \epsilon$; so that $y = \epsilon^6$. This clearly is a local minimum.

Figure 4.14: Graph of $y = (x - 1)^6$

4.10 L'Hôpital's Rule

If we are trying to determine the limit of a ratio of two functions that both limit to zero, calculus can help. For example, consider

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}.$$

This limit is of type " $\frac{0}{0}$ ", and there is an easy way to evaluate such limits, as well as limits of the type " $\frac{\infty}{\infty}$ ".

Theorem 4.39 — L'Hôpital's Rule. In general, suppose that $f(a) = g(a) = 0$ or $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = \infty$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

assuming the derivatives exist and $g'(a) \neq 0$.

This is known as L'Hôpital's * rule. Thus using this rule, we have

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \frac{\cos 0}{1} = 1.$$

Example 4.40 Find the following limits.

(a) $\lim_{x \rightarrow 0} \frac{xa^x}{a^x - 1}$ where $a > 1$.

*Guillaume de L'Hôpital (1661-1704) was a French mathematician who was known to be humble and generous, which was very rare among mathematicians in his era. He is mostly known for his work on differential calculus and published a book which includes the rule he came up with to calculate indeterminate limits. He worked alongside well-renowned mathematicians such as Johann Bernoulli. After L'Hôpital's death, Bernoulli claimed that many ideas in L'Hôpital's book were actually his. However, this claim is disputable.

We have

$$f(x) = xa^x, \quad g(x) = a^x - 1$$

and so $f(0) = g(0) = 0$. Now

$$f'(x) = a^x + xa^x \ln a = a^x(1 + x \ln a).$$

hence $f'(0) = 1$. Also

$$g'(x) = a^x \ln a$$

so $g'(0) = \ln a \neq 0$. Hence

$$\lim_{x \rightarrow 0} \frac{xa^x}{a^x - 1} = \frac{1}{\ln a}.$$

$$(b) \lim_{x \rightarrow 0} \frac{e^{3x} - 3x - 1}{\sin^2 5x} = \frac{f(x)}{g(x)}.$$

Now $f'(x) = 3e^{3x} - 3$ and so $f'(0) = 0$. Also $g'(x) = 5 \sin 10x$ so $g'(0) = 0$. We apply L'Hôpital's Rule again to get

$$f''(x) = 9e^{3x}, \quad f''(0) = 9, \quad g''(x) = 50 \cos 10x, \quad g''(0) = 50.$$

Hence

$$\lim_{x \rightarrow 0} \frac{e^{3x} - 3x - 1}{\sin^2 5x} = \frac{9}{50}.$$

In this case, we needed to repeatedly apply L'Hôpital's rule.

$$(c) \lim_{x \rightarrow 0^+} \frac{\ln x}{\cot x}.$$

This limit is of type " $\frac{\infty}{\infty}$ ". We have

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{\ln x}{\cot x} &= \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-\frac{1}{\sin^2 x}} \\ &= - \lim_{x \rightarrow 0} \frac{\sin^2 x}{x} \\ &= - \left(\lim_{x \rightarrow 0} \sin x \right) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 0. \end{aligned}$$

Example 4.41 Find $\lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}}$.

Solution: Note that this is a limit of the type " 1^∞ ", and we cannot make any conclusions as to what this is without a direct

proof. To compute this limit, we will use the rule

$$\lim_{x \rightarrow a} f(x) = e^{\ln(\lim_{x \rightarrow a} f(x))}$$

and the rule

$$\lim_{x \rightarrow a} \ln(f(x)) = \ln(\lim_{x \rightarrow a} f(x)).$$

So we will evaluate $\lim_{x \rightarrow 0} \ln(1+x)^{\frac{1}{x}}$ then apply the exponential to our answer. We have

$$\begin{aligned} \lim_{x \rightarrow 0} \ln(1+x)^{\frac{1}{x}} &= \lim_{x \rightarrow 0} \frac{\ln(1+x)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{1}{1+x}}{1} \quad (\text{using L'Hôpital's Rule}) \\ &= \lim_{x \rightarrow 0} \frac{1}{1+x} = 1 \end{aligned}$$

Now, we take the exponential of the answer so that

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e^{\lim_{x \rightarrow 0} \ln(1+x)^{\frac{1}{x}}} = e^1.$$

4.11 Curve Sketching

Using the results of this chapter, we can make a list of points to consider when needing to sketch the graph of a function.

- Where does the graph cross/touch the y and x axes?
- Are there any stationary points?
- If $y' > 0$ curve is increasing, if $y' < 0$ then it's decreasing.
- Are there any inflection points, $y'' = 0$?
- Odd or even (symmetry)?
- Periodic function?
- Limiting behaviour as $x \rightarrow \pm\infty$?
- Asymptotes in x and y .

Example 4.42 Sketch $y = \frac{x}{(1+x)^2}$.

Solution: First we find the points where the curve hits the axes. At $x = 0$, $y = 0$. At $y = 0$, $x = 0$. So $(0,0)$ is the

only place where the curve will touch/cross the axes. We also see that the denominator is zero at $x = -1$ so $x = -1$ is a vertical asymptote. Now, let's look at the limiting behaviour as $x \rightarrow \pm\infty$. We have

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{x}{(1+x)^2} &= \lim_{x \rightarrow \infty} \frac{x}{1+2x+x^2} \\ &= \lim_{x \rightarrow \infty} \frac{1/x}{1/x^2 + 2/x + 1} = \frac{0}{1} = 0\end{aligned}$$

So as x approaches ∞ from the left or right, y approaches 0. We also note that when x is positive, y is positive, and when x is negative, y is also negative.

Now, let's check to see if the curve has any stationary points. On differentiating, we have

$$y' = \frac{1-x}{(1+x)^3}$$

which is zero if $x - 1 = 0$ so $x = 1$ is a stationary point. The corresponding y value is $\frac{1}{4}$. Although we have not used all item in the list above, we have enough information to sketch this curve.

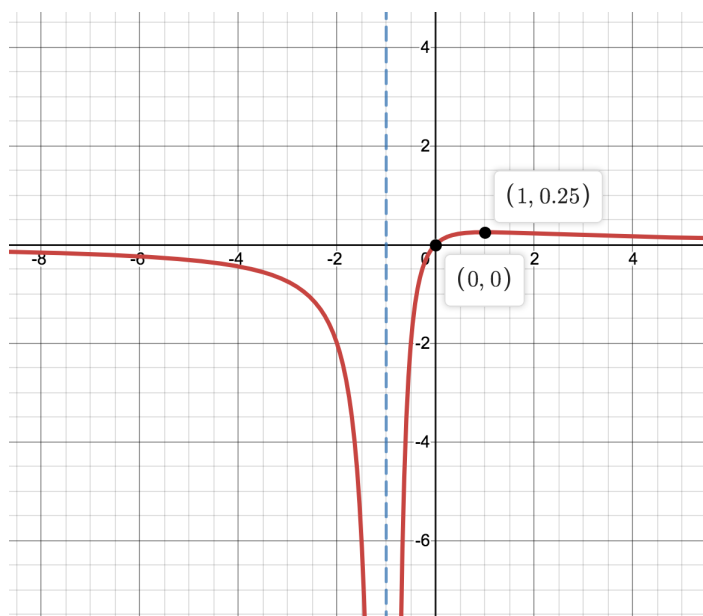


Figure 4.15: Graph of $y = \frac{x}{(1+x)^2}$.

Exercise 4.43 Sketch the following.

1. $y = x^3 + x^2 - 5x - 5$

$$2. y = 2 + \frac{1}{x}$$

$$3. y = \frac{4x}{x^2 + 9}$$

$$4. y = \frac{x^2 - 9}{2x - 4}$$

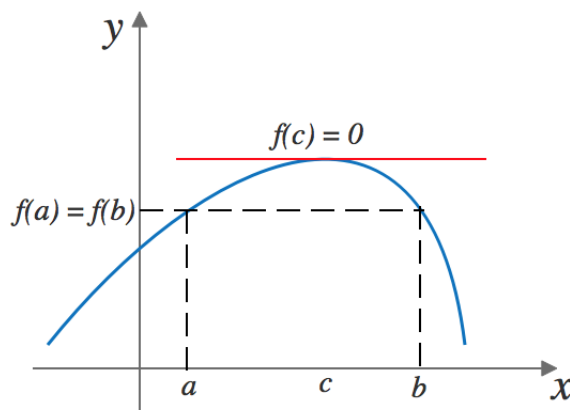
4.12 Rolle's Theorem and the Mean Value Theorem

The following theorems are behind many of the more powerful techniques of calculus and numerical methods.

Theorem 4.44 — Rolle's Theorem. Suppose $f(x)$ is continuous for all $x \in [a, b]$, differentiable for all $x \in (a, b)$, and $f(a) = f(b)$. Then there exists a value $c \in (a, b)$ such that

$$f'(c) = 0.$$

Figure 4.16 demonstrates Rolle's * Theorem.



Rolle's Theorem is an existence theorem, it tells us that a point exists inside an interval, it doesn't tell us exactly what that point is.

Example 4.45 Let $f(x) = x^2 - 7x + 12$, show that there is a stationary point in $(2, 5)$.

Solution: We can write this as $f(x) = (x - 2)(x - 5) + 2$ so $f(2) = f(5) = 2$. Since f is a polynomial, it is differen-

*Michel Rolle (1652-1719) was a French mathematician who is known for various contributions in mathematics. His family was financially unstable and so he was self-educated throughout most of his life. Sometimes he even called the theorem "which bears his name". However, like many results in differential calculus, Rolle's Theorem was initially discovered, without a proof, almost half a millennium before Rolle's time by the prominent Indian mathematician and astronomer Bhāskara II

tionable everywhere and so it is certainly continuous on $[2, 5]$ and differentiable on $(2, 5)$. So, using Rolle's Theorem, $f'(c) = 0$ for some $c \in (2, 5)$. If we wanted to go further, we could say that indeed $f'(x) = 2x - 7$ so $f'(7/2) = 0$.

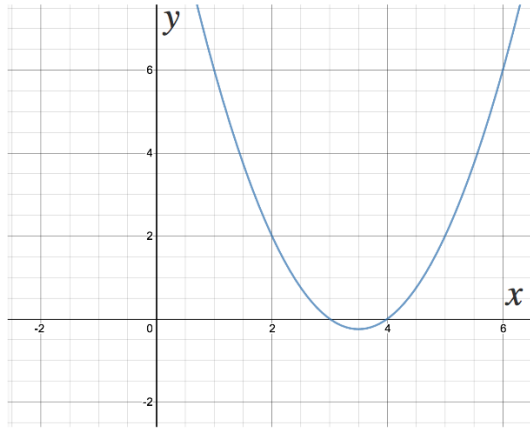


Figure 4.17: Graph of $f(x) = x^2 - 7x + 12$.

When applying a theorem, we have to make sure, and justify that the conditions of the theorem hold before we can use the concluding statement. Also, the above example simply asks us to show that there is a stationary point, it doesn't tell us to find it. Here, it is enough to use Rolle's Theorem, and this is particularly useful when we cannot find an *exact* root of $f'(x) = 0$.

Theorem 4.46 — The Mean Value Theorem (MVT).

If $f(x)$ is continuous for all $x \in [a, b]$ and differentiable for all $x \in (a, b)$ then there exists a $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

This theorem tells us that if we consider the gradient of the line connecting points a and b , then there is at least one point in the interval (a, b) where the gradient of the tangent line to f takes this value.

Example 4.47 Show that for all x there is a $0 < c < x$ such that

$$\cos c = \frac{\sin x}{x}.$$

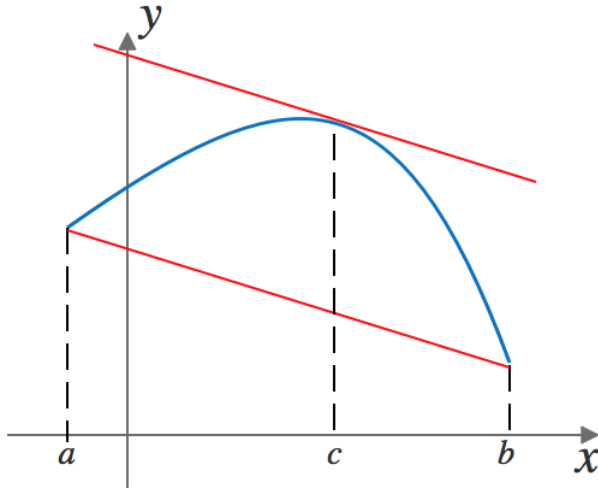


Figure 4.18: *Demonstrating the Mean Value Theorem*

Hence show that $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$.

Solution: Applying the MVT to $\sin x$ on the interval $(0, x)$ we see that there is a $0 < c < x$ such that

$$\cos c = \frac{\sin x - \sin 0}{x - 0} = \frac{\sin x}{x}.$$

Note that $\lim_{x \rightarrow 0^+} \cos x = 1$; thus

$$\lim_{x \rightarrow 0^+} \frac{\sin x}{x} = 1.$$

Corollaries of MVT

- (i) If $f'(x) = 0$ for all $x \in (a, b)$ then $f(x)$ is constant.
- (ii) If $f'(x) = g'(x)$ at each $x \in (a, b)$ then $f(x) = g(x) + C$. This can be shown by setting $h(x) = f(x) - g(x)$ and noting that $h'(x) = 0$; then applying (i).

5 Sequences and series

5.1 Sequences

A *sequence* is a list of real numbers

$$a_1, a_2, a_3, \dots = \{a_n\}_{n=1}^{\infty}$$

This may start at zero or any other number and may continue indefinitely (an infinite sequence) or may stop after a finite number of terms. We can think of a_n as being a general term in a sequence. In fact we can think of a sequence as a function from $\mathbb{N}$ to $\mathbb{R}$.

Simple examples:

- The sequence of even numbers

$$\{2, 4, 6, 8, 10, \dots\} = \{2n\}_{n=1}^{\infty}$$

- The sequence of odd numbers

$$\{1, 3, 5, 7, 9, \dots\} = \{2n - 1\}_{n=1}^{\infty}$$

- A constant sequence

$$\{2, 2, 2, 2, 2, \dots\} = \{2\}_{n=1}^{\infty}$$

- The harmonic sequence

$$\left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\} = \left\{\frac{1}{n}\right\}_{n=1}^{\infty}$$

- A geometric sequence; fix on a constant r called the *common ratio* and a the *first term*.

$$\{a, ar, ar^2, ar^3, \dots\} = \{ar^n\}_{n=0}^{\infty}$$

- An arithmetic sequence; fix constants a and b .

$$\{a, a + b, a + 2b, a + 3b, \dots\} = \{a + nb\}_{n=0}^{\infty}$$

Defining sequences

Not all terms need to be different, as in the constant series or for example

$$a_n = (-1)^n.$$

We can specify a sequence in a number of ways; for example:

(a) By specifying a general term, such as

$$a_n = 2^n + 1.$$

(Strictly speaking, we also need to specify the range of n).

(b) By giving a *recurrence formula* or *recursion formula*. This allows one to calculate in principle any term in the sequence. For example suppose that

$$a_n = na_{n-1}$$

and $a_0 = 1$. Then we can compute $a_4 = 4a_3 = 4 \cdot 3 \cdot a_2 = 4 \cdot 3 \cdot 2 \cdot a_1 = 4 \cdot 3 \cdot 2 \cdot 1 \cdot a_0 = 24$.

One can also define sequences by other means, for instance

$$\{3, 1, 4, 1, 5, 9, 2, 6, 5, 3\} = \text{digits of } \pi.$$

Example 5.1 *Fibonacci* * *sequence* Let $a_{n+2} = a_{n+1} + a_n$ for $n \in \mathbb{N}$ and $a_1 = a_2 = 1$. Compute $\{a_n\}_{n=1}^{10}$.

Solution: We can compute $a_3 = 1 + 1 = 2$, $a_4 = 2 + 1 = 3$, $a_5 = 3 + 2 = 5$, $a_6 = 5 + 3 = 8$, $a_7 = 8 + 5 = 13$, $a_8 = 13 + 8 = 21$, $a_9 = 21 + 13 = 34$, $a_{10} = 34 + 21 = 55$, $a_{11} = 55 + 34 = 89$.

*Leonardo Fibonacci was an Italian mathematician. He was known to love exploring things in nature and discovered the remarkable mathematical pattern, in many natural phenomena such as the number of petals in flowers and spirals in shells, which came to be known as the Fibonacci Sequence.

Exercise 5.2 Prove that every third number in the Fibonacci sequence is even, and that the others are odd.

5.2 Limits of Sequences

The theory of *limits of sequences*, is very similar to the theory of *limits of function* that you saw in Section 2.7. Moreover, the rules

for manipulating limits of sequences are precisely as for limits of functions, except we usually come across limits as $n \rightarrow \infty$ rather than as $x \rightarrow a$.

Definition 5.3 A number $L \in \mathbb{R}$ is a **limit** of the sequence $\{a_n\}_{n=1}^{\infty}$ if for every $\epsilon > 0$ there exists an $N(\epsilon)$ such that for all $n > N(\epsilon)$, we have

$$|a_n - L| < \epsilon.$$

A sequence that has a limit is said to **converge**. In such a case we write

$$a_n \rightarrow L \text{ as } n \rightarrow \infty$$

or equivalently

$$\lim_{n \rightarrow \infty} a_n = L.$$

If a sequence does not converge to a finite value we say that it **diverges**.

A sequence either converges or diverges, but not both.

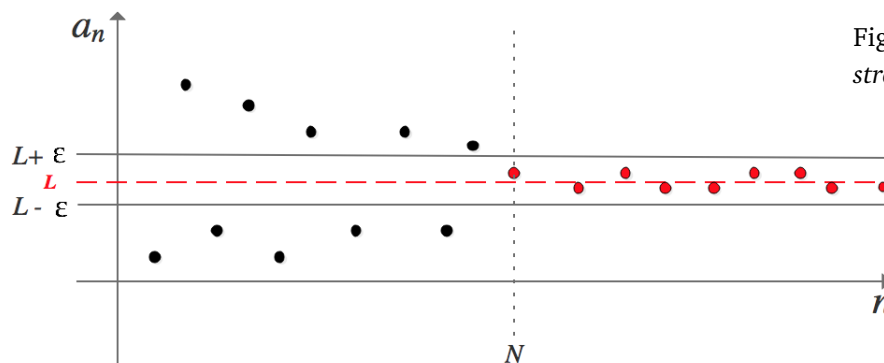


Figure 5.1: An example demonstrating Definition 5.3

Some remarks about the above definition:

- We say a_n tends to the limit L as n tends to infinity.
- It is important to have a good grasp of the above definition of convergence. You will work with this definition in other modules. The definition is saying that however small we make

the interval around L , all but a finite number of terms of the sequence will lie inside of the interval. To state this in another way: the ‘tail of the sequence’ will lie inside the interval. The smaller we make ϵ , the larger $N = N(\epsilon)$ has to be but, when we have convergence, there is always such a value $N(\epsilon)$.

- Another (equivalent) way of stating the definition: an infinite sequence a_n *converges to a limit* L if for any arbitrarily small interval $[L - \epsilon, L + \epsilon]$ containing L , then $L - \epsilon < a_n < L + \epsilon$ except for at most finitely many values of n .
- A divergent sequence does not need to grow unbounded. It can oscillate between two values, for instance.
- If a sequence $\{a_n\}_{n=1}^{\infty}$ is such that $a_n > 0$ for sufficiently large n and $\frac{1}{a_n} \rightarrow 0$ we say that $a_n \rightarrow \infty$. Similarly, if $a_n < 0$ for sufficiently large n we say $a_n \rightarrow -\infty$, a_n diverges to infinity.

Monotonic (increasing/decreasing) and bounded sequences

Definition 5.4 We say an infinite or finite sequence is *strictly increasing* if for all n

$$a_n < a_{n+1}.$$

It is *strictly decreasing* if for all n

$$a_n > a_{n+1}.$$

It is *increasing* if for all n

$$a_n \leq a_{n+1}.$$

It is *decreasing* if for all n

$$a_n \geq a_{n+1}.$$

Definition 5.5 We say a sequence is *bounded above* if there is a number M , called an *upper bound*, such that for all $n \in \mathbb{N}$

$$a_n \leq M.$$

It is *bounded below* if there is a number m , called a *lower bound*, such that for all $n \in \mathbb{N}$

$$a_n \geq m.$$

A sequence that is bounded above and below is referred to as a *bounded sequence*.

Example 5.6 Let $a_n = 1 + \frac{a_{n-1}}{3}$ and $a_1 = 0$. Determine the limit of the sequence a_n .

Solution: The set of the first few terms of the sequence is

$$\{0, 1, 1.3333, 1.4444, 1.48148, 1.493827, 1.4979\}.$$

In fact one can show that this sequence is monotonic increasing and converges to 1.5; this is a *fixed point of the recurrence*; note that if we solve

$$a = 1 + \frac{a}{3}$$

we obtain $a = 1.5$ which is in fact the limit.

Some remarks:

- $\{a_n\}_{n=1}^{\infty}$ being bounded does not imply a_n has a limit.
- $\{a_n\}_{n=1}^{\infty}$ has a limit does imply a_n is bounded.
- $\{a_n\}_{n=1}^{\infty}$ being unbounded implies a_n has NO limit.

Many infinite sequences do not have a limit, for example

$$\{n^2\} = 1, 4, 9, 16, \dots$$

has no limit, nor does

$$\{(-1)^n\} = 1, -1, 1, -1, 1, -1, \dots$$

Neither does

$$a_n = \begin{cases} 1 & n = 2^k, k \in \mathbb{R} \\ 0 & \text{otherwise.} \end{cases}$$

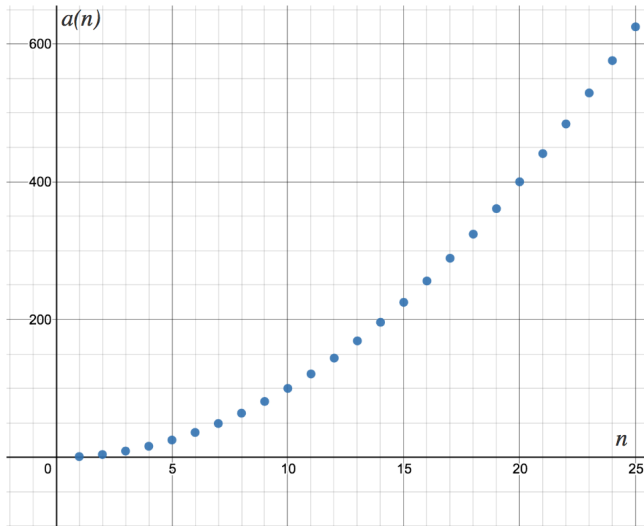


Figure 5.2: Graph of the sequence $a_n = n^2$

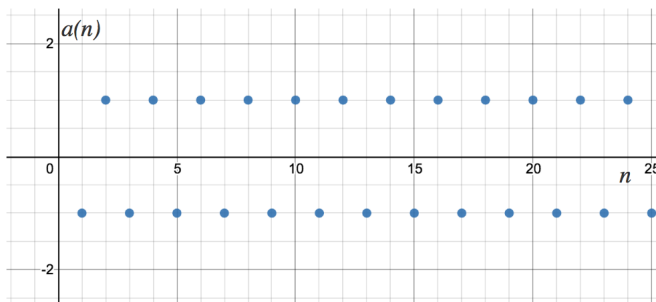


Figure 5.3: Graph of the sequence $a_n = (-1)^n$

Exercise 5.7 Sketch the graph of the sequence

$$a_n = \begin{cases} 1 & n = 2^k, k \in \mathbb{R} \\ 0 & \text{otherwise.} \end{cases}$$

Although the latter two are bounded they still do not have limits.

A fundamental result is the following:

Theorem 5.8 If an infinite sequence is monotonic increasing and bounded above then it must converge to a finite limit.

Sketch Proof: Suppose a_n is increasing and bounded above.

There must be an infinite number of values of the sequence in a finite interval. But owing to it being increasing, it cannot ‘return’ to a value after passing it; hence it must have a limit.

Theorem 5.9 — The Sandwich Theorem. Let $\{a_n\}_{n=1}^{\infty}$, $\{b_n\}_{n=1}^{\infty}$, and $\{c_n\}_{n=1}^{\infty}$ be sequences. Suppose that

$$a_n \leq b_n \leq c_n$$

and that

$$\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n.$$

Then the sequence b_n converges and its limit is the same as a_n and c_n .

Proof. Let

$$L = \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n.$$

The convergence of $\{a_n\}_{n=1}^{\infty}$ and $\{c_n\}_{n=1}^{\infty}$ to L implies that given any $\epsilon > 0$ there exists $N(\epsilon) \in \mathbb{N}$ such that for all $n > N(\epsilon)$, we have

$$|a_n - L| < \epsilon \quad \text{and} \quad |c_n - L| < \epsilon.$$

Now, since $a_n \leq b_n \leq c_n$, we have

$$a_n - L \leq b_n - L \leq c_n - L$$

and hence, for $n \geq N(\epsilon)$,

$$-\epsilon < a_n - L \leq b_n - L \leq c_n - L < \epsilon$$

and so $|b_n - L| < \epsilon$. Thus $\lim_{n \rightarrow \infty} b_n = L$. ■

Example 5.10 Suppose that $a_n \rightarrow 0$ as $n \rightarrow \infty$. Show that

$$\lim_{n \rightarrow \infty} a_n \cos n = 0.$$

Solution: We note that

$$-a_n \leq a_n \cos n \leq a_n$$

Moreover the outermost sequences converge to 0; hence by the Sandwich Theorem, $\lim_{n \rightarrow \infty} a_n \cos n = 0$.

Combining sequences and limits

Suppose that we have two sequences a_n and b_n such that $\lim_{n \rightarrow \infty} a_n$, $\lim_{n \rightarrow \infty} b_n$ both converge, and let $c \in \mathbb{R}$. Then in all cases, we have the following results *

*These rules form the famous Algebra of Limits Theorem that you will revisit in more detail in MTH1001.

$$(1) \lim_{n \rightarrow \infty} (ca_n) = c \lim_{n \rightarrow \infty} (a_n)$$

$$(2) \lim_{n \rightarrow \infty} (a_n b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) \left(\lim_{n \rightarrow \infty} b_n \right)$$

$$(3) \lim_{n \rightarrow \infty} (a_n + b_n) = \left(\lim_{n \rightarrow \infty} a_n \right) + \left(\lim_{n \rightarrow \infty} b_n \right)$$

$$(4) \text{ If } \lim_{n \rightarrow \infty} b_n \neq 0 \text{ then}$$

$$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n}.$$

Remark 5.11

Note that if $\lim_{n \rightarrow \infty} (a_n + b_n)$ converges to a limit, it is not necessary that $\lim_{n \rightarrow \infty} a_n$ or $\lim_{n \rightarrow \infty} b_n$ have limits.

For example consider $a_n = n + \frac{1}{n}$, $b_n = -n$. Both a_n and b_n do not converge but their sum does.

Comparing limits

Suppose that a_n and b_n are sequences with

$$a_n \leq b_n.$$

If the limits exist then

$$\lim a_n \leq \lim b_n.$$

Also, if $a_n \rightarrow \infty$ then $b_n \rightarrow \infty$ too.

If a_n does not converge, then either it diverges to $\pm\infty$ or we say it *oscillates* (for example $a_n = (-1)^n$ oscillates).

Limits of important sequences

(1) For any $p > 0$

$$\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$$

A special case of this is the harmonic sequence when $p = 1$.

Also

$$\lim_{n \rightarrow \infty} n^p = \infty.$$

- (2) A geometric sequence will converge if and only if its common ratio r satisfies $|r| < 1$. In other words:

$$\lim_{n \rightarrow \infty} r^n = \begin{cases} \infty & \text{if } |r| > 1 \\ 0 & \text{if } |r| < 1 \end{cases}$$

Question: what happens if $|r| = 1$?

- (3) For any $c > 0$ we have

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n^c} = 0$$

- (4) For any $p > 1$ and $c > 0$ we have

$$\lim_{n \rightarrow \infty} \frac{n^c}{p^n} = 0$$

- (5) For any $c \in \mathbb{R}$ we have

$$\lim_{n \rightarrow \infty} \frac{c^n}{n!} = 0$$

- (6) Suppose that $a > 0$. Then

$$\lim_{n \rightarrow \infty} \sqrt[n]{a} = 1.$$

- (7) The exponential as a limit is given by

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

and its generalisation for $a \in \mathbb{R}$ is given by

$$\lim_{n \rightarrow \infty} \left(1 + \frac{a}{n}\right)^n = e^a.$$

- (8) Another limit that is useful is

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1.$$

To prove this, we use the fact that $\sqrt[n]{n} = e^{\ln(n)/n}$ and then use the result $\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0$.

Example 5.12 Find the limits of the following sequences as

$n \rightarrow \infty$.

(a) $a_n = \frac{n+1}{n}$

We have

$$\lim_{n \rightarrow \infty} \frac{1 + 1/n}{1} = 1.$$

(b) $a_n = \frac{n^2 + 3n}{n(2n - 1)}$

We have

$$\lim_{n \rightarrow \infty} \frac{1 + 3/n}{2 - 1/n} = \frac{1}{2}$$

(c) $a_n = \frac{2^n + 2^{-n}}{2^{n+2}}$

We have

$$\lim_{n \rightarrow \infty} \frac{1 + 2^{-2n}}{2^2} = \frac{1}{4}$$

Example 5.13 For each of the following sequences, determine whether the limit exists and calculate it when it does.

(a) $\lim_{n \rightarrow \infty} 3n^2 + n + 2$

The general term is

$$3n^2 + n + 2 > n$$

and $n \rightarrow \infty$, hence the limit is infinity. So the sequence diverges (see Subsection Comparing Limits).

(b) $\lim_{n \rightarrow \infty} n3^{-n}$

This converges to zero (compare with geometric progression); note that

$$2^n > n$$

and so

$$0 < n3^{-n} < 2^n 3^{-n} = \left(\frac{2}{3}\right)^n$$

The latter sequence converges to zero; hence by the Sandwich Theorem the sequence converges to zero.

(c) $\lim_{n \rightarrow \infty} \frac{1 + 2n}{4 - 5n - 6n^2}$

We write the general term as

$$\frac{1 + 2n}{4 - 5n - 6n^2} = \frac{2/n + 1/n^2}{-6 - 5/n + 4/n^2}$$

Taking the limit, we have

$$\lim_{n \rightarrow \infty} \frac{2/n + 1/n^2}{-6 - 5/n + 4/n^2} = \frac{0}{6} = 0.$$

(d) $\lim_{n \rightarrow \infty} (-1)^n \frac{1}{e^{pn} + 1}$ for $p \in \mathbb{R}$.

This converges to 0 for $p > 0$, and oscillates for $p < 0$.

For $p > 0$ we note that e^{-pn} converges to 0 and so

$$(-1)^n \frac{1}{e^{pn} + 1} = (-1)^n \frac{e^{-pn}}{1 + e^{-pn}} \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

For $p < 0$ we note that e^{pn} converges to 0 and so

$$\frac{1}{e^{pn} + 1} = 1$$

Hence the sequence does not converge but oscillates.

Remark 5.14

Do not write $\lim_{n \rightarrow \infty} a_n = b_n$. This does not make sense.

Instead, you usually mean

$$\lim_{n \rightarrow \infty} (a_n - b_n) = 0.$$

5.3 Series

Calculations in applications often amount to computation of sums of a large number of terms; in particular integration can be seen as a limit of sums. These sums are called *series*.

Definition 5.15 Given a sequence $\{a_n\}_{n=1}^{\infty}$, the corresponding *series* is the sequence $\{s_n\}_{n=1}^{\infty}$ of partial sums for which the n -th term is

$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k.$$

We denote the *full series* by $\sum_{k=1}^{\infty} a_k$.

Note that just as a limit may not exist, it is possible that an infinite series may not be computable, particularly if the above limit does not exist. In this case we say the series *diverges*, otherwise it *converges*. Unlike sequences where we can often compute limits and hence show convergence, for series we can show convergence for a much larger set of series that one can compute the limits. For example, we can show that

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$$

Remark 5.16

Do not confuse sequences and series.

Example 5.17 1. The infinite sum

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots = \sum_{k=1}^{\infty} \frac{1}{k}$$

is a series for the sequence $\left\{\frac{1}{k}\right\}_{k=1}^{\infty}$.

2. A series can have alternating positive and negative terms

like $\sum_{k=1}^{\infty} \frac{(-1)^k}{k} = \sum_{k=1}^{\infty} \frac{\cos(k\pi)}{k}$.

Geometric series

The geometric series is an important series and is actually one of the few ‘easily’ computable series.

Definition 5.18 Let $a \neq 0$ and r be real numbers. A *geometric series* of common ratio r is that of the form

$$a + ar + ar^2 + \dots + ar^{k-1} + \dots = \sum_{k=1}^{\infty} ar^{k-1}.$$

This series is often written as $\sum_{k=0}^{\infty} ar^k$, starting at $k = 0$, to simplify the notation.

Theorem 5.19 If $|r| < 1$ then

$$\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}.$$

If $|r| \geq 1$, then $\sum_{k=1}^{\infty} ar^{k-1}$ is divergent.

Proof. When $r = 1$, the series is obviously divergent. When $r \neq 1$, let

$$s_n = \sum_{k=1}^n ar^{k-1} = a + ar + ar^2 + \dots + ar^{n-1}$$

Then

$$rs_n = r \left(\sum_{k=1}^n ar^{k-1} \right) = ar + ar^2 + ar^3 + \dots + ar^n.$$

Now, $rs_n - s_n = a - ar^n$, so $s_n(1 - r) = a(1 - r^n)$. Rearranging gives

$$s_n = \frac{a(1 - r^n)}{1 - r}.$$

Now, taking the limit as $n \rightarrow \infty$ gives

$$\sum_{k=1}^{\infty} ar^{k-1} = \frac{a}{1-r}.$$



Example 5.20 Evaluate the series $\sum_{k=1}^{10} \frac{2}{3^k}$ and $\sum_{k=1}^{\infty} \frac{2}{3^k}$.

Solution: This is a geometric series with $a = 2/3$, $r = 1/3$ and $n = 10$. Hence

$$\sum_{k=1}^{10} \frac{2}{3^k} = \frac{2}{3} \left(\frac{1/3^{10} - 1}{1/3 - 1} \right) = \cdots = \frac{3^{10} - 1}{3^{10}}$$

The infinite series is

$$\sum_{k=1}^{\infty} \frac{2}{3^k} = \frac{2}{3} \left(\frac{1}{(1 - 1/3)} \right) = 1.$$

Example 5.21 Does the series $\sum_{k=1}^{\infty} 2^{2k} 3^{1-k}$ converge?

Solution: We can write this series as the geometric series $\sum_{k=1}^{\infty} 3 \left(\frac{4}{3} \right)^k$. Since $r = 4/3 > 1$, we conclude that the series diverges.

Example 5.22 For what values of x does the series $\sum_{k=1}^{\infty} \frac{x^k}{2^k}$ converge?

Solution: Here x is variable, but this series is a geometric series with $r = \frac{x}{2}$. The series converges if and only if $\left| \frac{x}{2} \right| < 1$, i.e. if $|x| < 2$ and diverges otherwise. In the case $|x| < 2$, we have

$$\sum_{k=1}^{\infty} \frac{x^k}{2^k} = \frac{2x}{2 - x}.$$

Telescoping Sums

We will demonstrate what telescoping sums are using an example.

Example 5.23 Compute

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)}$$

Solution: Here, we will use partial fractions to transform the

sum into a difference. We write

$$\begin{aligned}
 \sum_{k=1}^N \frac{1}{k(k+1)} &= \sum_{k=1}^N \left(\frac{1}{k} - \frac{1}{k+1} \right) \\
 &= \sum_{k=1}^N \frac{1}{k} - \sum_{k=2}^{N+1} \frac{1}{k} \\
 &= 1 + \sum_{k=2}^N \left(\frac{1}{k} - \frac{1}{k} \right) - \frac{1}{N+1} \\
 &= 1 - \frac{1}{N+1}
 \end{aligned}$$

Taking the limit as $N \rightarrow \infty$, we have

$$\sum_{k=1}^{\infty} \frac{1}{k(k+1)} = \lim_{N \rightarrow \infty} 1 - \frac{1}{N+1} = 1$$

This is an example of a *telescoping sum*.

Definition 5.24 A finite sum $s_n = \sum_{k=1}^n a_k$ in which subsequent terms cancel each other, leaving only a finite number of terms is called a *telescoping sum*.

Example 5.25 Transform $\sum_{k=1}^{\infty} \sin(k)$ into a telescoping sum to show that the series diverges.

Solution: Using trigonometric identities, we have

$$\begin{aligned}
 \sum_{k=1}^N \sin(k) &= \sum_{k=1}^N \frac{1}{2} \csc(1/2) (2 \sin(1/2) \sin(k)) \\
 &= \frac{1}{2} \csc(1/2) \sum_{k=1}^N \left(\cos\left(\frac{2k-1}{2}\right) - \cos\left(\frac{2k+1}{2}\right) \right) \\
 &= \frac{1}{2} \csc(1/2) \left(\cos\left(\frac{1}{2}\right) - \cos\left(\frac{2N+1}{2}\right) \right)
 \end{aligned}$$

We can then see that taking the limit as $N \rightarrow \infty$, the sum has no limit and thus the series diverges.

The Harmonic Series**Theorem 5.26** The Harmonic Series

$$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \dots$$

diverges.

Proof. Let $a_n = \frac{1}{n}$. Consider the following inequalities

$$a_1 = 1 \geq 1,$$

$$a_1 + a_2 = 1 + \frac{1}{2} \geq 1 + \frac{1}{2},$$

$$a_1 + a_2 + a_3 + a_4 = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} \geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) = 1 + \frac{2}{2},$$

$$a_1 + \dots + a_8 = 1 + \frac{1}{2} + \dots + \frac{1}{7} + \frac{1}{8} \geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \dots + \left(\frac{1}{2^n} + \dots + \frac{1}{2^n}\right) = 1 + \frac{n}{2}.$$

This shows that we can make $\sum_{k=1}^n \frac{1}{k}$ as large as we like by choosing n large enough. Consequently the series cannot converge. ■

Exercise 5.27 Use a similar technique to show that

$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

converges.

However, often we are not concerned with the exact value of a series, but just whether the series converges or not.

5.4 Tests for Series Convergence and Divergence**Divergence test**

Suppose that we try to compute

$$\sum_{n=1}^{\infty} a_n$$

If the sequence $a_n \not\rightarrow 0$ then this series cannot converge.

Example 5.28 Show that

$$\sum_{n=1}^{\infty} \frac{n}{1+n}$$

does not converge.

Solution: Note that $\lim_{n \rightarrow \infty} \frac{n}{1+n} = 1 \neq 0$ and so the series diverges by the divergence test.

Note that the statement: if $\lim_{n \rightarrow \infty} a_n \neq 0$ then the series diverges, is a one way statement. Now, if $\lim_{n \rightarrow \infty} a_n = 0$ then the series *may converge*. To find out if it does converge, we must apply a *test*. We will look at some tests here.

Comparison test

Suppose that a_n and b_n are series such that $a_n, b_n \geq 0 \forall n \in \mathbb{N}$, and

$$a_n \leq b_n$$

.

- If $\sum_{n=1}^{\infty} b_n$ converges then $\sum_{n=1}^{\infty} a_n$ converges.
- If $\sum_{n=1}^{\infty} a_n$ diverges then $\sum_{n=1}^{\infty} b_n$ diverges.

We will prove the first item, and leave the second as an exercise: Let $C_N = \sum_{n=1}^N a_n$. Then

$$C_N \leq \sum_{n=1}^N b_n \leq \sum_{n=1}^{\infty} b_n$$

and so C_N is bounded. It is also increasing as $C_{N+1} = C_N + a_{N+1} \leq C_N$. Hence it converges to a limit.]

The comparison test is a powerful test, yet unfortunately it is not always easy to apply this test as we need to compare the series with a known series.

Example 5.29 For which values of p does the following series converge

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

When $p \leq 1$, we see that

$$n^p \leq n$$

so

$$\frac{1}{n} \leq \frac{1}{n^p}$$

and so the series diverges using the Comparison Test.

When $p \geq 2$, we have $p \geq 2$, we have

$$n^2 \leq n^p$$

so

$$\frac{1}{n^p} \leq \frac{1}{n^2}$$

and so the series converges using the Comparison Test.

In fact, it can be shown using the Integral Test, which is outside the scope of this chapter, that the series also converges for $1 < p < 2$. Thus $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$ and diverges otherwise.

As well as the Integral Test, there are other convergence tests that we will not see here such as the Limit Comparison Test and the Root Test.

Ratio test

If we apply the *Ratio Test* * to compare with a geometric series one can derive the following highly useful result: Suppose that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = k$$

Then

- If $k < 1$, then $\sum a_n$ converges.
- If $k > 1$, then $\sum a_n$ diverges.
- If $k = 1$, then the test is inconclusive.

Example 5.30 State whether the following series converge or diverge.

(a) $\sum_{n=1}^{\infty} \frac{n}{n^2 + 1}$

This one diverges. Note that the divergence test and ratio test are no good for this series; both inconclusive. We note

*The Ratio Test is mainly due to J.L.R. d'Alembert (1717-1783) who was a French mathematician. He was at the forefront of the mathematics of his time particularly in calculus and methods for Physics and was one of the first to use partial differential equations. At the end of his life, he devoted more time to philosophy and literature and became the Perpetual Secretary of the Académie Française.

that

$$n + \frac{1}{n} \leq 2n$$

for all n . Hence $\frac{n}{(n^2 + 1)} \geq \frac{1}{2n}$. Note that $\sum_{n=1}^{\infty} \frac{1}{2n}$ diverges (harmonic series) and so by the comparison test the series diverges.

(b) $\sum_{n=1}^{\infty} \frac{1}{3^n + 4^n}$.

This one converges. Again the divergence test is of no use.

We can try using the ratio test. We have

$$\frac{a_{n+1}}{a_n} = \frac{3^n + 4^n}{3^{n+1} + 4^{n+1}} = \frac{1 + (3/4)^n}{4(1 + (3/4)^{n+1})}.$$

Hence

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{4} < 1$$

and so the series converges by the ratio test.

(c) $\sum_{n=1}^{\infty} n^3$. This one diverges. Note that $\lim_{n \rightarrow \infty} n^3 = \infty \neq 0$ and so by the divergence test the series does not converge.

Leibniz' Test

Suppose we have an alternating series. This is a series of the form

$$\sum_{n=1}^{\infty} (-1)^n a_n.$$

Leibniz' Theorem states that if all the following hold

- (i) $a_n \geq 0$ for all n
- (ii) a_n is decreasing
- (iii) $a_n \rightarrow 0$ as $n \rightarrow \infty$

then

$$\sum_{n=1}^{\infty} (-1)^n a_n$$

converges.

Example 5.31 Show that $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ converges.

Solution: This is known as the *alternating harmonic series*. The terms $\frac{1}{n}$ are positive, decreasing and the limit as $n \rightarrow \infty$

is zero. Hence this series converges by Leibniz' Theorem. By contrast we know that $\sum \frac{1}{n}$ diverges.

However, this is the start of the problems with this series. By re-arranging the alternating harmonic series one can get the series to diverge;

$$-1 + 1/2 + 1/4 + 1/8 - 1/3 + 1/10 + 1/12 + 1/14 + 1/16 - 1/5 \dots$$

In order to avoid such bad behaviour we sometimes need a stronger notion of convergence.

Absolute convergence

Definition 5.32 Suppose that

$$\sum_{n=1}^{\infty} |a_n|$$

converges. Then

$$\sum_{n=1}^{\infty} a_n$$

must also converge.

We say the series *converges absolutely*.

The converse is not true; for example the alternating series

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

converges, but does not converge absolutely since $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

5.5 Power Series

Definition 5.33 A *power series* is a series of the form

$$\sum_{n=1}^{\infty} c_n(x-a)^n$$

where $\{c_n\}_{n=0}^{\infty}$ is a sequence of real numbers.

We often work with the special case where $a = 0$, and so use power series of the form

$$\sum_{n=1}^{\infty} c_n x^n.$$

A theorem tells us the values of x for which the power series converges. First, we will give a definition.*

*The Greek letter ρ is pronounced 'row' as in *row your boat*.

Definition 5.34 Let $\sum_{n=0}^{\infty} c_n(x-a)^n$ be a power series and let

$$\rho = \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right|.$$

The *radius of convergence*, R , of the series is equal to

- (i) $R = \frac{1}{\rho}$ when $0 < \rho < \infty$
- (ii) $R = 0$ when $\rho = \infty$
- (iii) $R = \infty$ when $\rho = 0$.

Theorem 5.35 Given a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$,

- (i) the series converges for all x that satisfy $|x-a| < R$ when $R = \frac{1}{\rho}$.
- (ii) the series converges just for $x = a$ when $R = 0$
- (iii) the series converges for all x when $R = \infty$

For case (i) in the above theorem, we need to check what happens for $x = a + R$ and $x = a - R$ individually.

Example 5.36 Find the radius of convergence for the following power series, and deduce the values of x for which the series converge.

(a) $\sum_{n=1}^{\infty} \frac{(x-1)^n}{n2^n}$

Solution: We need to work with $c_n = \frac{1}{n2^n}$. We evaluate

$$\begin{aligned}\rho &= \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{n2^n}{(n+1)2^{n+1}} \\ &= \lim_{n \rightarrow \infty} \frac{n}{2(n+1)} = \frac{1}{2}.\end{aligned}$$

Using Theorem 5.35, the power series converges when $|x-1| < 2$ and diverges when $|x-1| > 2$. To see when happens at the radius of convergence, we replace $x = 3$ and $x = -1$ in the series. In the first case, we get $\sum_{n=1}^{\infty} \frac{1}{n}$ which diverges (harmonic series), and in the second case, we get the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ which converges (alternating harmonic series). So we can conclude that the series converges for $x \in [-1, 3)$.

(b) $\sum_{n=0}^{\infty} \frac{x^n}{n!}$

Solution: We proceed as in the previous part. We have

$$\begin{aligned}\rho &= \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{n!}{(n+1)!} \\ &= \lim_{n \rightarrow \infty} \frac{1}{n+1} = 0.\end{aligned}$$

So the radius of convergence is $R = \infty$ and this the power series converges for all $x \in \mathbb{R}$. (This series is in fact a representation of the exponential function.)

(c) $\sum_{n=1}^{\infty} n!x^n$

Solution: Similarly, we have,

$$\begin{aligned}\rho &= \lim_{n \rightarrow \infty} \left| \frac{c_{n+1}}{c_n} \right| \\ &= \lim_{n \rightarrow \infty} \frac{(n+1)!}{n!} \\ &= \lim_{n \rightarrow \infty} (n+1) = \infty.\end{aligned}$$

Thus the radius of convergence is zero and so the series converges only for $x = 0$. Here, the coefficient c_n tends to infinity so fast that no small (but non zero) amount of x can keep the series converging.

Almost all functions we consider in this course can be written as power series near any given point in their domain. For example

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{3!} + \dots = \sum_{k=0}^{\infty} \frac{x^k}{k!}.$$

One can show that this series converges for all x as follows. We fix the value of x , consider as a series $\sum a_n$ and apply the ratio test

$$\begin{aligned}\left| \frac{a_{k+1}}{a_k} \right| &= \left| \frac{x^{k+1}k!}{x^k(k+1)!} \right| \\ &= \frac{|x|}{k+1} \rightarrow 0 \text{ as } n \rightarrow \infty.\end{aligned}$$

Hence the series converges for all values of x .

One can show moreover that if $\sum a_n x^n$ converges for some x_0 then the sum converges absolutely for all x with $|x| < |x_0|$.

5.6 Taylor and Maclaurin series

Definition 5.37 Suppose that $f(x)$ has derivatives of all orders up to N on an interval containing a as an interior point.

The **Taylor polynomial** of $f(x)$ at order $n \leq N$ about $x = a$ is a polynomial given by

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x - a)^n.$$

In the above definition, and in many sources, the ' (n) ' in the power denotes the n -th derivative, i.e. f differentiated n times.

The *infinite Taylor* * *expansion* is thus given by the following power series in $(x - a)$;

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x - a)^k$$

assuming all derivatives exist.

The special case when $a = 0$ is important, widely used, and has a name - The Maclaurin [†] Series. So, it is given by

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k$$

Example 5.38 Find the Maclaurin series of $f(x) = \sin(x)$ up to and including order 6.

Solution: By computing the derivatives, we find

$$\begin{aligned} f(x) &= \sin(x), & f'(x) &= \cos(x), & f''(x) &= -\sin(x), \\ f^{(3)}(x) &= -\cos(x), & f^{(4)}(x) &= \sin(x). \end{aligned}$$

So, after four derivatives, we are back to where we started, and the sequence of derivatives of $\sin(x)$ cycles through the pattern given above. At $x = 0$, we then get the following values for the derivatives

$$f^{(4j)}(0) = f^{(4j+2)}(0) = 0, \quad f^{(4j+1)}(0) = f^{(4j+3)}(0) = 1, \quad j \in \mathbb{N}.$$

*Brook Taylor (1685-1731) was a famous British mathematician. As well as being well known for his contributions to calculus and mechanics, he was also a talented artist and musician. He obtained a doctoral degree in Law from the University of Cambridge but later on specialised in mathematics for the rest of his life.

[†]Colin Maclaurin (1698-1746) was a Scottish mathematician. He got his degree at 14 years of age. He was known for spreading Newton's theory of fluxions (Newton's version of calculus), developed the integral test for series and worked on geometry. He is also the founder of actuarial studies.

Now, we substitute these into the formula for the Maclaurin series to obtain

$$f(x) \approx x - \frac{x^3}{3!} + \frac{x^5}{5!}.$$

The next non-zero term is of order 7, so in this example, the polynomial of order 6 is the polynomial of order 5.

General results of Taylor polynomials

1. The Taylor polynomial of the sum h of two functions f and g is given by the sum of their Taylor polynomials.
2. The Taylor polynomial of the product of two functions f and g is given by the terms of degree up to n of the product of their Taylor polynomials.
3. The Taylor polynomial of the composition h of two functions f and g , $h(x) = f(g(x))$ is given by the terms of degree up to n of the composition of their Taylor polynomials.

Example 5.39 Consider $f(x) = e^x$ and $g(x) = \sin(x)$. Find the Maclaurin polynomial of order 3 of

(a) $h(x) = f(x) + g(x)$

(b) $k(x) = f(x)g(x)$

Solution: Consider $f(x) = e^x$ and $g(x) = \sin(x)$. We calculate the Maclaurin polynomials of order 3 as

(a)

$$\begin{aligned} h(x) &\approx \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3!}\right) + \left(x - \frac{x^3}{3!}\right) \\ &= 1 + 2x + \frac{x^2}{2}. \end{aligned}$$

(b)

$$\begin{aligned} k(x) &\approx \left(1 + x + \frac{x^2}{2} + \frac{x^3}{3!}\right) \left(x - \frac{x^3}{3!}\right) \\ &= x + x^2 + \left(-\frac{1}{3!} + \frac{1}{2}\right)x^3 \\ &= x + x^2 + \frac{1}{3}x^3. \end{aligned}$$

Example 5.40 Let $f(x) = \frac{e^{2x}}{1+3x}$. Find the Maclaurin polynomial of order 3 of $f(x)$.

Solution: We have

$$e^{2x} \approx 1 + 2x + \frac{2^2 x^2}{2!} + \frac{2^3 x^3}{3!} = 1 + 2x + 2x^2 + \frac{4x^3}{3}$$

and

$$\frac{1}{1+3x} \approx 1 - 3x + 9x^2 - 27x^3.$$

So, on multiplying these two, we have

$$f(x) \approx 1 - x + 5x^2 - \frac{41}{3}x^3.$$

Example 5.41 Find the Maclaurin polynomial of $f(x) = e^{\sin(x)}$ up to order 4. This is a composition of two functions. We have

$$\sin(x) \approx x - \frac{x^3}{6}$$

and

$$e^x \approx 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24}.$$

So, we have

$$\begin{aligned} f(x) &\approx 1 + \left(x - \frac{x^3}{6}\right) + \frac{\left(x - \frac{x^3}{6}\right)^2}{2} + \frac{\left(x - \frac{x^3}{6}\right)^3}{6} + \frac{\left(x - \frac{x^3}{6}\right)^4}{24} \\ &= 1 + x + \frac{x^2}{2} - \frac{x^4}{8}. \end{aligned}$$

Since Taylor's polynomial is an approximation, there must be an *error*, or *remainder term*.

Definition 5.42 Let f be an n -times differentiable function, then

$$R_n(x) = f(x) - P_n(x)$$

is called the *n -th order remainder term* of $P_n(x)$.

The main result of this section is a theorem that allows us to estimate this error, this is known as *Taylor's Theorem*.

Theorem 5.43 — Taylor's Theorem. Let f be an $n + 1$ times differentiable function on some open interval I containing a . Then, for every x in the interval I there is a ξ between a and x such that

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}. \quad (5.1)$$

This theorem is called Taylor's Theorem as it provides an estimate of the error from using Taylor's polynomial, however formula 5.1 is called the Lagrange Remainder Formula*. Although, in many cases, it is not possible to compute the mystery point ξ precisely, Lagrange's formula allows you to *estimate* it. The best way we can estimate the remainder is by the following theorem.

*Joseph-Louis Lagrange (1736-1813) was an Italian-French mathematician who was known to have made great contributions in physics, number theory and analytic mechanics. His book on analytic mechanics, which he published during his stay at the Louvre in Paris, set the foundations for all other work in that field which followed.

Theorem 5.44 — Remainder Estimation. Let f be an $n + 1$ times differentiable function on some open interval I containing a . If there is a constant M such that

$$|f^{(n+1)}(\xi)| \leq M,$$

for all ξ between a and x , then

$$|R_n f(x)| \leq \frac{M |x-a|^{n+1}}{(n+1)!}.$$

So for $|x-a|$ small enough

$$R_n(x) \rightarrow 0$$

as $n \rightarrow \infty$. Hence, in this case

$$f(x) = \lim_{n \rightarrow \infty} P_n(x)$$

and the function is equal to its Taylor series.

Corollary 5.45 — Taylor series. If, for a given a, x and f we have $R_n \rightarrow 0$ then one can write the function as the Taylor series: For any x in I which also contains a ,

for each n we have

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

Example 5.46 Find the Taylor series of e^x about $x = 0$ to any order in n , with remainder term. Hence show that the function is equal to its Taylor series for all values of x .

Solution: Note that $f^{(n)}(x) = e^x$ for all n . Hence

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \frac{e^h x^{n+1}}{(n+1)!}.$$

for some h between 0 and x . Note that $e^h < e^x$ which is bounded, and hence

$$\lim_{n \rightarrow \infty} \frac{e^h x^{n+1}}{(n+1)!} = 0$$

meaning that e^x is equal to its Taylor series.

We can use the remainder term to estimate the numerical error. For example, to find e to get an error $< 10^{-6}$. We have

$$R_n(1) = \frac{e^h}{(n+1)!}$$

One can show that $e < 3$ from first principles. Hence

$$\frac{1}{(n+1)!} < R_n(1) < \frac{3}{(n+1)!}$$

Hence if we choose $n+1 = 10$, we get

$$10^{-6} < \frac{3}{10!}.$$

Hence

$$1 + 1 + \frac{1}{2} + \cdots + \frac{1}{9!} = 2.718282.$$

which is accurate to within 10^{-6} .

Binomial expansion

This is the Maclaurin series expansion of

$$f(x) = (1+x)^m.$$

Note that

$$f^{(k)}(x) = m(m-1)(m-2)\cdots(m-k+1)(1+x)^{m-k}.$$

Hence

$$(1+x)^m = 1 + mx + \frac{m(m-1)}{2}x^2 + \frac{m(m-1)(m-2)}{3!}x^3 + \dots \\ + \frac{m(m-1)(m-2)\dots(m-k+1)}{k!}x^k + \dots$$

for any $k \in \mathbb{N}$. This series converges (ratio test) for $|x| < 1$.

We define the binomial coefficients by

$$\binom{m}{0} = 0, \quad \binom{m}{1} = m, \quad \binom{m}{2} = \frac{m(m-1)}{2!}, \quad \binom{m}{k} = \frac{m(m-1)\dots(m-k+1)}{k!}$$

and then

$$(1+x)^m = \sum_{k=0}^{\infty} \binom{m}{k} x^k.$$

Note that

- If $m \in \mathbb{N}$ then

$$\binom{m}{k} = 0$$

if $k > m$.

- If $m = -1$ then

$$\binom{-1}{k} = (-1)^k$$

- If $m = \frac{1}{2}$ then

$$(1+x)^{1/2} = 1 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{16} - \frac{5x^4}{128} + \dots$$

- If $m = -\frac{1}{2}$ then

$$(1+x)^{-1/2} = 1 - \frac{x}{2} + \frac{3x^2}{8} - \frac{5x^3}{16} + \frac{35x^4}{128} + \dots$$

- Note that $\frac{d}{dx} \arcsin(x) = (1-x^2)^{-1/2}$ and so

$$\frac{d}{dx} \arcsin(x) = 1 + \frac{x^2}{2} + \frac{1 \cdot 3}{2 \cdot 4}x^4 + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6}x^6 \dots$$

Hence

$$\arcsin(x) = x + \frac{1}{2} \frac{x^3}{3} + \frac{1 \cdot 3}{2 \cdot 4} \frac{x^5}{5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6} \frac{x^7}{7} + \dots$$

We can use this to estimate π pretty fast;

$$\frac{\pi}{6} = \arcsin(1/2) = \frac{1}{2} + \frac{1}{2 \cdot 3 \cdot 2^3} + \frac{1 \cdot 3}{2 \cdot 4 \cdot 5 \cdot 2^5} + \frac{1 \cdot 3 \cdot 5}{2 \cdot 4 \cdot 6 \cdot 7 \cdot 2^7} + \dots$$

6 Complex Numbers

6.1 The Argand plane

The system of complex numbers has multiple aspects. It comprises:

1. A set,

$$\mathbb{C} = \{x + iy : x, y \in \mathbb{R}\},$$

the set of complex numbers.

2. A geometric representation, this being a function $\mathbb{C} \rightarrow \mathbb{R}^2$ mapping $x + iy \mapsto (x, y)$. This representation is called the *Argand* diagram* or the *Argand plane*.

* Jean-Robert Argand, 1768 - 1822, a Swiss - French amateur mathematician

3. Rules of arithmetic by which we may add and multiply complex numbers.

From a purely set-theoretic perspective, we may treat i as a meaningless symbol. Let $x, y, a, b \in \mathbb{R}$. The definition of two complex numbers $x + iy$ and $a + ib$ being equal, written $x + iy = a + ib$, is that $x = a$ and $y = b$. The mapping $x + iy \mapsto (x, y)$ thus establishes a bijection between $\mathbb{C}$ and $\mathbb{R}^2$. The quantity x is called the *real part* and y is called the *imaginary part* of the complex number $x + iy$.

For typographical reasons, we may write $x + yi$ in place of $x + iy$, especially when y is given as a constant. As a case in point, $3 + 4i$ looks better than $3 + i4$. We write $x + i$ for $x + 1i$. Where a complex number has imaginary part 0, we write x for $x + 0i$. Likewise, if a number has real part 0, we write y for $0 + iy$. Finally, we write $x - iy$ for $x + (-y)i$. Thus 14 , $8i$, $2 - 9i$ are representations of the complex numbers $14 + 0i$, $0 + 8i$, and $2 + (-9)i$, respectively.

We define two functions $\Re : \mathbb{C} \rightarrow \mathbb{R}$ and $\Im : \mathbb{C} \rightarrow \mathbb{R}$ by $\Re(x + iy) = x$ and $\Im(x + iy) = y$, where $x, y \in \mathbb{R}$.

For the complex number $3 + 4i$, the real part is 3 and the imaginary part is 4. That is, $\Re(3 + 4i) = 3$ and $\Im(3 + 4i) = 4$. We

plot this point and $1 + 2i$ on the Argand diagram in Figure 6.1. The horizontal axis is called the real axis and vertical axis is called the imaginary axis.

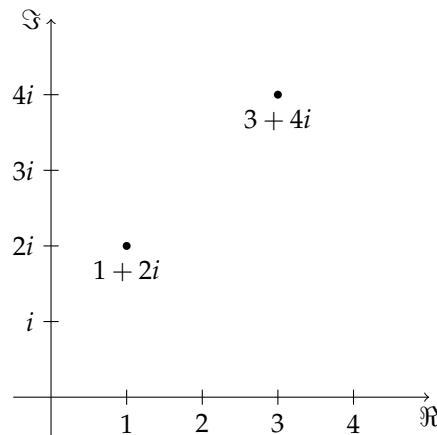


Figure 6.1: Two points in the Argand plane

Remark 6.1

The set $\mathbb{R}$ of real numbers is identified with a subset of $\mathbb{C}$ as we write x for $x + 0i$. Thus we can have $5 \in \mathbb{C}$. More generally, $x \in \mathbb{C}$ for every real number $\mathbb{R}$, whence $\mathbb{R} \subseteq \mathbb{C}$.

6.2 Complex arithmetic

6.2.1 Addition

We define an operation $+$ on $\mathbb{C}$. Given $z, w \in \mathbb{C}$, we define

$$z + w = (\Re(z) + \Re(w)) + i(\Im(z) + \Im(w)).$$

Formally, $+$ can be treated as a function $+: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ with the rule

$$(z, w) \mapsto (\Re(z) + \Re(w)) + i(\Im(z) + \Im(w)).$$

Geometrically, addition of complex numbers corresponds to addition of vectors in the Argand plane, as shown in Figure 6.2. We have

$$(4 + i) + (1 + 2i) = 5 + 3i.$$

One may prove that addition of complex numbers satisfies many of the expected properties of addition of real numbers:

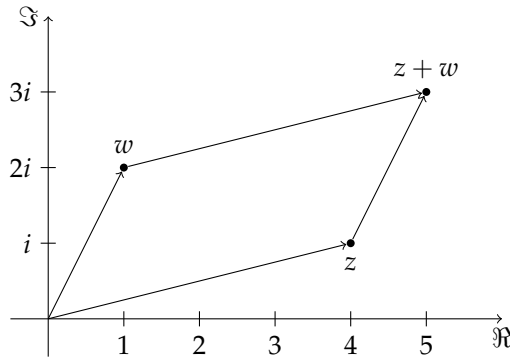


Figure 6.2: Addition of complex numbers in the Argand plane

- [commutativity] $z + w = w + z$, for $z, w \in \mathbb{C}$.
- [associativity] $(u + v) + w = u + (v + w)$, for $u, v, w \in \mathbb{C}$.
- [identity] $0 + z = z = z + 0$, for every $z \in \mathbb{C}$.
- [inverse] for every $z \in \mathbb{C}$, there exists a (unique) $w \in \mathbb{C}$ for which $z + w = 0 = w + z$. We write $-z$ for w .

Example 6.2 We'll prove associativity. Assume $u, v, w \in \mathbb{C}$. Then there exist $a, b, c, d, x, y \in \mathbb{R}$ for which $u = a + ib$, $v = c + di$, and $w = x + iy$. By definition,

$$\begin{aligned}
 (u + v) + w &= ((a + c) + (b + d)i) + w \\
 &= ((a + c) + x) + ((b + d) + y)i \\
 &= (a + (c + x)) + (b + (d + y))i \\
 &= u + ((c + x) + (d + y)i) \\
 &= u + (v + w),
 \end{aligned}$$

where we use associativity of addition of real numbers to justify the transition from the second to the third line.

Example 6.3 We prove existence and uniqueness of the additive inverse. Let $z \in \mathbb{C}$. Then $z = x + iy$, for $x, y \in \mathbb{R}$. Write $u = -x + i(-y)$. Then

$$z + u = (x + (-x)) + i(y + (-y)) = 0 + 0i = 0.$$

By commutativity, $u + z = 0$. This shows that u is an additive inverse of z .

For uniqueness: suppose $v \in \mathbb{C}$ is an additive inverse of z .

Write $v = a + ib$ with $a, b \in \mathbb{R}$. By definition, $z + v = 0$, but

$$z + v = (x + a) + i(y + b).$$

By definition of equality of complex numbers, $x + a = 0$ and $y + b = 0$. It follows that $a = -x$ and $b = -y$. Thus $v = u$, using our previous notation. This shows uniqueness of the additive inverse.

Exercise 6.4 Prove that addition of complex numbers is commutative.

6.2.2 Multiplication

The multiplication function $\cdot : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ is defined by the rule

$$(x + iy) \cdot (a + ib) = (xa - yb) + i(xb + ya),$$

for $x + iy, a + ib \in \mathbb{C}$ with $x, y, a, b \in \mathbb{R}$.

If we take $w = i$ and consider $w \cdot w$, we have $w = 0 + 1i$, so

$$w \cdot w = (0 + 1i) \cdot (0 + 1i) = (0 - 1) + (0 + 0)i = -1.$$

Put another way, $i^2 = -1$. This gives some meaning to the previously meaningless symbol i .

As with ordinary multiplication, we write z^0 for 1 and z^n for $z \cdot z^{n-1}$, if $n \in \mathbb{N}$.

The geometric interpretation of complex multiplication is not clear as for addition. We will come to it in the next section.

For simplicity, we write zw for $z \cdot w$.

Example 6.5 If $z = 3 + 5i$ and $w = 2 + 4i$, then

$$zw = (3 \times 2 - 5 \times 4) + (3 \times 4 + 5 \times 2)i = -14 + 22i.$$

If $u = 2 - 4i$, then

$$zu = (3 \times 2 - 5 \times (-4)) + (3 \times (-4) + 5 \times 2)i = 26 - 2i.$$

One may, from the definition prove:

- [commutativity] $zw = wz$, for all $z, w \in \mathbb{C}$.
- [associativity] $(uv)w = u(vw)$, for all $u, v, w \in \mathbb{C}$.

- [identity] $1 \cdot z = z = z \cdot 1$, for every $z \in \mathbb{C}$.
- [inverse] if $z \neq 0$, then there exists a (unique) $w \in \mathbb{C}$ for which $zw = 1 = wz$. We write z^{-1} or $\frac{1}{z}$ for w .
- [distributivity] $u(v + w) = uv + uw$, for all $u, v, w \in \mathbb{C}$.

Notation 6.1. Let $z \in \mathbb{C}$ and let $n \in \mathbb{N}$. We define z^{-n} to be $(z^{-1})^n$, where z^{-1} is the multiplicative inverse of z , as above.

The definition of multiplication can be recovered by taking $i^2 = -1$ as a rule and using the above rules of arithmetic in the expression $(x + iy) \cdot (a + ib)$. We have

$$\begin{aligned}(x + iy) \cdot (a + ib) &= x(a + ib) + iy(a + ib) \\ &= xa + ixb + iya + i^2yb \\ &= (xa - yb) + i(xb + ya),\end{aligned}$$

as before.

Exercise 6.6 Write $(3 - 2i)(5 + 4i)$ in the form $x + iy$ for $x, y \in \mathbb{R}$.

Exercise 6.7 Prove that multiplication of complex numbers is (1) commutative, (2) associative.

Example 6.8 Having introduced addition and multiplication of complex numbers, it is meaningful to write $z = x + iy$ where x, y need not be real numbers! Take $x = 2 + i$ and $y = 3 + 5i$. Let $z = x + iy$. Then

$$\begin{aligned}z &= x + iy = (2 + i) + i(3 + 5i) \\ &= (2 + i) + (-5 + 3i) \\ &= -3 + 4i.\end{aligned}$$

So $\Re(z) = -3$ and $\Im(z) = 4$. Observe it's *not true* that $\Re(z) = x$ and $\Im(z) = y$.

6.2.3 Complex conjugate

For a complex number z , its *complex conjugate*, written $\bar{z}$ is the complex number $\Re(z) - \Im(z)i$. Thus, the complex conjugate of $2 + 3i$ is

$$\overline{2 + 3i} = 2 - 3i.$$

Likewise, the complex conjugate of $5 - 6i$ is

$$\overline{5 - 6i} = 5 + 6i.$$

Geometrically, taking the conjugate of a complex number represents reflection in the real axis. This is shown in Figure 6.3.

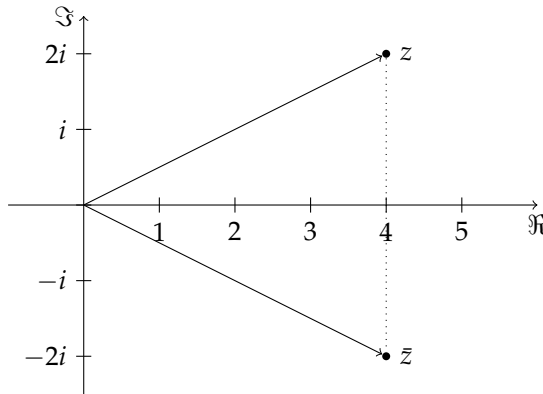


Figure 6.3: Complex conjugation

Note that the conjugate of a conjugate is the original number. That is, $\bar{\bar{z}} = z$, for every $z \in \mathbb{C}$.

6.2.4 Modulus

Let $z = x + iy \in \mathbb{C}$ with $x, y \in \mathbb{R}$. The *absolute value*, also called the *modulus*, of z , written $|z|$, is the real number $\sqrt{x^2 + y^2}$. Geometrically, it represents the length of the line segment between 0 and z in the Argand plane, as illustrated in Figure 6.4.

We have $|3 + 4i| = \sqrt{3^2 + 4^2} = 5$. Also

$$|3 - 4i| = \sqrt{3^2 + (-4)^2} = 5.$$

More generally, for every complex number $|z| = |\bar{z}|$.

We also see that $|z| \geq 0$ with equality if and only if $z = 0$.

Lemma 6.9 Let $z \in \mathbb{C}$. Then $z\bar{z} = |z|^2$.

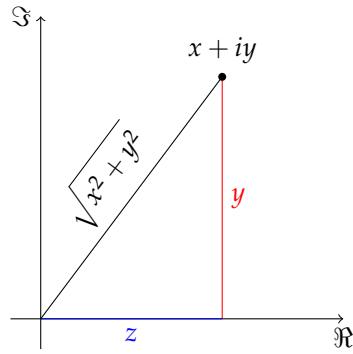


Figure 6.4: The modulus of a complex number

Proof. Write $z = x + iy$ with $x, y \in \mathbb{R}$. Then

$$z\bar{z} = (x + iy)(x - iy) = x^2 + y^2 = |z|^2.$$

■

Indeed, as a first application of this result, we find a way to compute the multiplicative inverse of a complex number. Take $z = 3 + 4i$. Then $\bar{z} = 3 - 4i$ and $|z|^2 = 3^2 + 4^2 = 25$. Applying the formula above, we have

$$(3 + 4i)(3 - 4i) = 25.$$

Rearranging this,

$$\frac{1}{3 + 4i} = \frac{1}{25}(3 - 4i).$$

We've discovered that the multiplicative inverse of $z = 3 + 4i$ is $z^{-1} = 25^{-1}(3 - 4i)$. It is instructive to compare this with the process one goes through when simplifying a surd. More generally, we have the following result.

Corollary 6.10 For every non-zero complex number z ,

$$z^{-1} = \frac{1}{|z|^2}\bar{z}.$$

Proof. Let z be a non-zero complex number. Then $|z| \neq 0$. By the previous Lemma, $z\bar{z} = |z|^2$. Multiplying by $|z|^{-2}$ on both sides,

$$z(|z|^{-2}\bar{z}) = 1.$$

Thus $|z|^{-2}\bar{z}$ is the inverse of z .

■

Exercise 6.11 Compute the modulus and the inverse of $5 + 12i$.

Exercise 6.12 Prove that for all $z, w \in \mathbb{C}$,

1. $\bar{\bar{z}} = z$,
2. $\overline{z + w} = \bar{z} + \bar{w}$,
3. $\overline{zw} = \bar{z}\bar{w}$,
4. $|z| = |\bar{z}|$,
5. $|zw| = |z| |w|$,
6. $|0| = 0$ and if $z \neq 0$, then $|z|$ is a positive real number.

Remark 6.13

Complex numbers cannot be compared, at least not in the ordinary sense. The real number system has the property that if $x, y \in \mathbb{R}$ then either $x < y$, $y < x$, or $x = y$. The relation ‘less than’ preserves (in an appropriate sense) addition and multiplication. If $0 < x$ and $0 < y$ then $0 < x + y$ and $0 < xy$. There isn’t a similar ordering for complex numbers. Do not make the mistake of writing $z < w$ for complex numbers z and w .

The best we can do is compare their moduli.

6.3 The polar form

6.3.1 The argument of a complex number

As is clear from the Argand diagram, a non-zero complex number z can be specified by its modulus and its direction. We define the argument of z to be the angle θ that the line from 0 to z makes with the positive real axis, when measured anticlockwise, and the sign of the angle changes when measured clockwise. See Figure 6.8. The problem with this definition is that this angle is

not uniquely defined. For example, Figure 6.5 shows an argument for the complex number $1 + i$. Here, $\theta = \pi/4$. However, we could add a multiple of 2π to get other values for the argument: $9\pi/4$, $-7\pi/4$, and $17\pi/4$ are all valid arguments with this definition.

To overcome this difficulty, we impose a choice of interval for the argument θ of the complex number z . We require $-\pi < \theta \leq \pi$. This gives what is sometimes known as the *principal value* of the argument.

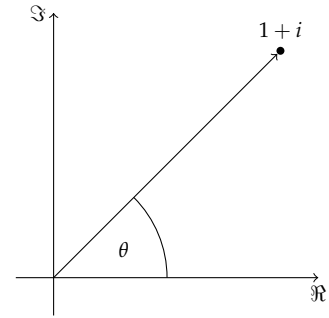


Figure 6.5: An argument for the point $1 + i$

Definition 6.14 Let z be a non-zero complex number. The **argument** of z , written $\arg(z)$, is the unique angle θ in the range $-\pi < \theta \leq \pi$ for which the line segment between 0 and z makes an angle θ with the positive real axis, when measured anticlockwise.

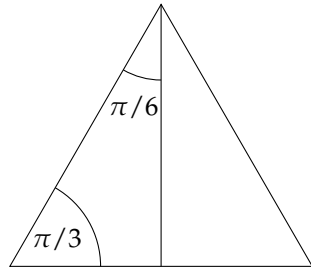
Example 6.15 We'll consider the arguments of complex numbers in each of the four quadrants: $\arg(1 + i) = \frac{\pi}{4}$, $\arg(-1 + i) = \frac{3\pi}{4}$, $\arg(1 - i) = -\frac{\pi}{4}$, and $\arg(-1 - i) = -\frac{3\pi}{4}$. These can be seen by drawing diagrams.

A useful triangle

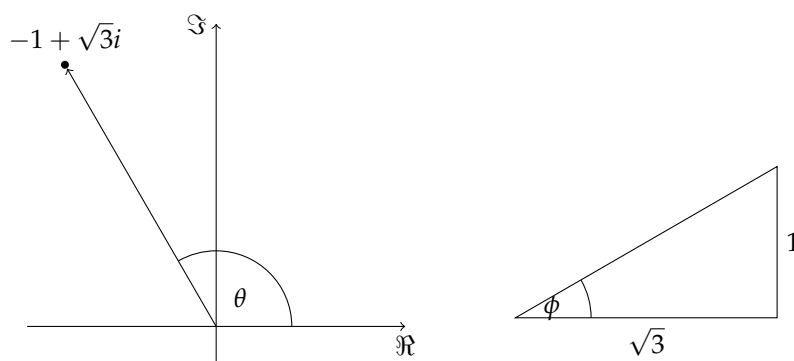
The angles $\pi/3$ and $\pi/6$ have nice sines and cosines. They can be computed by reference to the equilateral triangle with side length 1, as shown in Figure 6.6. The base of the right triangle is $1/2$, the hypotenuse is 1 and the height (by Pythagoras' Theorem) is $\sqrt{3}/2$. This gives

$$\begin{aligned}\cos \pi/3 &= 1/2, & \sin \pi/3 &= \sqrt{3}/2 \\ \cos \pi/6 &= \sqrt{3}/2, & \sin \pi/6 &= 1/2.\end{aligned}$$

Example 6.16 What is the argument of $z = -1 + \sqrt{3}i$? Figure 6.7 shows this situation. In this situation, we can decompose the argument $\theta = \arg(z)$ into two angles: $\theta = \pi/2 + \phi$, where ϕ is the angle that the line segment between 0 and z

Figure 6.6: Computing sines and cosines of $\pi/3$ and $\pi/6$

makes with the positive imaginary axis. But then $\tan \phi = 1/\sqrt{3}$ and we infer $\phi = \arctan(1/\sqrt{3}) = \pi/6$. Hence $\arg(z) = \pi/2 + \pi/6 = 2\pi/3$.

Figure 6.7: Finding $\arg(-1 + \sqrt{3}i)$

Remark 6.17

You might be tempted to assume that if $z = x + iy$, then $\arg(z) = \arctan(y/x)$. This isn't always true. In the previous example,

$$\arctan(\sqrt{3}/(-1)) = \arctan(-\sqrt{3}) = -\pi/3,$$

which is not the same as $\arg(z)$. The problem is that the $\arctan$ function is also not only defined up to multiples of 2π but, by convention always lies in the range

$$-\frac{\pi}{2} < \arctan(t) < \frac{\pi}{2}.$$

This is the range of principal values of $\arctan$. We see that using $\arctan$ to compute the argument of a complex number is only valid when the argument lies in the interval $(-\pi/2, \pi/2)$. Otherwise, draw a diagram and make the necessary adjustments.

6.3.2 The polar form

Let $z = x + iy$ be a non-zero complex number with $x, y \in \mathbb{R}$.

Let $\theta = \arg(z)$. Then

$$x = |z| \cos \theta \quad \text{and} \quad y = |z| \sin \theta.$$

This is shown in Figure 6.8.

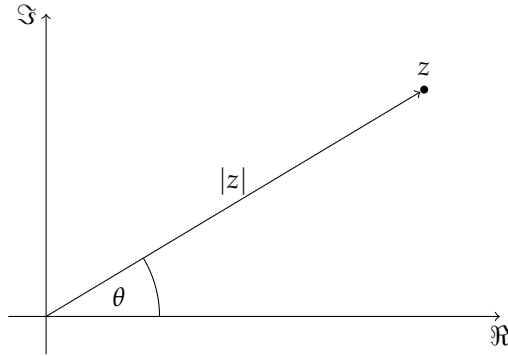


Figure 6.8: Argument and modulus of a complex number

Let $r = |z|$. We may write z in the form

$$z = r(\cos \theta + i \sin \theta).$$

This is called the *polar form* of a complex number.

Example 6.18 Let $r \geq 0$, $s \geq 0$, $\theta, \phi \in \mathbb{R}$. Write $z = r(\cos \theta + i \sin \theta)$ and $w = s(\cos \phi + i \sin \phi)$. Then

$$\begin{aligned} zw &= r(\cos \theta + i \sin \theta)s(\cos \phi + i \sin \phi) \\ &= rs((\cos \theta \cos \phi - \sin \theta \sin \phi) + i(\cos \theta \sin \phi + \sin \theta \cos \phi)) \\ &= rs(\cos(\theta + \phi) + i \sin(\theta + \phi)), \end{aligned}$$

where we use trigonometric formula for the sum of angles.

In particular, if z, w are non-zero complex numbers, then $|zw| = |z||w|$ and (!) $\arg(zw) = \arg(z) + \arg(w)$. Well not quite. It's possible that $\arg(z) + \arg(w)$ lies outside the interval $(-\pi, \pi]$. We may need to add or subtract a multiple of 2π .

From this example, we extract a result.

Lemma 6.19 Let z and w be non-zero complex numbers.

Then $|zw| = |z||w|$ and $\arg(zw) = \arg(z) + \arg(w) + 2\pi n$, for some $n \in \{-1, 0, 1\}$.

Example 6.20 Let $z = 1 + i$ and let $w = 1 + \sqrt{3}i$. Write zw in polar form.

We have $|z| = \sqrt{2}$ and $|w| = 2$. By a diagram or otherwise, $\arg(z) = \pi/4$ and $\arg(w) = \pi/3$. Certainly, $|zw| = 2\sqrt{2}$. Now $\arg(z) + \arg(w) = 7\pi/12$. This is in the range of principal values for the argument. We deduce

$$zw = 2\sqrt{2} (\cos(7\pi/12) + i \sin(7\pi/12)).$$

Example 6.21 Let $z = -1 + i$ and let $w = 1 + \sqrt{3}i$. Write zw in polar form.

As before, $|z| = \sqrt{2}$ and $|w| = 2$. Here, $\arg(z) = 3\pi/4$ and $\arg(w) = \pi/3$. Now $|zw| = 2\sqrt{2}$. We have

$$\arg(zw) = \arg(z) + \arg(w) + 2n\pi = 13\pi/12 + 2n\pi.$$

As $13\pi/12 > \pi$, we must subtract 2π to give

$$\arg(zw) = 13\pi/12 - 2\pi = -9\pi/12 = -3\pi/4.$$

The polar form is $zw = 2\sqrt{2}(\cos(-3\pi/4) + i \sin(-3\pi/4))$.

Example 6.22 Give the polar form of z^2 , where $z = 1 + i$.

Note that $z^2 = z \cdot z$. We have $|z| = \sqrt{2}$ and $\arg z = \pi/4$. So $|z^2| = |z|^2 = 2$ and $\arg(z^2) = 2 \arg(z) = \pi/2$. Thus

$$z^2 = 2(\cos(\pi/2) + i \sin(\pi/2)).$$

6.3.3 Geometric interpretation of multiplication

Lemma 6.19 gives a geometric interpretation of multiplication. Multiplying two complex numbers corresponds to multiplying their moduli and adding their arguments. The situation is shown in Figure 6.9 where we plot

$$z_1 = r_1 (\cos \theta_1 + i \sin \theta_1), \quad z_2 = r_2 (\cos \theta_2 + i \sin \theta_2),$$

and

$$z_1 z_2 = r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)).$$

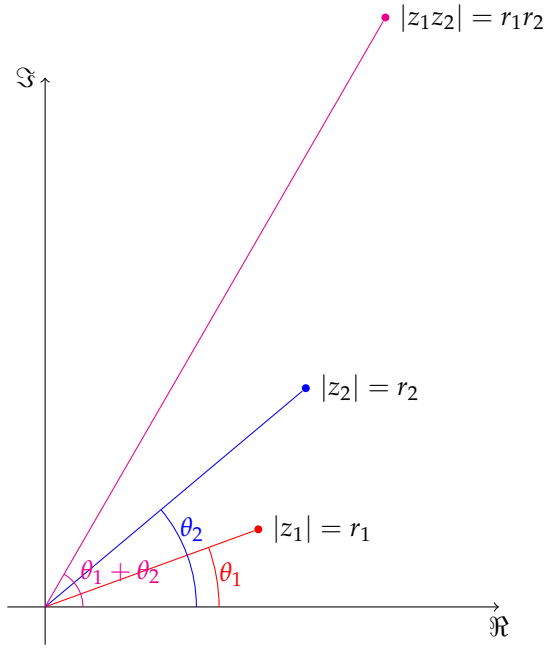


Figure 6.9: Multiplication of complex numbers in the Argand plane

6.3.4 De Moivre's theorem

De Moivre* made the observation that multiplication of a complex number by itself can be iterated. He thereby derived a formula for the polar form of a power of a complex number.

*Abraham de Moivre (1667 – 1754), French mathematician who helped develop the theory of probability. From his twenties, he lived as a Huguenot exile in London.

Theorem 6.23 — de Moivre. Let $r > 0$, let $\theta \in \mathbb{R}$, and let $n \in \mathbb{Z}$. Then

$$(r(\cos \theta + i \sin \theta))^n = r^n (\cos n\theta + i \sin n\theta).$$

Proof. We start by proving the result for $n \in \mathbb{N}$. We use proof by induction. The result is trivial in the case $n = 1$.

Let $k \in \mathbb{N}$ and assume $(r(\cos \theta + i \sin \theta))^k = r^k (\cos k\theta + i \sin k\theta)$.

Write $z = r(\cos \theta + i \sin \theta)$. Then

$$\begin{aligned} z^{k+1} &= (r(\cos \theta + i \sin \theta))(r(\cos \theta + i \sin \theta))^k \\ &= (r(\cos \theta + i \sin \theta)) \left(r^k (\cos k\theta + i \sin k\theta) \right) \\ &= r \cdot r^k (\cos(\theta + k\theta) + i \sin(\theta + k\theta)) \\ &= r^{k+1} (\cos((k+1)\theta) + i \sin((k+1)\theta)). \end{aligned}$$

This proves the induction step.

By induction, we have proved the result for every $n \in \mathbb{N}$.

It remains to prove the result for negative integers. ■

Exercise 6.24 Complete the proof of de Moivre's theorem by showing that if $n \in \mathbb{N}$, then

$$(r(\cos \theta + i \sin \theta))^{-n} = r^{-n}(\cos(-n\theta) + i \sin(-n\theta)).$$

Your proof should use the definition of z^{-n} , as given in Notation 6.1.

Using de Moivre's theorem, we can recover trigonometric identities involving powers.

Example 6.25 For every $\theta \in \mathbb{R}$,

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta, \quad \sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta.$$

By an application of the binomial theorem,

$$\begin{aligned} (\cos \theta + i \sin \theta)^3 &= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta \\ &= (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta). \end{aligned}$$

But, $(\cos \theta + i \sin \theta)^3$ can also be expressed by de Moivre's theorem. We deduce

$$\cos 3\theta + i \sin 3\theta = (\cos^3 \theta - 3 \cos \theta \sin^2 \theta) + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta).$$

By definition of equality of complex numbers (i.e. by comparing the real and imaginary parts of the above expression), we derive the claimed results.

Example 6.26 Adapting the above example:

1. Derive similar formulae for $\cos 4\theta$ and $\sin 4\theta$.
2. Thereby show that $\cos(\pi/12)$ is a solution of the equation

$$16x^4 - 16x^2 + 1 = 0.$$

Solution: For simplicity, we denote $c = \cos \theta$ and $s = \sin \theta$.

By the binomial theorem,

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= c^4 + 4ic^3s - 6c^2s^2 - 4ics^3 + s^4 \\ &= (c^4 - 6c^2s^2 + s^4) + i(4c^3s - 4cs^3). \end{aligned}$$

But this is also equal to $\cos 4\theta + i \sin 4\theta$, by de Moivre's theorem. By definition of equality, we deduce

$$\begin{aligned}\cos 4\theta &= \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta, \\ \sin 4\theta &= 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta.\end{aligned}$$

But $\sin^2 \theta = 1 - \cos^2 \theta$. Substituting this in our formula for $\cos 4\theta$ gives

$$\begin{aligned}\cos 4\theta &= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta) + (1 - \cos^2 \theta)^2 \\ &= \cos^4 \theta + 6 \cos^4 \theta - 6 \cos^2 \theta + 1 - 2 \cos^2 \theta + \cos^4 \theta \\ &= 8 \cos^4 \theta - 8 \cos^2 \theta + 1.\end{aligned}$$

Take $\theta = \pi/12$. Then $\cos 4\theta = \cos \pi/3 = 1/2$. Write $x = \cos \theta$. Substituting this in the above formula gives

$$\frac{1}{2} = 8x^4 - 8x^2 + 1.$$

Rearranging gives

$$16x^4 - 16x^2 + 1 = 0,$$

as required.

6.3.5 The logarithm of a complex number

We have seen that for real numbers, the natural logarithm is the inverse function of the exponential so that $\exp(\ln x) = x$. In contrast with the real logarithm function which is only defined for positive real numbers, the complex logarithm is defined for all nonzero complex numbers, but at a price: the function is not single-valued. This has to do with the periodicity or, equivalently, with the multivaluedness of the argument, $\arg(z)$, since we can also add multiples of 2π to it.

By analogy with the real natural logarithm, we define the complex logarithm as an inverse to the complex exponential function, $\exp(\ln(z)) = z$. In other words, we say that a logarithm of a nonzero complex number z , is any complex number w such that $e^w = z$. We define the function $\ln(z)$ by

$$w = \ln(z) \quad \text{if} \quad e^w = z. \quad (6.1)$$

From the periodicity of the exponential function, $e^{z+2\pi i} = e^z$, it follows that if $w = \ln(z)$ so is $w + 2\pi ik$ where $k \in \mathbb{Z}$. So, if $w = \ln(z)$ we also have $w + 2\pi ik = \ln(z)$. Therefore we see that $\ln(z)$ is a multivalued function.

Consider now the polar form $z = re^{i\theta}$. This can be rewritten as $z = e^{\ln(re^{i\theta})}$ by direct substitution into (6.1), which gives

$$z = e^{\ln(r)+i\theta}.$$

Now taking logarithms on both sides yields

$$\ln(z) = \ln(r) + i\theta$$

where r and θ are real. As θ is ambiguous (adding $2\pi\theta$ does not change $z = re^{i\theta}$), so is $\ln(z)$. Using the modulus and argument of z , the complex logarithm can also be written in the form

$$\ln(z) = \ln|z| + i \arg(z),$$

where the ambiguity in the imaginary part is manifest.

6.4 Root extraction

6.4.1 Equality of complex numbers in polar form

Suppose $z = r(\cos \theta + i \sin \theta)$ and $w = s(\cos \phi + i \sin \phi)$ with $r, s > 0$. We have

$$\begin{aligned} \Re(z) &= r \cos \theta, & \Re(w) &= s \cos \phi, \\ \Im(z) &= r \sin \theta, & \Im(w) &= s \sin \phi. \end{aligned}$$

If $z = w$, then $\Re(z) = \Re(w)$ and $\Im(z) = \Im(w)$.

We infer

$$r \cos \theta = s \cos \phi, \quad r \sin \theta = s \sin \phi.$$

Squaring and adding,

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta = s^2 \cos^2 \phi + s^2 \sin^2 \phi.$$

As $1 = \cos^2 \theta + \sin^2 \theta$ and $1 = \cos^2 \phi + \sin^2 \phi$, we deduce $r^2 = s^2$. But $r, s > 0$, so $\boxed{r = s}$.

Substituting in the previous equations and dividing through by r (which we can do as $r \neq 0$),

$$\cos \theta = \cos \phi, \quad \sin \theta = \sin \phi.$$

By standard results on trigonometric functions, we infer $\phi = \theta + 2n\pi$, for some $n \in \mathbb{Z}$.

Conversely, it's clear that if $r = s$ and $\phi = \theta + 2n\pi$, then $z = w$. We have proved a result on the equality of complex numbers in polar form.

Lemma 6.27 Let z, w be non-zero complex numbers. Write $z = r(\cos \theta + i \sin \theta)$ and $w = s(\cos \phi + i \sin \phi)$ with $r, s > 0$ and $\theta, \phi \in \mathbb{R}$. We have $z = w$ if and only if $r = s$ and $\phi = \theta + 2n\pi$, for some $n \in \mathbb{Z}$.

6.4.2 Complex polynomials

A complex polynomial of degree n is an expression of the form,

$$P(z) = c_0 + c_1 z + c_2 z^2 + \cdots + c_n z^n$$

here, c_i ($i = 0, \dots, n$ and $c_n \neq 0$) are complex constants and $z = x + iy$ is a complex variable.

The Fundamental Theorem of Algebra asserts that every non constant, single variable polynomial with complex coefficients, $P(z)$ has at least one root. Note that this includes real polynomials since every real number is a complex number with imaginary part equal to 0. In general, an n degree polynomial will have n roots (in some cases these are repeated roots so that the number of solutions is less than n). Moreover, any polynomial can be factorised as follows

$$P(z) = c_n (z - a_1) (z - a_2) \cdots (z - a_n)$$

where $a_1, a_2, \dots, a_n$ are the (complex) roots. Even if the c_i coefficients are real, the roots can still be complex. However, in this case, the roots occur in complex pairs, i.e. if a_1 is complex, $\overline{a_1}$ will also be a root. A simple example is

$$P(z) = z^2 + 1 = (z - i)(z + i)$$

where the roots $z = i$ and $\bar{z} = -i$ are complex conjugates.

It follows that if $P(z)$ is a polynomial with real coefficients of degree n and n is odd, that there will be at least one real root.

Example 6.28 Find all $z \in \mathbb{C}$ for which $z^2 = 1 + i$.

First note that every solution to this equation must be non-zero. Write z in polar form. We seek to find all z in the form $z = r(\cos \theta + i \sin \theta)$ that solve the equation. The polar form of $1 + i$ is

$$1 + i = \sqrt{2}(\cos \pi/4 + i \sin \pi/4).$$

By de Moivre's theorem, we must solve

$$r^2(\cos 2\theta + i \sin 2\theta) = \sqrt{2}(\cos \pi/4 + i \sin \pi/4).$$

From our result on equality of complex numbers in polar form, this equation holds if and only if

$$r^2 = \sqrt{2}, \quad 2\theta = \pi/4 + 2n\pi,$$

for some $n \in \mathbb{Z}$. The former equation holds if and only if $r = 2^{1/4}$. The latter holds if and only if $\theta = \pi/8 + n\pi$, for some $n \in \mathbb{Z}$. That is, θ is one of

$$\dots, -\frac{15\pi}{8}, -\frac{7\pi}{8}, \frac{\pi}{8}, \frac{9\pi}{8}, \frac{17\pi}{8}, \dots$$

It may seem as though there are infinitely many solutions to the equation. But, by our result on equality of complex numbers in polar form, we know that two solutions are equal if the θ values differ by a multiple of 2π . So, for instance,

$$2^{1/4} \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right) = 2^{1/4} \left(\cos \frac{17\pi}{8} + i \sin \frac{17\pi}{8} \right).$$

We'll choose the convention that θ is the principal value of the argument of z . That is, we choose $\theta \in (-\pi, \pi]$. The only values of θ in this range are $\pi/8$ and $-7\pi/8$. We have the solutions*

$$\begin{aligned} z_1 &= 2^{1/4} \left(\cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right), \\ z_2 &= 2^{1/4} \left(\cos \left(-\frac{7\pi}{8} \right) + i \sin \left(-\frac{7\pi}{8} \right) \right), \end{aligned}$$

*There is no significance to my labelling of one of the solutions as z_1 and the other as z_2 .

These solutions are plotted in Figure 6.10. Note that the argument of z_1 is half the argument of $1 + i$. The points z_1 and z_2 are antipodal to each other on the circle of radius $2^{1/4}$, centred at the origin.

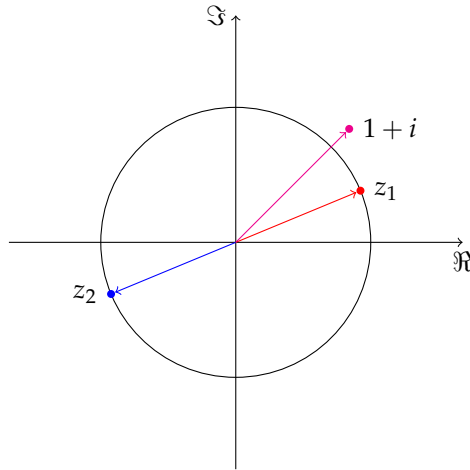


Figure 6.10: Square roots of $1 + i$

Exercise 6.29 Find all $z \in \mathbb{C}$ for which $z^2 = 1 + i\sqrt{3}$.

Example 6.30 Find all $z \in \mathbb{C}$ for which $z^3 = 1 + i$.

As before, write $z = r(\cos \theta + i \sin \theta)$ with $r > 0$ and $\theta \in \mathbb{R}$. We need to solve

$$r^3 (\cos 3\theta + i \sin 3\theta) = 2^{1/2} (\cos \pi/4 + i \sin \pi/4).$$

That is, to solve $r^3 = 2^{1/2}$ and $3\theta = \pi/4 + 2n\pi$, for $n \in \mathbb{Z}$.

The solutions are $r = 2^{1/6}$ and $\theta = \pi/12 + 2n\pi/3$, for $n \in \mathbb{Z}$. The only values of θ that lie in the interval $(-\pi, \pi]$ are

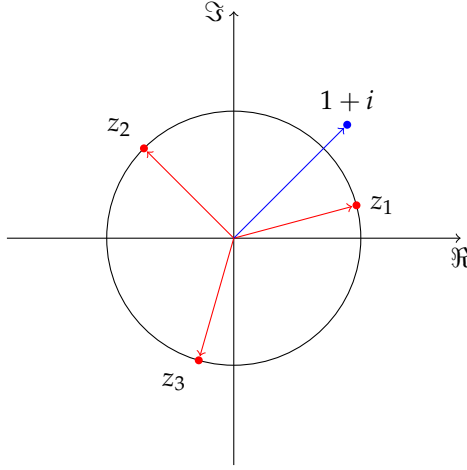
$$\theta_1 = \pi/12, \quad \theta_2 = 9\pi/12, \quad \theta_3 = -7\pi/12.$$

Thus, the complex cube roots of $1 + i$ are

$$z_k = 2^{1/6} (\cos \theta_k + i \sin \theta_k),$$

for $k = 1, 2, 3$. This is shown in Figure 6.11.

Exercise 6.31 Find all complex cube roots of -2 .

Figure 6.11: Cube roots of $1 + i$

6.5 Distances and the triangle inequality

Let $z = x + iy$ be a complex number with $x, y \in \mathbb{R}$. By definition, $|z| = \sqrt{x^2 + y^2}$. Now $\Re(z) = x$, so $\Re(z)^2 = x^2 \leq x^2 + y^2$. Taking square roots of both sides, we deduce $|\Re(z)| \leq |z|$. Likewise, $|\Im(z)| \leq |z|$.

Moreover, note that $z + \bar{z} = (x + iy) + (x - iy) = 2x$. That is, $z + \bar{z} = 2\Re(z)$. Likewise, $z - \bar{z} = 2\Im(z)$. Using these results, we will prove the so-called triangle inequality.

Lemma 6.32 — Triangle inequality. Let $z, w \in \mathbb{C}$. Then $|z + w| \leq |z| + |w|$.

Proof. As both sides of the inequality are non-negative, it suffices to prove $|z + w|^2 \leq (|z| + |w|)^2$. But

$$\begin{aligned} |z + w|^2 &= (z + w)\overline{(z + w)} \\ &= (z + w)(\bar{z} + \bar{w}) \\ &= z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w} \\ &= |z|^2 + z\bar{w} + \overline{z\bar{w}} + |w|^2 \\ &= |z|^2 + 2\Re(z\bar{w}) + |w|^2. \end{aligned}$$

By our previous result, $\Re(z\bar{w}) \leq |z\bar{w}| = |z||w|$. This gives

$$|z + w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2,$$

proving the result. ■

The distance between two complex numbers is the modulus of their difference.

The geometrical intuition behind the triangle inequality is made clear by first introducing the notion of distance between two complex numbers.

Definition 6.33 Let $\mathbb{R}_{\geq 0}$ denote the set of non-negative real numbers. The *distance function*

$$d : \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{R}_{\geq 0}$$

is given by the rule $d(z, w) = |w - z|$.

For example, let $z = 3 + i$ and $w = 2 + 5i$, then the distance between z and w is

$$d(z, w) = |(3 + i) - (2 + 5i)| = |1 - 4i| = \sqrt{1^2 + 4^2} = 5.$$

We can prove important properties of the distance function.

Lemma 6.34 1. For all $z, w \in \mathbb{C}$, we have $d(z, w) \geq 0$.

Moreover, $d(z, w) = 0$ if and only if $z = w$.

2. For all $z, w \in \mathbb{C}$, we have $d(z, w) = d(w, z)$.

3. For all $u, w, z \in \mathbb{C}$, we have $d(u, z) \leq d(u, w) + d(w, z)$.

The first two properties are restatements about equivalent properties concerning the modulus function. Likewise, the last property is a restatement of the triangle inequality. To prove it, let u, w, z be complex numbers. Write $a = u - w$ and $b = w - z$. Then

$$d(u, w) + d(w, z) = |a| + |b| \geq |a + b|,$$

by the triangle inequality. But

$$|a + b| = |(u - w) + (w - z)| = |u - z| = d(u, z),$$

giving the desired result.

If u, w, z are treated as the vertices of a triangle, then this last

property asserts that the length of any side of the triangle is at most the sum of the lengths of the other two sides.

Remark 6.35

The triangle inequality is used heavily in the branch of mathematics known as analysis.